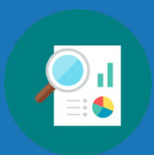


Unit 13 Overview: Algebraic Expressions

Unit Narrative



In this unit, students begin by applying the order of operations learned to evaluate algebraic expressions using substitution to evaluate algebraic expressions that include grouping symbols and exponents. As the unit progresses, students apply their knowledge of the properties of operations to rewrite algebraic expressions, generate equivalent expressions, and determine whether equations and inequalities are true.

Section 1 focuses on students exploring how to use substitution to evaluate algebraic expressions with grouping symbols and exponents. Students apply the properties of operations to rewrite algebraic expressions and generate equivalent expressions. The complexity of the values students encounter progresses from positive integers to negative integers and then to rational numbers.

Section 2 focuses on making connections between algebraic expressions and written descriptions. In Lesson 5, students develop an understanding of how to write algebraic expressions from bar models, mathematical written descriptions, and real-world written descriptions. Next, students extend their skills to write real-world written descriptions from algebraic expressions in Lesson 6.

Section 3 provides an opportunity for students to engage with adding and subtracting linear expressions with rational coefficients in meaningful ways. In Lesson 7, students use algebra tiles to explore the concept of combining like terms and connect prior knowledge from Unit 6 of forming zero pairs to eliminate terms. Next, in Lessons 8 and 9, students apply these skills to add and subtract linear expressions with rational coefficients.

In Section 4, students continue to explore the concept of equivalence in algebraic expressions. Students build on their understanding of evaluating expressions from Section 1 to determine whether two expressions are equivalent using properties of operations. Students also explore how to use substitution to determine whether equations or inequalities comparing the value of two expressions are true.

Unit At-A-Glance

Section 1: Evaluating Expressions and Applying Properties of Operations	Benchmarks Addressed	Lesson 1	Evaluating Expressions
	MA.6.AR.1.3 MA.6.AR.1.4	Lesson 2	Evaluating Algebraic Expressions Using Substitution
		Lesson 3	Applying Properties of Operations
		Lesson 4	Generating Equivalent Algebraic Expressions
Section 2: Representing Contexts with Algebraic Expressions	Benchmarks Addressed	Lesson 5	Writing Algebraic Expressions
	MA.6.AR.1.1	Lesson 6	Writing Descriptions of Real-World Contexts From Algebraic Expressions
Section 3: Adding and Subtracting Linear Expressions	Benchmarks Addressed	Lesson 7	Adding and Subtracting Linear Expressions With Integer Coefficients
	MA.7.AR.1.1	Lesson 8	Adding Linear Expressions with Rational Coefficients
		Lesson 9	Subtracting Linear Expressions with Rational Coefficients
Section 4: Equivalent Expressions	Benchmarks Addressed	Lesson 10	Equivalent Linear Expressions
	MA.6.AR.2.1 MA.7.AR.1.2	Lesson 11	What Values Makes an Equation or Inequality True?
		Lesson 12	Equivalent Expressions Using Properties of Operations



Differentiation

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Engagement <i>The “WHY” of learning</i>	- In the Lesson 1 Warm-Up, student interest is ignited as they are asked to determine the correct value of an expression that has led to controversy on social media. - The real-world problems embedded throughout the unit help students gain an appreciation for the utility of the mathematics they are learning. The contexts presented will also engage students if they are related to their interests. If needed, you can modify the contexts in the problems based on the interests of your students. math expression is dividing the internet. What is the value of $8 \div 2(2 + 2)$?
Representation <i>The “WHAT” of learning</i>	- Throughout the unit, allow students to use visual models, such as colored counters, area models, bar models, and number lines the relationships explored. - Students engage in partner, small group, and whole-class discussions throughout the unit. These discussions provide an auditory entry point and help students build confidence and understanding. - Algebra tiles come in all forms, and so do the representations for their expressions in students' minds. Consider this representation from a digital Desmos Activity. Both are “correct” representations of $3x + 6$, just in different forms. - Do you think Gabriel and Kayla are representing the same algebraic expression? Why or why not? - Which represents $3(x + 2)$ and which represents $3x + 6$? Is one more “correct”? Why or why not? Gabriel Kayla
Action & Expression <i>The “HOW” of learning</i>	- The Wrap-Ups throughout the unit allow students to demonstrate their understanding, confidence, and mastery of the learning targets using different methods, such as through reflection, solving example problems, or drawing a visual model. In the Lesson 1 Wrap Up, students are asked to write letters to two students explaining the mistakes in their work. Consider providing alternative methods for students who struggle expressing themselves in writing, such as using an audio recorder, like FlipGrid, or speech-to-text software. - If a component requires a written explanation, consider allowing the student to draw a diagram using a variety of models (algebra tiles, sketches, etc.) or explain their reasoning verbally initially instead of providing a written explanation. Additional entry points for expressing their understanding will provide more information regarding students' understanding and progression.

ELL Information

Close Reading

Close Reading is an effective decoding strategy for building literacy for students early in their language acquisition stages. Like the Three Reads Language Routine, students intentionally target specific pieces of information in order to scaffold their decoding process and reduce anxiety for word problems.

Consider opportunities to incorporate Close Reading:

- Lesson 1: 13.1.2 Guided Instruction
- Lesson 2: 13.2.2 Collaboration
- Lesson 4: 13.4.4 Your Turn!

Mark Up Text

In Lessons 5 and 6, students will be solving real-world problems. This topic is text and reading heavy. When students translate words into numbers, algebraic expressions, or equations, encourage them to use a highlighter to highlight phrases and words to call attention to action words and the action of a story in real-world contexts.

The teacher will use a word problem where students will be able to identify which parts would need to be represented in numbers and how to write those numbers in an expression or equation.

The teacher will provide students with the following symbols:

- ✓ - background knowledge; they know what to do.
- ? - confusing information; is this needed to solve the problem?
- ! - marks where a variable is needed to represent a number.
- + - extra info in word problem; not needed.

Using a small piece of text, the teacher will model for students how you would mark up before releasing them to do the same. Students will read the text slowly and carefully, focusing on one sentence at a time and marking text based on the key. Then they will turn to a partner to discuss each sentence and compare where they made marks. For the first few examples, have groups share their decisions out to the class. Then guide students through turning the marked-up text into an expression. ELL students may find this a helpful intermediary step, so encourage them to use this strategy even as you release them to work independently.

All glossary terms are available in multiple languages, including English, Spanish, Haitian Creole, and American Sign Language, as support for ELL students. Consider pairing the following videos with strategies from “Additional Differentiated Support.”

- | | | |
|------------------------------------|--|---|
| • Associative property of addition | • Associative property of multiplication | • Coefficient |
| • Commutative property of addition | • Commutative property of multiplication | • Distributive property of multiplication over addition |
| • Equation | • Expression | • Order of operations |

Name _____

Use CLOSE READING with word problems!

Problem: _____

1st Read: What is this problem about?

2nd Read: What numbers are important in this problem and why?

3rd Read: Create a model. At the end of each sentence, stop and add to or change your model as needed.

4th Read: What equation would represent the problem and the unknown?

Additional Differentiated Support

Scaffolding Resources	On-Ramp
Notice the examples used within the 7th Grade benchmarks addressing equivalent expressions. - MA.7.AR.1.1: $(7x - 4) - (2 - \frac{1}{2})$ is equivalent to $\frac{15}{2}x - 6$. - MA.7.AR.1.2: Are the expressions $\frac{4}{3}(6 - x) - 3x$ and $8 - \frac{5}{3}x$ equivalent? Rational coefficients expressed as fractions are an expectation of these benchmarks and spiraled throughout the unit. Fluency with fractions can become a barrier to developing a working understanding of equivalence. While there may be moments to substitute fractional coefficients with integers or decimals, to maintain the focus on equivalence, notice both examples in the AR benchmarks require fluency with fractions. This directly relates to the 7.NSO benchmarks addressed in Unit 11 where procedural fluency in operations with rational numbers is the expectation. - Assign small groups during Try It or Your Turn!, specifically for students struggling with fluency in fractions. - Represent integers (2) as fractions ($\frac{4}{2}$) where appropriate, breaking down the anxiety and barrier of fractions. Use resources from Units 1 or 11 to address potential gaps in understanding from prior units. Pair – Share Group students in pairs and have them take turns verbally stating each of the properties of operations and an expression. The student duplicating the graph can ask clarifying questions to complete the graph. - The other student will try to write an equivalent expression using the property described. - The student writing an equivalent expression can ask clarifying questions. - Both students compare expressions to ensure they are equivalent. - Students trade roles and repeat with another property and expression.	On-Ramp to 6th Grade provides video and practice tools for students to scaffold the following skills: - Adding Two-Digit and One-Digit Numbers - Addition Fluency - Subtraction Fluency - Evaluate Expressions - Write and Interpret Expressions - True or False Equations - Real-World Addition/Subtraction Multi-Step Problems - Real-World Multiplication/Division Multi-Step Problems - Real-World Multiple Operations Multi-Step Problems - Adding Decimals - Subtracting Decimals - Recognize and Generate Equivalent Fractions Using Models - Commutative Property of Multiplication - Associative Property of Multiplication - Distributive Property - Multiplying Decimals



Section Overview

Section 1: Evaluating Expressions and Applying Properties of Operations (Lessons 1-4)

Benchmarks Addressed	Learning Targets	
MA.6.AR.1.3 Evaluate algebraic expressions using substitution and order of operations. - Clarification 1: Within this benchmark, the expectation is to perform all operations with integers. - Clarification 2: Refer to Properties of Operations, Equality and Inequality (Appendix D). MA.6.AR.1.4 Apply the properties of operations to generate equivalent algebraic expressions with integer coefficients. - Clarification 1: Properties include associative, commutative and distributive. - Clarification 2: Refer to Properties of Operations, Equality, and Inequality (Appendix D).	Lesson 1	- I can apply order of operations to an algebraic expression. - I can evaluate expressions that include exponents.
	Lesson 2	I can evaluate algebraic expressions using substitution.
	Lesson 3	I can apply the associative, commutative, and distributive properties to generate equivalent expressions.
	Lesson 4	I can apply properties of operations to generate equivalent expressions.

Section 2: Representing Contexts with Algebraic Expressions (Lessons 5-6)

Benchmark Addressed	Learning Targets	
MA.6.AR.1.1 Given a mathematical or real-world context, translate written descriptions into algebraic expressions and translate algebraic expressions into written descriptions.	Lesson 5	- I can translate a written description into an algebraic expression. - I can translate a real-world description into an algebraic expression.
	Lesson 6	I can create a real-world context from an algebraic expression.

Section 3: Adding and Subtracting Linear Expressions (Lessons 7-9)

Benchmark Addressed	Learning Targets	
MA.7.AR.1.1 Apply properties of operations to add and subtract linear expressions with rational coefficients. <i>Clarification 1: Instruction includes linear expressions in the form $ax \pm b$ or $b \pm ax$, where a and b are rational numbers.</i> <i>Clarification 2: Refer to Properties of Operations, Equality, and Inequality (Appendix D).</i>	Lesson 7	I can use algebra tiles to add and subtract linear expressions.
	Lesson 8	I can add linear expressions with rational coefficients.
	Lesson 9	- I can subtract linear expressions with rational coefficients. - I can apply properties of operations to add and subtract linear expressions with rational coefficients.

Section 4: Equivalent Expressions (Lessons 10-12)

Benchmark Addressed	Learning Targets	
MA.6.AR.2.1 Given an equation or inequality and a specified set of integer values, determine which values make the equation or inequality true or false. <i>Clarification 1: Problems include the variable in multiple terms or on either side of the equal sign or inequality symbol.</i> MA.7.AR.1.2 Determine whether two linear expressions are equivalent. <i>Clarification 1: Instruction includes using properties of operations accurately and efficiently.</i> <i>Clarification 2: Instruction includes linear expressions in any form with rational coefficients.</i> <i>Clarification 3: Refer to Properties of Operations, Equality, and Inequality (Appendix D).</i>	Lesson 10	I can verify using substitution that linear expressions are equivalent.
	Lesson 11	I can determine if a given integer value makes an equation or inequality true.
	Lesson 12	- I can use properties of operations to write equivalent forms of an algebraic expression. - I can determine whether two linear expressions are equivalent.



Vertical Alignment

Grade 6 Accelerated Unit 13
Algebraic Expressions



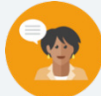
Grade 7 Accelerated Unit 1 and 14
Equations and Inequalities
Equivalent Algebraic Expressions

Prior Course(s)	Course Level	Future Course(s)
MA.5.AR.2.1 Translate written real-world and mathematical descriptions into numerical expressions and numerical expressions into written mathematical descriptions. MA.5.AR.2.2 Evaluate multi-step numerical expressions using order of operations. MA.5.AR.2.3 Determine and explain whether an equation involving any of the four operations is true or false.	MA.6.AR.1.1 Given a mathematical or real-world context, translate written descriptions into algebraic expressions and translate algebraic expressions into written descriptions. MA.6.AR.1.3 Evaluate algebraic expressions using substitution and order of operations. MA.6.AR.1.4 Apply the properties of operations to generate equivalent algebraic expressions with integer coefficients. MA.6.AR.2.1 Given an equation or inequality and a specified set of integer values, determine which values make the equation or inequality true or false. MA.7.AR.1.1 Apply properties of operations to add and subtract linear expressions with rational coefficients. Example: $(7x - 4) - (2 - \frac{1}{2})$ is equivalent to $\frac{15}{2}x - 6$. MA.7.AR.1.2 Determine whether two linear expressions are equivalent. Example: Are the expressions $\frac{4}{3}(6 - x) - 3x$ and $8 - \frac{5}{3}x$ equivalent?	MA.7.AR.2.2 Write and solve two-step equations in one variable within a mathematical or real-world context, where all terms are rational numbers. MA.8.AR.1.2 Apply properties of operations to multiply two linear expressions with rational coefficients. MA.8.AR.1.3 Rewrite the sum of two algebraic expressions having a common monomial factor as a common factor multiplied by the sum of two algebraic expressions.

13.1 Evaluating Expressions

Lesson Introduction

 Benchmark	MA.6.AR.1.3 Evaluate algebraic expressions using substitution and order of operations.		 Learning Targets	- I can apply order of operations to an algebraic expression. - I can evaluate expressions that include exponents.	
 Benchmark Coherence	Building On		Working Toward		
	MA.5.AR.2.1 Translate written real-world and mathematical descriptions into numerical expressions and numerical expressions into written mathematical descriptions.		MA.7.NSO.2.1 Solve mathematical problems using multi-step order of operations with rational numbers including grouping symbols, whole-number exponents, and absolute value.		
 Essential Questions	- How do you evaluate an expression by applying the order of operations? - How do you evaluate an expression when it includes an exponent? - What information is needed in order to “evaluate” an algebraic expression?				
 Lesson Narrative	In this first lesson of the unit, students extend their understanding of evaluating numeric expressions using the order of operations to evaluate algebraic expressions. Students connect numeric expressions with and without grouping symbols to mathematical models, deepening their understanding of these symbols. Students were introduced to evaluating exponents in Unit 6 and evaluating numeric expressions involving exponents in Unit 11. This lesson will be the first time they are evaluating exponents within algebraic expressions that include multiple operations.				
 Component Summary	13.1.1 Warm-Up (~5 min.)	Evaluate a numeric expression to resolve a disagreement (<i>SE p. 263</i>)	13.1.4 Guided Instruction (10 min.)	Evaluate numeric expressions with exponents (<i>SE p. 267</i>)	
	13.1.2 Guided Instruction (15 min.)	Extend evaluating numeric expressions using the order of operations to algebraic expressions (<i>SE p. 264</i>)	13.1.5 Try It (10 min.)	Evaluate algebraic expressions with exponents (<i>SE p. 268</i>)	
	13.1.3 Collaboration (5-10 min.)	Explore how exponents can be written out as products (<i>SE p. 266</i>)	13.1.6 Wrap-Up (~5 min.)	Find and explain errors in students’ work (<i>SE p. 269</i>)	
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> - Order of operations - Variable - Evaluate - Value - Parentheses <u>Additional Terms:</u> N/A		(optional) Colored pencils		N/A

	Common Misconceptions	Things to Consider	
 Teacher Tips	- Students may substitute the variable as a digit instead of multiplying it by the coefficient. For example, in $5x$, where $x = 4$, students may write 54 instead of $5(4)$. - Students may prioritize left to right operations in lieu of parenthesis. For example, during the second expression in 11.1.2 Collaboration, $3 \times (5 + 4)$, students may multiply 3×5 first.	- This lesson has several key glossary terms necessary for comprehension. Consider having students add to interactive notebooks or create a Frayer Model for key vocabulary. - Consider modeling examples and allowing students to demonstrate their mastery with chips to make sense of expressions. - There are many misconceptions about the order of operations. The teaching of the mnemonic device of PEMDAS leads students to perceive that the order of PEMDAS is rigid, where it is not. For example, in the expression $30 - 4(3 + 7) + 12 \div 3$, there are options as to where to begin. Students actually have a choice and may first simplify the $3 + 7$ in the parentheses, distribute the 4 to the 3 and to the 7, or perform $12 \div 3$ before doing any other computation—all without affecting an accurate outcome. For this reason, the lessons do not empathize this mnemonic device. Encourage your students to consider different entry points and discuss them. - If time is an issue, consider assigning part of all of the 13.1.5 Try It for homework.	
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard	
	Compare and Connect For 13.1.5 Try It, consider using the Compare and Connect routine by encouraging students to compare how the values of the expressions change depending on the order in which the operations are performed.	MA.K12.MTR.4.1: Discussion and Reflection 13.1.2 Guided Instruction opens by presenting students with three hypothetical evaluations of an expression and calls on them to find which evaluation is correct. Clarifications for this standard include recognizing an error and suggesting how to fix it. Encourage students to not only find the error but formalize how to prevent it from happening	
 Differentiate Instruction	For 13.1.5 Try It, consider placing students in differentiated groups during this activity. Students who have demonstrated a good grasp of the content can be assigned to work independently to complete the questions, while students who may still be struggling can work through the content with guidance from the teacher before trying independently.		
Lesson Notes:			

13.1.1 Warm-Up: Evaluating Numeric Expressions

Component Overview: Students examine two different values to determine which one correctly represents the expression, then work to find the value of another expression independently.



13.1.1 Warm-Up: Evaluating Numeric Expressions

This math expression is dividing the internet.

What is the value of $8 \div 2(2 + 2)$?



Poll the Class

Engage students with a Poll the Class routine to determine whose answer they agreed with. Consider having students write their responses on a whiteboard and holding it up or show either one or two (using their hand) to indicate who they agree with. Encourage students to discuss how they know who is correct and the error the other person made.

- Do you agree with Alex or Ms. B? Explain or show why using order of operations.

Ms. B

$$8 \div 2(2 + 2)$$

$$8 \div 2(4)$$

$$4(4)$$

$$16$$

- Find the value of the expression. Show your work.

$$4 - 2 \times 6 \div 3 + 7$$

$$4 + (-2) \times 6 \div 3 + 7$$

$$4 + (-12) \div 3 + 7$$

$$4 + (-4) + 7$$

$$4 - 4 + 7$$

$$0 + 7$$

$$7$$



Students may not have social media, but they could recreate the activity for homework by polling their friends and family.

13.1.2 Guided Instruction: Applying Order of Operations to Algebraic Expressions

Component Overview: Activate students' prior knowledge both from previous units and from this lesson in the 13.1.1 Warm-Up. Position students as sense makers in this error analysis scenario in order to elicit increasingly precise and mathematical responses on evaluating expressions.



13.1.2 Guided Instruction: Applying Order of Operations to Algebraic Expressions

Recall from 5th grade that an order of operations is used to evaluate numeric expressions with multiple operations. Grouping symbols such as parentheses are used to indicate which operations should be performed first. The following order is used.

- Step 1: Perform operations within grouping symbols.
- Step 2: Evaluate exponents.
- Step 3: Multiply and divide in order from left to right ($\rightarrow$).
- Step 4: Add and subtract in order from left to right ($\rightarrow$).

1. Three students completed the warm-up. The steps in each student's process are shown.

Leila's Work	Fabian's Work	Jaylah's Work
$4 - 2 \times 6 \div 3 + 7$	$4 - 2 \times 6 \div 3 + 7$	$4 - 2 \times 6 \div 3 + 7$
$2 \times 6 \div 3 + 7$	$4 - 12 \div 3 + 7$	$4 - 12 \div 3 + 7$
$12 \div 3 + 7$	$4 - 4 + 7$	$4 - 4 + 7$
$4 + 7$	$4 - 11$	$0 + 7$
11	-7	7

- a. Complete the statements about which student correctly applied the given steps of the order of operations.

- ☐ Leila
- ☐ Fabian
- ☒ Jaylah

evaluated the numerical expression correctly.

She first performed all

- ☒ multiplication and division
- ☐ addition and subtraction

in

order from left to right and then performed all

- ☐ multiplication and division
- ☒ addition and subtraction

in order from left to right.



Error Analysis

Engage students in an error analysis of each student's work from question 1. Consider having students partner up to analyze the student work and determine the errors before participating in a whole class discussion to address the misconceptions displayed in each response.



Challenge students to not only name the errors each student made but to also write an explanation for how to fix and prevent them from making the same mistakes in the future.

13.1.2 Guided Instruction: Applying Order of Operations to Algebraic Expressions (continued)

- b. With your partner, discuss the work of each student, Leila, Fabian, and Jaylah. Two of these students made a mistake somewhere. Circle the mistakes that you and your partner found.

Leila went from left to right.

Fabian did addition before subtraction.

2. Omari evaluated Expression A and Expression B. His work is shown.

Expression A	Expression B
$12 + 24 \div 6 - 2 \times 2$	$12 + 24 \div (6 - 2 \times 2)$
$12 + 4 - 2 \times 2$	$12 + 24 \div (6 - 4)$
$12 + 4 - 4$	$12 + 24 \div (2)$
$16 - 4$	$12 + 12$
12	24

- a. With your partner, discuss the two original expressions that Omari evaluated and his corresponding answers.
- b. Omari's partner gave him a thumbs up on both of his answers. Why did Omari arrive at two different values when evaluating these expressions?
Omari had two different answers because Expression B contains parentheses that are not in Expression A. Following order of operations Expression B is completed differently than Expression A.

Like numeric expressions, algebraic expressions can be evaluated using order of operations when given a specific value for the **variable(s)**.

In an algebraic expression, a **variable** is a letter that represents a number. It is called a **variable** because the value used to replace the letter can **vary**.



Turn and Talk

Before answering question 2, have students participate in a Turn and Talk routine to formalize their answers before circling the mistakes. Encourage students to notice the differences between the two expressions and consider how/why that might impact the value of the expressions.



- What is the same about the expressions? What is different?
- How do parentheses impact the order of operations? How do they impact the value of the expression?



Consider having students create a foldable, Frayer Model, or other type of interactive to allow students to synthesize the vocabulary outlined in this section. Examples, nonexamples, and verbal descriptions will provide multiple pathways for students to engage with the content in meaningful ways that encourage conceptual understanding over rote memorization.

13.1.2 Guided Instruction: Applying Order of Operations to Algebraic Expressions (continued)

3. When evaluating algebraic expressions, first substitute the given **number** for the variable(s). Then, apply the given steps for order of operations.



In mathematics, **evaluate** is an instruction that indicates the answer is a **value**.

4. Evaluate $\frac{n}{6} + 2$ when $n = 12$.

$$\frac{12}{6} + 2 = 2 + 2 = 4$$

In numeric expressions, multiplication can be represented in different ways. Each expression below is a way to represent "three times five."

$$3 \times 5$$

$$3 \cdot 5$$

$$3(5)$$

In algebraic expressions, the multiplication symbol, $\times$, could be confused with the variable x . Therefore, writing a number next to a variable is a better way to note multiplication when using variables.

$3x$ means "three times x ."

Multiplication of a number and a variable could also be represented using the multiplication dot, $3 \cdot x$, or parentheses, $3(x)$, but $3x$ will be the most common way.

5. Explain why $3 \cdot 5$ cannot be written as 35.

The dot in the problem represents multiplication. You need to multiply 3 times 5.

6. What is the value of $3x$ if $x = 5$?

$$3 \cdot 5 = 15$$



When substituting values into algebraic expressions, it is helpful to put the substituted value in **parentheses** when the variable is being multiplied by a known value.

7. Find the value of $7m - (3 + b - 12) \div 2$ if $m = 4$ and $b = 15$. Show your work.

$$7(4) - (3 + 15 - 12) \div 2$$

$$28 - (18 - 12) \div 2$$

$$28 - (6) \div 2$$

$$28 - 3 = 25$$



- How do the parentheses impact the evaluation of the expression?
- How would the value change if the parentheses were moved around the first part of the expression?

13.1.3 Collaboration: Order of Operations

Component Overview: Read the red boxes out loud to students, then have them work in partner pairs to find the numeric value of the given expression. After reviewing their work, engage the class in a discussion about the differences between their work and what is presented.



13.1.3 Collaboration: Order of Operations



An order of operations was first noted in First Year Algebra by Webster Wells and Walter Wilson Hart. The textbook was published in 1912.

- The problem below is an example from First Year Algebra.

EXAMPLE. Find the numerical value of the expression,

$$4ab + \frac{6c}{b} - d^3,$$

when $a = 4$, $b = 3$, $c = 5$, $d = 2$.

- What is the numerical value of the expression? Show your work.

$$\begin{aligned} &4(4)(3) + \frac{6(5)}{3} - 2^3 \\ &48 + \frac{30}{3} - 8 \\ &48 + 10 - 8 \\ &50 \end{aligned}$$

- Check your work with your partner.



Notice and Wonder

Consider implementing a Notice and Wonder routine for question 1. Ask students to identify what they notice and wonder about their work and the work from the textbook.



What are some differences between numeric and algebraic expressions? The textbook changed d^3 to $2 \cdot 2 \cdot 2$. The order of operations in the textbook did not include exponents, so each exponent was rewritten as multiplication.

- When evaluating numeric expressions, how many values are possible?
- When evaluating algebraic expressions, how many values are possible?
- What information is needed in order to evaluate an algebraic expression?

13.1.3 Collaboration: Order of Operations

c. The work shown in the textbook is below.

$$4ab + \frac{6c}{b} - d^3 = 4 \cdot 4 \cdot 3 + \frac{6 \cdot 5}{3} - 2^3 = 4 \cdot 4 \cdot 3 + \frac{6 \cdot 5}{3} - 2 \cdot 2 \cdot 2$$

$$= 48 + 10 - 8 = 50.$$

What is the difference between your work and the work from the book?

Emphasis should be placed on writing 2^3 as $2 \cdot 2 \cdot 2$. This is an illustration of how the order of operations was originally written without exponents. Instead, it was written out as a product.



An order of operations became more important when programmers needed a set of rules to program a computer to find values of numeric expressions. In 1883, Ada Lovelace described in her book about Charles Babbage's analytical engine, the first computer, that codes could be created so the device could handle letters and symbols along with numbers.



Convey to students that the original computers were what is known today as calculators. Their only purpose was to do mathematical operations. Students who have seen *Hidden Figures* may be able to relate to this because the Katherine Johnson, Mary Jackson, and Dorothy Vaughn were referred to as computers. In the film, their jobs appeared to be threatened by the installment of computers which were machines that would do the calculations they were doing by hand.

13.1.4 Guided Instruction: Evaluating Expressions with Exponents

Component Overview: Guide students through the instructions at the beginning of this component. Engage students in a discussion about why evaluating exponents should be done before other multiplication and division in the expression. Then have students work independently to evaluate the numeric expressions. Finally, engage students in a discussion to debrief the questions and consolidate their understandings.



13.1.4 Guided Instruction: Evaluating Expressions with Exponents

The work in the book, First Year Algebra, showed 2^3 written out as multiplication. This method can get complex as shown in the table.

$(-3)^3 + 2^6 \div -16 + 5$
Exponents written as multiplication
$((-3)(-3)(-3)) + (2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2) \div -16 + 5$

$(-3)^3 + 2^6 \div -16 + 5$
Evaluate expressions with exponents first
$-27 + 64 \div -16 + 5$

- When writing $(-3)^3$ and 2^6 using multiplication, both were written inside a set of parentheses.
- In order of operations, parentheses indicate that the operations inside should be done **before** other steps.
- Therefore, evaluating exponents is done before other multiplication and division when following the order of operations.

- Evaluate $(-3)^3 + 2^6 \div -16 + 5$.

$$\begin{aligned} & -27 + 64 \div -16 + 5 \\ & -27 + (-4) + 5 \\ & -31 + 5 = -26 \end{aligned}$$

- Evaluate each expression, then determine if the expressions are equivalent.

Expression A	Expression B	Equivalent?
$(-3)^4 = 81$	$-(3^4) = -81$	No
$-(9)^2 = -81$	$(-9)^2 = 81$	No
$(-5^2) = -25$	$(-5)^2 = 25$	No
$26 - 4^3$ $= 26 - 64$	$26 - (-4)^3$ $= 26 + 64$	No
$17 + (-2^4)$ $= 17 + (-16)$	$17 - (2^4)$ $= 17 - 16$	Yes



This activity provides an entry point for visual learners through the use of color-coding the expressions in the table.



Students may need a brief review of evaluating exponents, particularly with negative bases. The second example in 13.1.4 Guided Instruction provides an opportunity to monitor to assess formatively how the students approach this within the order of operations. Guide students in order to address common misconceptions as needed with specific students or the whole class through modeling.

13.1.4 Guided Instruction: Evaluating Expressions with Exponents (continued)

3. Describe a strategy that can be used when evaluating expressions with exponents that include a negative sign or a subtraction sign.

Answers will vary. Guide students to even/odd exponents, the placement of a negative in front of the parentheses as opposed to inside.

4. Elizabeth evaluated $b + a^3 \cdot 7 - 9$ when $a = -4$ and $b = 17$. Her work is below.

- a. Circle Elizabeth's error.

$$17 + (-4)^3 \cdot 7 - 9$$

$$17 + (-64) \cdot 7 - 9$$

$$(-47) \cdot 7 - 9$$

$$-329 - 9$$

$$-338$$

- b. Which step of the order of operations did she not apply correctly?
- Ⓐ Evaluate expressions with exponents.
 - Ⓑ Perform operations inside grouping symbols.
 - Ⓒ Perform multiplication and division in order from left to right.
 - Ⓓ Perform addition and subtraction in order from left to right.



The order of performing computations is first to work within grouping symbols using the order of operations. Then, simplify terms with exponents. Next, while reading from left to right, perform multiplication and division in the order in which it appears. Finally, while reading from left to right, perform addition and subtraction in the order in which it appears.



Solidify understanding by asking these questions:

- What are some differences between numeric and algebraic expressions? (Numeric expressions contain all known values while algebraic expressions include some unknown value(s).)
- When evaluating numeric expressions, how many answers are possible?
- What information is needed in order to "evaluate" an algebraic expression? (Specific values for the variable(s).)



While reviewing the order of operations in the vocabulary box, remind students that up to this point it has been shown that these are not hard and fast rules but should be used to guide them when they determine a value. Although there are some things where the order does not matter there are others where the order does matter. It may be helpful to create an anchor chart or reference cards with examples for students to refer to as they build their understanding.

13.1.5 Try It: Discovering the Importance of Left to Right

Component Overview: Students work independently to evaluate expressions in different ways, working to determine how changing grouping symbols or the order in which operations are performed changes the value of the expression. After discussing question 1, use your discretion to determine if students can work independently or would do better checking in with a partner regularly.



13.1.5 Try It: Discovering the Importance of Left to Right

1. Mia is evaluating the expression, $x^3 \div r \cdot (k^2 - 5)$ for $k = 3, r = -8$ and $x = 4$. Her first few steps are shown:

$$\begin{aligned} 4^3 \div -8 \cdot (3^2 - 5) \\ 64 \div -8 \cdot (9 - 5) \\ 64 \div -8 \cdot 4 \end{aligned}$$

- a. Mia thinks that she could finish her work by either finding $64 \div -8$ or $-8 \cdot 4$ first. Show the work for both possibilities to determine if Mia is correct.

$$\begin{array}{ll} 64 \div -8 \cdot 4 & 64 \div -8 \cdot 4 \\ (64 \div -8) \cdot 4 & 64 \div (-8 \cdot 4) \\ -8 \cdot 4 = -32 & 64 \div -32 \\ & -2 \end{array}$$

The answers are not the same.

- b. Was Mia correct when she stated that both methods would work? Use the steps of the order of operations to justify whether or not Mia is correct.

Multiplication and division should be done from left to right.

2. Answer the following problems.

- a. Evaluate $r \cdot W \div x \div m$ when $m = 2, r = -4, W = -60$ and $x = -10$. Show your work.

$$\begin{aligned} -4 \cdot (-60) \div -10 \div 2 \\ 240 \div -10 \div 2 \\ -24 \div 2 \\ -12 \end{aligned}$$



Compare and Connect

Each question in this section provides an opportunity for students to compare how changing the order in which operations are performed changes the value of the expression. Consider modeling a Think Aloud for question 1b to model the thinking students should be doing. Then have students engage in this Compare and Connect with a partner before completing question c for 2.



Consider splitting the class into differentiated groups to complete this section after modeling the Think Aloud to answer question 1. Guide students who may need more support through the thinking for question c.

13.1.5 Try It: Discovering the Importance of Left to Right (continued)

- b. Evaluate $(r \cdot W) \div (x \div m)$ when $m = 2$, $r = -4$, $W = -60$ and $x = -10$. Show your work.

$$(-4 \cdot -60) \div (-10 \div 2)$$

$$240 \div -5$$

$$-48$$

- c. How did the use of parentheses change the answers?

The parentheses changed the order so that $-10 \div 2$ was divided first instead of $-240 \div -10$ being first.

The parentheses around $r \cdot W$ did not change what was done without parentheses.



- What happens when you use parentheses in the expression? How do they change the value of the expression? Why does it change the value?
- How does the value of the expression change when you rearrange the order of the operations?

13.1.6 Wrap-Up: Error Analysis

Component Overview: In this formative assessment, students demonstrate their understanding of evaluating algebraic expressions. Students analyze two students' work to identify errors. Then they explain how to fix the errors in a note to the students.



13.1.6 Wrap-Up: Error Analysis

- Nadine and Nora evaluated the expression $r^3 \div x \cdot c$ for $c = 9$, $r = -6$ and $x = -3$. Both students made errors in their work shown. Circle their errors.

Nadine's Work	Nora's Work
$r^3 \div x \cdot c$ $(-6)^3 \div (-3) \cdot 9$ $-216 \div (-3) \cdot 9$ $-216 \div -27$ 8	$r^3 \div x \cdot c$ $(-6)^3 \div (-3) \cdot 9$ $-18 \div (-3) \cdot 9$ $6 \cdot 9$ 54



Consider allowing students to record their answers using an app like FlipGrid. This will be beneficial for students who may understand the content but struggle with written response.

- Write a short explanation to each student to explain the mistake and what she needs to remember about evaluating expressions in the future.

To Nadine: Nadine, The order of operations states to multiply and divide from left to right. In the step $-216 \div (-3) \cdot 9$ you should divide $-216 \div (-3)$ then multiply $72 \cdot 9$.	To Nora: Nora, $(-6)^3$ is a shortcut way of writing $-6 \cdot -6 \cdot -6$, where: $-6 \cdot -6 = 36$ $36 \cdot -6 = -216$
--	--

13.2 Evaluating Algebraic Expressions Using Substitution

Lesson Introduction

 Benchmark	MA.6.AR.1.3 Evaluate algebraic expressions using substitution and order of operations.		 Learning Target	I can evaluate algebraic expressions using substitution.	
 Benchmark Coherence	Building On		Working Toward		
	MA.5.AR.2 Demonstrate an understanding of equality, the order of operations, and equivalent numerical expressions.		MA.7.NSO.2.1 Solve mathematical problems using multi-step order of operations with rational numbers including grouping symbols, whole-number exponents, and absolute value.		
 Essential Questions	- How do you evaluate algebraic expressions using substitution? - How does order of operations influence the outcome when simplifying an expression?				
 Lesson Narrative	Lesson 1 positioned students to evaluate algebraic expressions using substitution and order of operations. This lesson is designed to consolidate student understanding by analyzing samples of student work for errors. Students engage in error analysis to solidify their own conceptual understanding before applying to practice.				
 Component Summary	13.2.1 Warm-Up (~5 min.)	Create an exponential expression with the largest value (<i>SE p. 270</i>)	13.2.3 Your Turn! (15 min.)	Practice evaluating algebraic expressions (<i>SE p. 272</i>)	
	13.2.2 Collaboration (20 min.)	Analyze errors and evaluate algebraic expressions (<i>SE p. 271</i>)	13.2.4 Wrap-Up (~5 min.)	Self-assessment of understanding of evaluating algebraic expressions (<i>SE p. 273</i>)	
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> Evaluate		(Optional) Calculators (Optional) colored pencils		If differentiating by ability level, consider planning for how to collect informal data throughout the lesson to inform which expressions to assign to students. It may be helpful to use student work and explanations during 13.2.2 Collaboration as formative data.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may substitute the variable as a digit instead of multiplying it by the coefficient. For example, in $5x$, where $x = 4$, students may write 54 instead of $5(4)$. - Students may evaluate exponents as the base multiplied by the exponent instead of the base multiplied by itself the exponent number of times. - Students may struggle remembering the rules for operations with integers.	- Encourage students to substitute one variable at a time and to use parentheses and/or colored pencils. This may help students keep track of their work and evaluate expressions more accurately. - Encourage students to evaluate one operation at a time, underline the calculation at each step, and show all work.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share Use Think-Pair-Share in 13.2.2 Collaboration to give all students enough time to think about a prompt and form a response before they are expected to try to verbalize their thinking. Students will have an opportunity to share their thinking in a low-stakes way with one partner so that when they share with the class, they can feel calm and confident as well as say something meaningful that might advance everyone’s understanding. Additionally, the teacher can listen in so that they can purposefully select students to share with the class.	MA.K12.MTR.4.1: Discussion & Reflection During 13.2.2 Collaboration, students will be engaging in discourse to find and correct an error representing a common misconception that students make when evaluating algebraic expressions. This standard emphasizes students reflecting on their own work and the work of others to establish a culture in which mistakes are viewed as an opportunity for learning.
 Differentiate Instruction	Consider having students preview this lesson by watching the On-Ramp video on evaluating expressions or have the video available to watch for students to opt-in throughout this lesson, particularly during 13.2.3 Your Turn!	
Lesson Notes:		

13.2.1 Warm-Up: Order Explorer

Component Overview: Students work independently to use digits one through five to create an exponential expression with the largest possible value. After students have completed questions 1-3, bring the class together to review the expressions students created and their strategies for determining the largest value.



13.2.1 Warm-Up: Order Explorer

- Place a digit in each box to create an expression with the largest possible value. Use any of the digits 1 to 5, once each at most.

$$\boxed{4}^{\boxed{5}} + \boxed{2} \times \boxed{3}$$

- What is the value of your expression?
1,030
- On a scale of 1 to 10, how confident are you that your expression is the largest possible value?
Answers will vary.
- Explain your strategy for choosing and placing the numbers as you did.
I calculated using both 4^5 and 5^4 and once I realized that 4^5 was larger I then multiplied by the next largest numbers.



The goal of 13.2.1 Warm-Up is to be an open-ended activity for the student. The teacher should resist guiding students with how to approach the task. Instead, the questions below can be used to facilitate a deeper conversation around the type of thinking and reasoning students used to determine where to place each digit within the expression.



- Did you try multiple expressions to determine which would be the largest?
- Which digit did you choose not to use? Why?
- Where did you place the largest digit, five?
- Why did you choose to place it there?
- How did you decide where to place the remaining digits to create an expression with the largest possible value?

13.2.2 Collaboration: Evaluating Algebraic Expressions by Substituting Integers

Component Overview: Students work with a partner to find the error in student work provided. Engage students in a Think-Pair-Share routine before having them complete the written response for 1b. Consider implementing a whole-class discussion before students move on to question 2 to solidify understanding and address any questions students may have after completing question 1. Students then work independently to evaluate Expressions 2a-2c before checking their work with a partner.



13.2.2 Collaboration: Evaluating Algebraic Expressions by Substituting Integers

1. Brae evaluated the expression $7b + 12 - (5 - 3)^2$ for $b = -2$. Her work is shown.
 - a. Evaluate her work and determine where she made mistakes.

$$\begin{aligned}
 &7b + 12 - (5 - 3)^2 \\
 &7 - 2 + 12 - (5 - 3)^2 \\
 &7 - 2 + 12 - (-2)^2 \\
 &7 - 2 + 12 + 4 \\
 &5 + 12 + 4 \\
 &17 + 4 \\
 &21
 \end{aligned}$$

- b. Help Brae understand and learn from her mistakes. Write a brief note to Brae explaining what her mistakes were and how she could learn from the mistakes.

Means

I

Start

To

Acquire

Knowledge

Experience

Skills

When Brae substituted -2 in for $7b$ she forgot to put the -2 in parentheses. This made it look like $7 - 2$ instead of 7 times -2 . Brae should be careful and use parentheses when substituting a value in.

When Brae found $5 - 3$ she wrote -2 instead of 2 .

- c. Read the note your partner wrote and offer suggestions on ways to improve their note.



Think-Pair-Share

Read the directions for Part b to the class, then have students engage in a Think-Pair-Share with a partner. Encourage students to use the following format to answer the question completely:

1. Name the error(s).
2. Explain the error. Why do you think Brae made this mistake?
3. What does Brae need to remember to prevent making this mistake in the future?

Providing a structure for this discussion and the following response will help students think through the misconceptions presented in this question.

13.2.2 Collaboration: Evaluating Algebraic Expressions by Substituting Integers (continued)

- d. Using the word bank below, complete the three reminders that could help Brae.

inside outside parentheses

- When substituting values into algebraic expressions, it is helpful to put the substituted value(s) in *parentheses*.
 - When evaluating expressions with exponents, if the negative is *outside* the parentheses, the negative indicates subtraction, or the opposite of the value found by raising the value inside to a power.
 - If the negative is *inside* the parentheses, it indicates that the number is a negative number being raised to a power.
- e. Verify your answers with your partner and, if necessary, resolve any differences.



Consider a Read Aloud with a small group for students who struggle with reading comprehension. Alternatively, nominate reading captains for each small group that read for their respective groups.

13.2.2 Collaboration: Evaluating Algebraic Expressions by Substituting Integers (continued)

2. Complete the following problems.

- a. Evaluate the expression $-8m - 4q^2 + \frac{s}{3}$ where $m = 4$, $q = -3$, and $s = -24$.

$$\begin{aligned} & -8(4) - 4(-3)^2 + \frac{-24}{3} \\ & -32 - 4(9) - 8 \\ & -32 - 36 - 8 = -76 \end{aligned}$$

- b. Evaluate the expression $3s - 4q - m^2$ where $m = 4$, $q = -3$, and $s = -24$.

$$\begin{aligned} & 3(-24) - 4(-3) - (4)^2 \\ & -72 + 12 - 16 \\ & -76 \end{aligned}$$

- c. Evaluate the expression $-m^3 - 3q + (s - q)$ where $m = 4$, $q = -3$, and $s = -24$.

$$\begin{aligned} & -4^3 - 3(-3) + (-24 - (-3)) \\ & -64 + 9 + (-24 + 3) \\ & -64 + 9 + (-21) \\ & -64 - 21 + 9 \\ & -85 + 9 = -76 \end{aligned}$$

- d. All of the expressions should have the same result. Check your work with your partner.



- How can you break this question down into steps to make it easier to solve?
- What happens when you substitute each variable for a number?
- How can you use grouping symbols like parentheses to keep track of substituting variables for numbers?

13.2.3 Your Turn!

Component Overview: Students work independently to evaluate expressions by substituting variables.



13.2.3 Your Turn!

1. What is the value of $5(6x + 7) - 3y^2$ if $x = -1$ and $y = 5$?

$$\begin{aligned} &5(6(-1) + 7) - 3(5)^2 \\ &5(-6 + 7) - 3(25) \\ &5(1) - 75 \\ &5 - 75 = -70 \end{aligned}$$

2. Evaluate the expression $3d^4 - \frac{w}{3}$ where $d = 4$ and $w = -75$.

$$\begin{aligned} &3(4)^4 - \frac{(-75)}{3} \\ &3(256) + \frac{75}{3} \\ &768 + 25 = 793 \end{aligned}$$

$$\begin{aligned} &4^4 = 4 \cdot 4 \cdot 4 \cdot 4 \\ &\quad \swarrow \quad \searrow \\ &\quad 16 \quad 16 \\ &\quad \swarrow \quad \searrow \\ &\quad \quad 256 \\ &256 \cdot 3 = 768 \end{aligned}$$

3. Find the value of $-2(b + 4) + 3a - (7 - a)^5$ if $a = -3$ and $b = 5$.

$$\begin{aligned} &-2(5 + 4) + 3(-3) - (7 - (-3))^5 \\ &-2(9) + (-9) - (7 + 3)^5 \\ &-2(9) + (-9) - (10)^5 \\ &-18 + (-9) - 100,000 \\ &= -100,027 \end{aligned}$$

$$\begin{aligned} &10 \cdot 10 \cdot 10 \cdot 10 \cdot 10 \\ &\quad \swarrow \quad \searrow \\ &\quad 100 \cdot 100 \cdot 10 \\ &\quad \swarrow \quad \searrow \\ &\quad 10000 \cdot 10 = 100,000 \end{aligned}$$

4. Evaluate the expression $c^3 - 5k + (h - k)$ where $c = -9$, $h = 29$, and $k = -17$.

$$\begin{aligned} &(-9)^3 - 5(-17) + (29 - (-17)) \\ &(-9)^3 - 5(-17) + (29 + 17) \\ &-729 + 85 + 46 \\ &-729 + 131 \\ &= -598 \end{aligned}$$

$$\begin{aligned} &-9 \cdot -9 \cdot -9 \\ &81 \cdot -9 = -729 \\ &729 - 131 = 598 \end{aligned}$$



Consider strategies for differentiating this activity based on your students and classroom.

- Provide On-Ramp videos for students to opt-in for additional support before engaging in the practice.
- Conduct an open small group in which students can come and go as they need assistance.
- Allow student choice. Students can choose any two questions to solve before finding a partner who did the other two questions to then explain (not copy) their respective answers.



Each of the questions in 13.2.3 Your Turn! highlight some common misconceptions and errors that students may make.

Question 1 – Look for students who multiply -3 and 5 before squaring.

Question 2 – Look for students who will drop the negative from 75 .

Question 3 – Look for students who distribute the subtraction sign in front of $(7 - a)$ to the 7 before finding the value inside the parentheses.

Consider having students do the questions, and then check their work and answers with a partner and to work through any differences before completing question 5.

13.2.3 Your Turn! (continued)

5. Evaluate each expression.

a. $8 \div p - 16 \div (-5 + 7)$ where $p = 4$

$$8 \div 4 - 16 \div (2)$$

$$2 - 8 = -6$$

b. $13 + \frac{6}{y} - y^3$ if $y = -3$

$$13 + \frac{6}{-3} - (-3)^3$$

$$13 + (-2) - (-27)$$

$$13 - 2 + 27$$

$$11 + 27 = 38$$

c. $4g - 12 + (h - 7) \cdot 3^2$ where $g = -7$ and $h = 6$

$$4(-7) - 12 + (6 - 7) \cdot 9$$

$$-28 - 12 + (-9)$$

$$-28 - 12 - 9$$

$$= -49$$

d. $w \div t \cdot H \div y^2$ where $H = -3$, $t = 7$, $w = 84$, and $y = -2$

$$84 \div 7 \cdot (-3) \div (-2)^2$$

$$12 \cdot (-3) \div 4$$

$$-36 \div 4 = -9$$



Consider providing a four-function calculator to students who struggle with computation. A four-function calculator is not programmed to perform the order of operations so the student will still have to determine which operation to perform first. Allowing the use of a calculator may give students the chance to master the learning target without getting caught up in complex calculations. Have students write and say the calculations they enter in the calculator to reinforce their computational skills.

13.2.4 Wrap-Up: Reflection

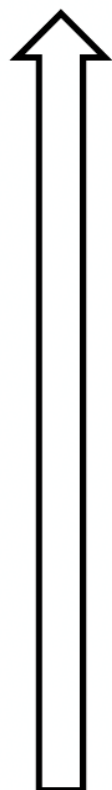
Component Overview: Students complete a self-assessment of their understanding of solving algebraic expressions. Consider using this assessment to inform small group remediation targets.



13.2.4 Wrap-Up: Reflection

Color in the arrow up to the statement that best describes your current understanding.

Answers will vary.



I'm so confident, I can explain evaluating algebraic expressions using substitution to someone else!

I can get to the right answer, but I don't understand well enough to explain it yet.

I understand some of this, but I don't understand all of it yet. I checked what I do not understand yet.

- ☐ Working with negative numbers
- ☐ Finding values raised to an exponent
- ☐ Remembering to multiply and divide from left to right
- ☐ Remembering to use parentheses when I substitute a value

I tried hard, but I am finding this challenging. I checked what I do not understand yet.

- ☐ Working with negative numbers
- ☐ Finding values raised to an exponent
- ☐ Remembering to multiply and divide from left to right
- ☐ Remembering to use parentheses when I substitute in a value

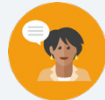


Students could describe their understanding of each learning goal verbally by coloring in the scale or by using a coding system (i.e. 0=not at all, 1=partial, 2=full).

13.3 Applying Properties of Operations

Lesson Introduction

 Benchmark	MA.6.AR.1.4 Apply the properties of operations to generate equivalent algebraic expressions with integer coefficients.		 Learning Target	I can apply the associative, commutative, and distributive properties to generate equivalent expressions.	
 Benchmark Coherence	Building On			Working Toward	
	MA.5.AR.2 Demonstrate an understanding of equality, the order of operations, and equivalent numerical expressions.			MA.7.AR.1.2 Determine whether two linear expressions are equivalent.	
 Essential Questions	- Which operations does each property (associative, commutative, distributive) apply to? - How can you use the properties of operations to write equivalent expressions? - How can you show that two algebraic expressions are equivalent?				
 Lesson Narrative	This lesson begins by revisiting the associative, commutative, and distributive properties that students are familiar with from prior grades. The lesson primarily focuses on students exploring these properties for numerical expressions to ensure students recall the properties clearly. Students' connections of these properties to algebraic expressions is scaffolded throughout the lesson.				
 Component Summary	13.3.1 Warm-Up (~5 min.)	Rewrite expressions using the distributive property of multiplication over addition (<i>SE p. 274</i>)	13.3.3 Guided Instruction (20-25 min.)	Apply the properties of operations to write equivalent expressions (<i>SE p. 276</i>)	
	13.3.2 Exploration (15 min.)	Apply the properties of operations to write equivalent expressions to represent the areas of rectangles (<i>SE p. 275</i>)	13.3.4 Wrap-Up (~5 min.)	Demonstrate understanding of using the properties of operations to write equivalent expressions (<i>SE p. 279</i>)	
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> - Equivalent - Commutative Property of Multiplication - Commutative Property of Addition - Associative Property of Multiplication - Associative Property of Addition <u>Additional Terms:</u> - Distributive property		(Optional) Colored pencils		Prepare an exemplar math journal or interactive notebook entries to show students your expectations.

 Teacher Tips	Common Misconceptions - Students may struggle to recognize that expressions like $-4x + 3y$ and $3y - 4x$ are equivalent expressions by the commutative property. - Students may not realize that the number of elements inside of the parentheses is not limited to two elements when applying the distributive property.	Things to Consider Consider using word associations to help students understand the properties of operations. For example, relating commutative property to commuting, as in moving from one location to another. Similarly, the teacher can relate associative property to associates or associating with people in groups.
	Literacy and Language Routines Take Turns In 13.3.3 Guided Instruction, students can take turns explaining, justifying, agreeing and disagreeing, or asking clarifying questions as they complete the tables of equivalent expression. Students' understanding is deepened as they construct logical arguments and critique the reasoning of others.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.1.1: Participate in Effortful Learning Throughout 13.3.3 Guided Instruction, students collaborate to extend their understanding of the properties of operations. Students work together to support each other as they build their understanding.
	 Differentiate Instruction	

Consider Frayer Models, interactive notebooks, word walls, and other vocabulary strategies to assist students in their language acquisition development with the properties of addition and multiplication.

Definition

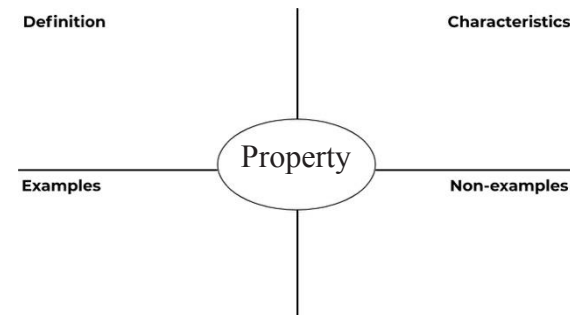
Characteristics

Examples

Non-examples

Property

Lesson Notes:



13.3.1 Warm-Up: Mental Math

Component Overview: In this Warm-Up, students have a chance to connect the distributive property to mental mathematics.



13.3.1 Warm-Up: Mental Math

Ulrich finds the product of $5 \cdot 107$ in his head. Ulrich wrote the following explanation of his thinking:

Because 107 is equal to $100 + 7$, I can write $5 \cdot 107$ as $5(100 + 7)$. I know that $5 \cdot 100$ is 500 and that $5 \cdot 7$ is 35. Then, I add 500 and 35 to get 535.

1. Use Ulrich's reasoning to write each in the form $a(b + c)$. Then, use that form to find the product.

Multiplication	$5 \cdot 104$	$5 \cdot 97$	$5 \cdot 130$
$a(b + c)$	$5(100 + 4)$	$5(90 + 7)$	$5(100 + 30)$
Product	$500 + 20 = 520$	$450 + 35 = 485$	$500 + 150 = 650$



Consider pairing students with different strengths for this activity and providing space for them to compare strategies for collective learning.



- How can you rewrite 104, 97, and 130 using the method that Ulrich used?
- How does using 100 make multiplying the two numbers easier to do using mental math?

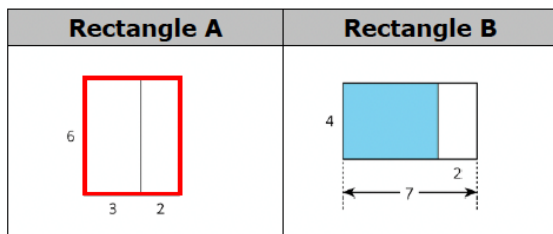
13.3.2 Exploration: Representing Area of a Rectangle

Component Overview: The 13.3.2 Exploration uses the area model to help students visualize the distributive property. The exploration also leads students to seeing equivalence in numeric expressions, then connecting that understanding to be applied to algebraic expressions.



13.3.2 Exploration: Representing Area of a Rectangle

1. Work with your partner to answer the questions.



This component provides a visual entry point by connecting the area model to the distributive property. Highlight this connection and encourage students to regularly use the area model.

- a. Rectangle A is divided into two smaller rectangles. Select all the expressions that could be used to determine the area of Rectangle A.

- ☐ $6 \cdot 5$
☐ $6 \cdot 3 \cdot 2$
☒ $6(3 + 2)$
☐ $6 \cdot 3 + 2$
☐ $6 + 3 + 2$
☒ $6 \cdot 3 + 6 \cdot 2$

- b. With your partner, discuss which expressions are in the form $a(b + c)$.
 $6(3 + 2)$

- c. Rectangle B is divided into two smaller rectangles. Select all the expressions that could be used to determine the area of the blue rectangle inside Rectangle B.

- ☒ $4 \cdot 5$
☐ $4 \cdot 7 \cdot 2$
☒ $4(7 + 2)$
☒ $4 \cdot 7 + 4 \cdot 2$
☐ $4 \cdot 7 + 4 \cdot 2$
☐ $4 \cdot 2 - 4 \cdot 7$

13.3.2 Exploration: Representing Area of a Rectangle (continued)

- d. With your partner, discuss which expressions are in the form $a(b + c)$.

$$4(7 - 2) = 4(7 + (-2))$$

2. Complete the following problems.

- a. Circle all of the expressions that have the same value as $-4(x + 7)$ if $x = -13$.

$-4x + 7$	$(x + 7) \cdot -4$	$-4x + -4 \cdot 7$	$4(x - 7)$	$-28 - 4x$
-----------	--------------------	--------------------	------------	------------

- b. With your partner, discuss whether or not the expressions you circled would change if the x -value were a different number. Summarize your decision and reason.

The expressions are equivalent for any value of x .



- What do you already know about substituting variables in algebraic expressions?
- Why do you think the three expressions circled resulted in the same value?
- Why do you think that the same expressions will be circled for any value of x ?

13.3.3 Guided Instruction: Applying Properties of Operations

Component Overview: Students formally define the distributive, commutative, and associative properties. Students practice applying each property to evaluate numeric and algebraic expressions.



13.3.3 Guided Instruction: Applying Properties of Operations

The **distributive property of multiplication over addition** is a property that applies to all rational numbers. The distributive property “distributes” a factor to all terms within a set of parentheses.

- Complete the table. The first row has been completed.

$a(b + c)$	$ab + ac$	Value
$-8(50 + 9)$	$-8(50) + -8(9)$	-472
$17(20 + 3)$	$17(20) + 17(3)$	$340 + 51 = 391$
$(100 - 7) \cdot -6$	$100 \cdot -6 + -7 \cdot -6$	$-600 + 42 = -558$
$(40 + 72)29$	$40 \cdot 29 + 29 \cdot 72$	$1160 + 2088 = 3248$
$-11(48 - 96)$	$-11(48) - -11(96)$	$-528 + 1056 = 528$

- In two of the rows the expression had a subtraction sign instead of an addition sign. Discuss with your partner how $a(b - c)$ can be rewritten using addition. Write a summary of your discussion.
 $a(b + (-c))$
- Use the expressions below to show that the distributive property of multiplication over addition works with rational numbers.

$a(b + c)$	$ab + ac$	Equivalent?
$\frac{1}{2}(-14 + 50)$	$\frac{1}{2}(-14) + \frac{1}{2}(50)$	Yes
$1.2(12 + 50)$	$1.2 \cdot 12 + 1.2 \cdot 50$	Yes

Mathematicians determine a property exists when expressions are equivalent, no matter what number is used. A property can then be used with variables.



Consider your students when planning your approach for 13.3.3 Guided Instruction.

- Teacher-led:** For students unfamiliar with the properties or who struggled with the 13.3.1 Warm-Up, consider modeling each property and providing Frayer Models to complete.
- Small groups:** Nominate student captains that pull small groups and act as co-teachers for the lesson. Alternatively, students can group on their own and report their findings back to the teacher after each property.
- Hybrid:** Utilize instructional routines like Think-Pair-Share or Collect and Display to facilitate guided instruction that positions students to activate what they know about properties and apply that knowledge to new situations within this activity.

13.3.3 Guided Instruction: Applying Properties of Operations (continued)

Expressions are **equivalent** if they have the same value, no matter what number is substituted in for each of the variables.

4. Write an equivalent expression for $8(x + 2)$.

$$8x + 16$$

5. Write an equivalent expression for $\frac{1}{2}(x - 16)$.

$$\frac{1}{2}x - 8 \text{ or } \frac{1}{2}x + (-8)$$

The **commutative property** states that the order in which an operation is performed does not change the value of the expression.

6. Use the pairs of expressions below to determine the operations for which the commutative property is true.

Operation	Example Expressions		Equivalent?
Addition	$\frac{5}{12} + \frac{3}{4}$	$\frac{3}{4} + \frac{5}{12}$	Yes
	$\frac{5}{12} + \frac{9}{12} = \frac{14}{12}$	$\frac{9}{12} + \frac{5}{12} = \frac{14}{12}$	
Subtraction	$\frac{3}{4} - \frac{11}{2}$	$\frac{11}{2} - \frac{3}{4}$	No
	$\frac{3}{4} - \frac{22}{4} = \frac{-19}{4}$	$\frac{22}{4} - \frac{3}{4} = \frac{19}{4}$	
Multiplication	$\frac{5}{6} \cdot \frac{2}{3} = \frac{10}{18}$	$\frac{2}{3} \left(\frac{5}{6}\right) = \frac{10}{18}$	Yes
	$\frac{7}{12} \div \frac{1}{4}$	$\frac{1}{4} \div \frac{7}{12}$	No
	$\frac{7}{12} \cdot \frac{4}{1} = \frac{28}{12}$	$\frac{1}{4} \cdot \frac{12}{7} = \frac{12}{28}$	

The commutative property only applies to **addition** and **multiplication**.



Connect other known words to each of the properties.

- Distributive property:** fairly distributing objects like candy or money
- Associative property:** associating with different people at different times depending on the situation
- Commutative property:** commuting from one place to another (switching from home to school)



Applying the properties using rational numbers helps to extend student understanding that these properties do work for all numbers while also supporting students learning of operations with rational numbers. Exploring whether the commutative property works for all four operations helps students understand why there is no commutative property for subtraction and division.

The Associative property is similarly explored.

13.3.3 Guided Instruction: Applying Properties of Operations (continued)

7. Use the commutative property of multiplication to write an expression equivalent to $5 \cdot \frac{3}{20} \cdot 4\frac{4}{5}$.

$$4\frac{4}{5} \cdot 5 \cdot \frac{3}{20} \quad 5 \cdot 4\frac{4}{5} \cdot \frac{3}{20} \quad \frac{3}{20} \cdot 5 \cdot 4\frac{4}{5} \quad \frac{3}{20} \cdot 4\frac{4}{5} \cdot 5 \quad 4\frac{4}{5} \cdot \frac{3}{20} \cdot 5$$

8. Rewrite $x + 9$ using the commutative property.
 $9 + x$
9. Rewrite $(-7)(x)$ using the commutative property.
 $(x)(-7)$



Commutative Property of Addition

$$a + b = b + a$$

Commutative Property of Multiplication

$$a \times b = b \times a$$

The **associative property** states that the way in which numbers are grouped does not change the value of the expression.

10. Use the pairs of expressions below to determine the operations for which the associative property is true.

Operation	Example Expressions	Equivalent?
Addition	$\left(\frac{8}{5} + \frac{3}{10}\right) + \frac{17}{10}$ $\frac{16}{10} + \frac{3}{10} + \frac{17}{10} = \frac{36}{10}$	Yes
Subtraction	$\frac{7}{2} - \left(\frac{3}{4} - \frac{1}{4}\right)$ $\frac{14}{4} - \left(\frac{3}{4} - \frac{1}{4}\right)$ $= \left(\frac{14}{4} - \frac{2}{4}\right)$ $= \frac{12}{4}$	No



Consider an inquiry-based approach to position students as sense makers.

- Do properties ever apply to subtraction or division? Why or why not? Give an example.
- How can you show that two numerical expressions are equivalent?
- How can you show that two algebraic expressions are equivalent?

13.3.3 Guided Instruction: Properties of Operations (continued)

Multiplication	$\frac{2}{3} \cdot \left(\frac{3}{5} \cdot \frac{1}{9} \right)$ $\frac{2}{3} \cdot \frac{3}{45} = \frac{6}{135}$	$\left(\frac{2}{3} \cdot \frac{3}{5} \right) \cdot \frac{1}{9}$ $\left(\frac{6}{15} \right) \cdot \frac{1}{9} = \frac{6}{135}$	Yes
Division	$\left(\frac{1}{5} \div \frac{1}{10} \right) \div \frac{1}{2}$ $\left(\frac{1}{5} \cdot \frac{10}{1} \right) \div \frac{1}{2}$ $= 2 \div \frac{1}{2}$ $= 2 \cdot \frac{2}{1} = 4$	$\frac{1}{5} \div \left(\frac{1}{10} \div \frac{1}{2} \right)$ $\frac{1}{5} \div \left(\frac{1}{10} \cdot \frac{2}{1} \right)$ $= \frac{1}{5} \div \frac{1}{5}$ $= \frac{1}{5} \cdot \frac{5}{1} = 1$	No

The associative property applies only to **addition** and **multiplication**.

- Rewrite $(3.25 + 9.8) + 7.2$ using the associative property.
 $3.25 + (9.8 + 7.2)$
- Rewrite $(x + 15) + -11$ using the associative property.
 $x + (15 + (-11))$
- Rewrite $(-2)(8 \cdot x)$ using the associative property.
 $(-2 \cdot 8) \cdot x$
- Rewrite $6((x + 3) + 2)$ using the associative property.
 $6(x + (3 + 2))$



Associative Property of Addition

$$(a + b) + c = a + (b + c)$$

Associative Property of Multiplication

$$(a \times b) \times c = a \times (b \times c)$$



Students often confuse commutative and associative properties. Consider prompting students to turn-and-talk with a partner about the difference between commutative and associative properties. Listen as they discuss, monitoring for any misconceptions that may need to be clarified with the group before closing out this component.

13.3.4 Wrap-Up: Ticket Out the Door

Component Overview: Students demonstrate their mastery of the learning goal by applying the associative, commutative, and distributive properties to generate equivalent expressions.



13.3.4 Wrap-Up: Ticket Out the Door

- Write an equivalent expression that uses each given property to complete the table.

Answers will vary.

Expression	Property	Equivalent Expression
$-2(4 + (x + -7))$	Commutative Property	$-2(4 + (-7 + x))$
$-2(4 + (x + -7))$	Associative Property	$-2((4 + x) + -7)$
$-2(4 + x + -7)$	Distributive Property	$-8 + (-2x) + 14$

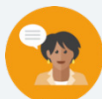


To challenge students who finish early, consider having them create additional equivalent expressions using the properties. This can serve as a primer for the next lesson in which students will focus on identifying and generating equivalent expressions.

13.4 Generating Equivalent Algebraic Expressions

Lesson Introduction

 Benchmark	MA.6.AR.1.4 Apply the properties of operations to generate equivalent algebraic expressions with integer coefficients.		 Learning Target	I can apply properties of operations to generate equivalent expressions.
 Benchmark Coherence	Building On		Working Toward	
	MA.5.AR.2 Demonstrate an understanding of equality, the order of operations, and equivalent numerical expressions.		MA.7.AR.1.2 Determine whether two linear expressions are equivalent.	
 Essential Questions	How can you use properties to generate that two algebraic expressions are equivalent?			
 Lesson Narrative	This lesson begins by using the area model, which students are familiar with, to model the distributive property. The lesson primarily focuses on applying the properties of operations (associative, commutative, and distributive) reviewed in the previous lesson to generate equivalent expressions.			
 Component Summary	13.4.1 Warm-Up (~5 min.)	Analyze a puzzle for a pattern (<i>SE p. 280</i>)	13.4.3 Try It (5 min.)	Apply properties of operation to generate equivalent algebraic expressions (<i>SE p. 282</i>)
	13.4.2 Guided Instruction (20 min.)	Review, extension, and application of properties to generate equivalent expressions (<i>SE p. 281</i>)	13.4.4 Your Turn! (10 min.)	Additional practice applying of properties of operation to generate equivalent algebraic expressions through area model and matching exercises (<i>SE p. 283</i>)
	13.4.5 Wrap-Up (~5 min.)	Demonstrate understanding of applying properties of operation to generate equivalent algebraic expressions through error analysis (<i>SE p. 284</i>)		
 Resources	Glossary		Materials	
	Key Terms: N/A Additional Terms: - Distributive property - Associative property - Commutative property		(Optional) Colored pencils	
			Additional Preparation (Optional) Determine student groupings before class (Optional) Prepare sentence stems, if needed	

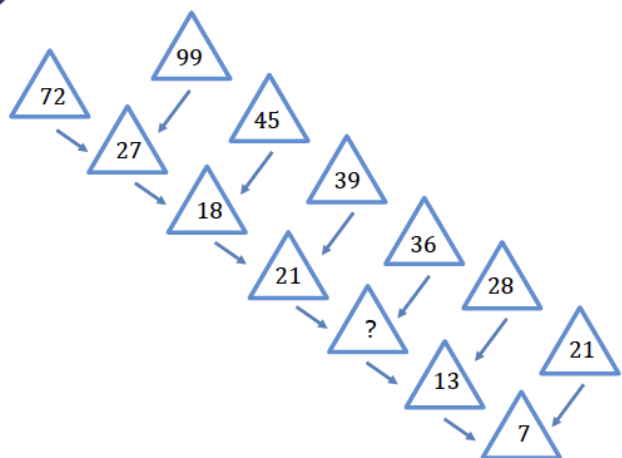
 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may not distribute the term outside of the parentheses to all terms in the parentheses, or students may distribute the term outside the parenthesis to all terms in the expression regardless of where they are. - Students may incorrectly apply properties to contexts with subtraction and division.	Properties are not new information, but not all students will be entering this lesson on the same level. Consider what information can be used prior to this lesson to better inform groups and differentiate throughout.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share Engage students in a Think-Pair-Share routine during 13.4.4 Your Turn!. Ask students to first complete question 2 on their own, then share their responses with a partner and subsequently make any changes to their expressions before sharing out as a class. This routine encourages discourse and solidifies understanding.	MA.K12.MTR.5.1: Patterns and Structures In 13.4.4 Your Turn!, students use the structure of the expressions and properties to determine if different expressions are equivalent. Students also have to logical order their ideas answer the questions.
 Differentiate Instruction	Reflection and error analysis are positive contributors to self-regulation, math confidence, and conceptual understanding. However, not all students reflect and analyze the same way. Consider differentiating the ways in which students can engage in demonstrating their progress and mastery in this lesson. 13.4.2 Guided Instruction: Comparing Makayla’s and Soledad’s approaches with a partner 13.4.4 Your Turn!: Articulating their reflection and progress verbally with a partner or telling a family member at home and recording their conversation to submit as homework 13.4.5 Wrap-Up: Recording themselves using an app like FlipGrid, explaining their answer to someone at home or a friend in a different math class	
Lesson Notes:		

13.4.1 Warm-Up: Missing Number

Component Overview: Students analyze a puzzle for a pattern that can be used to determine missing numbers.



13.4.1 Warm-Up: Missing Number



Students may see the following pattern:

$$99 - 72 = 27; 45 - 27 = 18$$

This pattern works until the final step.

If this error is observed, prompt students to change the way they are thinking about operations and instead pay attention to the digits of each value, using an arrow to help make connections.

1. What is the first number you tried for the missing number?
Answers will vary.
2. Did your first number work for the rest of the pattern?
Answers will vary.
3. What strategy did you try if your first number did not work?

Add the digits.

$$\begin{aligned} 7 + 2 + 9 + 9 &= 27 \\ 2 + 7 + 4 + 5 &= 18 \\ 1 + 8 + 3 + 9 &= 21 \\ 2 + 1 + 3 + 6 &= 12 \\ 1 + 2 + 2 + 8 &= 13 \\ 1 + 3 + 2 + 1 &= 7 \end{aligned}$$


- What do you notice about the relationship between the numbers? Does that work for every number?
- How else can you use the digits to determine the missing number? What strategy did you use?
- Is guess and check an efficient strategy?

13.4.2 Guided Instruction: Generating Equivalent Algebraic Expressions

Component Overview: Students use the properties of operations to write equivalent expressions. Students have several opportunities to practice applying the commutative, associative, and distributive properties. Students also check two samples of student work to see if they correctly applied the properties to write an equivalent expression by substituting a variable into each expression.



13.4.2 Guided Instruction: Generating Equivalent Algebraic Expressions

1. Consider the following expressions.

$$8 \cdot x + 8 \cdot 9$$

$$8(x + 9)$$

$$8x + 9$$

- a. Which of these expressions are equivalent?
 $8 \cdot x + 8 \cdot 9$ and $8(x + 9)$
- b. Which property of operation justifies that the expressions are equivalent?
Distributive

2. Write two equivalent expressions. Name the property or properties you used to create each equivalent expression.

$$8 \cdot x + 9 \cdot 8 \text{ Commutative}$$

$$8(9 + x) \text{ Commutative}$$

$$8 \cdot 9 + 8 \cdot x \text{ Commutative}$$

$$x \cdot 8 + 8 \cdot 9 \text{ Commutative}$$

3. For each expression, use the associative and commutative properties to generate an equivalent expression.

Original Expression	Associative Property	Commutative Property
$(3x + 5) + 4z$	$3x + (5 + 4z)$	Answers will vary $4z + (3x + 5)$ $(5 + 3x) + 4z$
$8 \cdot (-1 \cdot 13d)$	$(8 \cdot -1) \cdot 13d$	Answers will vary $(-1 \cdot 13d) \cdot 8$ $8 \cdot (13d \cdot -1)$

4. Rewrite $-11b + 3(10k + 1)$ using the distributive property.

$$-11b + 30k + 3$$



Focus attention on assisting students with writing equivalent algebraic expressions. The goal of this section is for students to understand there are many ways to rewrite expressions using the properties. Encourage students to think flexibly, particularly when writing expressions using each property.



- How do $(8 + 9) + 7 + 4 = 7 + (8 + 9) + 4$ and $(9 + 8) + 4 + 7$ both demonstrate the commutative property?
- How do $(8 + 9) + 7 + 4 = 8 + 9 + (7 + 4)$ and $8 + (9 + 7) + 4$ both demonstrate the associative property?
- Which properties does $(8 + 9) + 7 + 4 = (8 + 4) + (7 + 9)$ show?
- Is it possible for expressions to demonstrate multiple properties?
- When you are rewriting expressions using the properties, is there only one way to do so?

13.4.2 Guided instruction: Generating Equivalent Algebraic Expressions (continued)

5. Rewrite $-11b + 3(10k + 1)$ using the commutative property of multiplication.

$$\begin{aligned} &-11b + (10k + 1) \cdot 3 \text{ or} \\ &b \cdot -11 + 3(10k + 1) \text{ or} \\ &-11b + 3(k \cdot 10 + 1) \end{aligned}$$

6. Rewrite $-11b + 3(10k + 1)$ using the commutative property of addition.

$$\begin{aligned} &-11b + 3(1 + 10k) \text{ or} \\ &3(10k + 1) + (-11b) \end{aligned}$$

7. Makayla and Soledad both rewrote $(-5h + 9) \cdot -14$ using the commutative property of multiplication. Verify that their expressions are equivalent by substituting a value in for h . Show your work in the table.

Makayla	Soledad
$-14(-5h + 9)$	$(h \cdot -5 + 9) \cdot -14$
$h = 1$	$h = 1$
$-14(-5(1) + 9)$ $-14(-5 + 9)$ $-14(4) = -56$	$(1 \cdot -5 + 9) \cdot -14$ $(-5 + 9) \cdot -14$ $4 \cdot -14$ $= -56$



Engage students in a discussion to provide an auditory entry point and check for understanding.



- Why must the variable h be the same value in both Makayla's and Soledad's expressions?



The benchmark MA.6.AR.1.4 states that students are to generate equivalent expressions. Note that the word "simplify" is not used in the instructions. Equivalence is different and has more useful mathematical applications than "simplifying."

13.4.3 Try It: Writing Equivalent Algebraic Expressions

Component Overview: Students practice writing equivalent algebraic expressions.



13.4.3 Try It: Writing Equivalent Algebraic Expressions

1. Select the expressions that are equivalent to $(2x + 7) + (5y - 3)$.

- ☐ $(5y - 3) + (7 + 2x)$
- ☐ $(3 - 5y) + (7 + 2x)$
- ☐ $7 + (5y + 2x) - 3$
- ☐ $(7 + 5y) + (2x - 3)$
- ☐ $5y + (7 + 2x) - 3$

2. Write two equivalent algebraic expressions for each expression below.

a. $(4 + 9) \cdot x$

$13x$

$x(4 + 9)$

b. $3t \cdot 4$

$12t$

$t \cdot 3 \cdot 4$

c. $-7x + 3 - 14$

$-7x - 11$

$-7x + (3 - 14)$

d. $3(-9p - 4) + 7$

$-27p - 5$

$-27p - 12 + 7$



Consider having students work in pairs or small groups to complete this component. Through discussions, students will formalize their understanding of using the properties of operations to write equivalent expressions.



Have students rewrite the algebraic expressions before and after applying the properties of operations using different colored pencils so that they can see the parts of the expressions that were changed.

13.4.4 Your Turn!

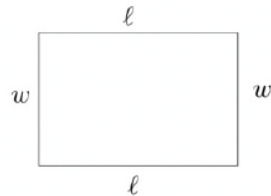
Component Overview: Students work independently to evaluate student work to determine if the expressions created accurately represent the area of a rectangle and then write to the student explaining why their expression is not equivalent to the others. Students then match equivalent expressions in a table, creating equivalent expressions for those who do not have matches.



13.4.4 Your Turn!

- The students in Ms. Justino's class are writing expressions for the perimeter of a rectangle of side length, ℓ , and width, w . The expressions for four students are in the table.

Student	Expression
De'Andre	$2(\ell + w)$
Summer	$\ell + w + \ell + w$
Abm	$2\ell + w$
Luzarie	$2\ell + 2w$



Ms. Justino told the group that one of the expressions is not equivalent. Write a short explanation to the student to show why their expression is not equivalent to the other expressions.

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

*Abm – Your expression is not equivalent to $2(\ell + w)$, $\ell + w + \ell + w$, and $2\ell + 2w$.
 $\ell + w + \ell + w = 2\ell + 2w$
Your expression does not have the same coefficient for w .*



Providing sentence stems will help students who may struggle with written responses. Additionally, allowing students to pair up and write joint responses may elicit more precise responses.

13.4.4 Your Turn! (continued)

2. Match equivalent expressions by drawing an arrow. If an expression does not have a match, write an equivalent expression to match it in one of the blank boxes provided.

Column A	Column B
$-8(x + 3)$	$2 + (-8x + 3)$
$-8x \cdot 3$	Answers will vary $2(8x - 3)$ $(-3 \cdot 8x) \cdot 2$
$(8x - 3) \cdot 2$	$3 \cdot -8x$
$(3 + -8x) + 2$	$-24 - 8x$
$2 + -8x$	Answers will vary $-8x + 2$ $2 - 8x$



Engage students in a Think-Pair-Share routine. Ask students to first complete question 2 on their own, then share their response with a partner and subsequently make any changes to their expressions before sharing out as a class. This routine encourages discourse and solidifies understanding.



- How can you find an equivalent expression?
- Did you and your partner create the same expressions? If not, are they both equivalent? How do you know?
- How can you use the properties of operations to find and create equivalent expressions?

13.4.5 Wrap-Up: Error Analysis

Component Overview: Students analyze a sample of student work to identify any errors in writing equivalent expressions. Students explain the error and correct the mistakes.



13.4.5 Wrap-Up: Error Analysis

Tangela was asked to circle all of the expressions equivalent to $6(y + 1)$.

Her work is shown below.

$6y + 1$	$6y + 7$	$6(y) + 1(y)$	$6(y) + 6(1)$	$6y + 6$
----------	----------	---------------	---------------	----------

- Determine if Tangela correctly identified all of the equivalent expressions. If she is missing an expression, circle the expression(s) she is missing.

No, Tangela identified two expressions that are not equivalent to $6(y + 1)$.

- If Tangela made an error, then explain her error and list the equivalent expressions.

$6y + 1$ and $6(y) + 1(y)$ are not equivalent expressions to $6(y + 1)$ as Tangela did not apply the distributive property correctly. If Tangela would have applied the distributive property correctly, she would have gotten then expression $6 \cdot y + 6 \cdot 1$, which is equivalent to:

*$6(y) + 6(1)$ and
 $6y + 6$*



Consider allowing students who may struggle with written response to explain their answers verbally or record them on an app, like FlipGrid, for assessment.



Challenge early finishers to determine which properties were used to write each equivalent expression.

13.5 Writing Algebraic Expressions

Lesson Introduction

 Benchmark	MA.6.AR.1.1 Given a mathematical or real-world context, translate written descriptions into algebraic expressions and translate algebraic expressions into written descriptions.		 Learning Targets	- I can translate a written description into an algebraic expression. - I can translate a real-world description into an algebraic expression.
 Benchmark Coherence	Building On MA.5.AR.2.1 Translate written real-world and mathematical descriptions into numerical expressions and numerical expressions into written mathematical descriptions.		Working Toward MA.7.AR.2.2 Write and solve two-step equations in one variable within a mathematical or real-world context, where all terms are rational numbers.	
 Essential Questions	- How do you write an algebraic expression from a written description? - How are expressions and equations similar? How are they different? - How can you use models to represent algebraic expressions? - How can algebraic expressions represent real-world situations?			
 Lesson Narrative	The lesson focuses on writing algebraic expressions for mathematical and real-world contexts. This is the first experience that students will have with writing expressions including a variable to represent an unknown value. The focus should be on understanding the action of the written description and selecting the appropriate mathematical operation based on that action. The use of bar models will help students make sense of the action and operation.			
 Component Summary	13.5.1 Warm-Up (~5 min.)	Matching and creating diagrams of numerical expressions (<i>SE p. 285</i>)	13.5.4 Guided Instruction (10 min.)	Write algebraic expressions based on real-world scenarios (<i>SE p. 289</i>)
	13.5.2 Guided Instruction (15 min.)	Represent algebraic expressions using bar models (<i>SE p. 285</i>)	13.5.5 Try It (5-10 min.)	Students write algebraic expressions to represent situations (<i>SE p. 290</i>)
	13.5.3 Collaboration (10 min.)	Match written descriptions to bar models and expressions with a partner (<i>SE p. 288</i>)	13.5.6 Wrap-Up (~5 min.)	Reflection (<i>SE p. 290</i>)
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - Algebraic expression <u>Additional Terms:</u> - Equation		N/A	If allowing students to use rods or other manipulatives to form bar models, have those ready prior to the lesson.

 Teacher Tips	Common Misconceptions - Students may want to use key words to write the expressions and skip the use of the bar models. - Students may struggle with the order of the terms in subtraction expressions. - Students may struggle with the order of the dividend and the divisor in division expressions. - Students may struggle to differentiate between the parts and the whole in bar models.	Things to Consider - Although this lesson addresses writing expressions, not equations, it is important for students to think about the action versus key words. These models will carry over to help students write equations to match written descriptions and also to writing scenarios to match an expression or equation. - Bar models force students to think about the meaning of the words and the actions indicated. This will help students write the algebraic expression rather than strictly translating key words which often leads to having subtraction or division expressions written in the reverse order. - If time is an issue, consider assigning 13.5.5 Try It for homework.														
	Literacy and Language Routines MLR 7: Compare and Connect To foster students' meta-awareness of different mathematical approaches and representations, teachers should demonstrate thinking out loud as they create and interpret bar models. Emphasize that bar models can be constructed in multiple ways to represent the same expression. Students should be prompted to reflect and respond as they discuss different approaches to creating bar models.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.4.1: Discussion and Reflection Students will be working to create and interpret bar models to represent expressions. When working with a partner, encourage students to clearly communicate their ideas and to collaborate with their partner to analyze each other's ideas. Students will also justify their responses and recognize and correct errors in collaboration with their partner.														
	 Differentiate Instruction	Bar models drive the instruction throughout Lesson 5. Consider providing additional scratch paper and colored pencils or markers for students to engage with more practice as they work on each component in the lesson. Consider having groups construct their own anchor charts on bar models for additional scaffolding. <table><tr><td>Equal Part</td><td>Equal Part</td><td>Equal Part</td><td>Equal Part</td><td>Equal Part</td><td>Equal Part</td><td>Equal Part</td><td>Equal Part</td></tr></table> Whole								Equal Part	Equal Part	Equal Part	Equal Part	Equal Part	Equal Part	Equal Part
Equal Part	Equal Part	Equal Part	Equal Part	Equal Part	Equal Part	Equal Part	Equal Part									
Lesson Notes:																

13.5.1 Warm-Up: Bell Work

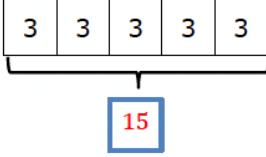
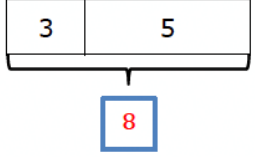
Component Overview: Students engage with visual models as they orient to the content of the lesson. Students begin by calculating the value of given diagrams before producing diagrams of their own for addition and multiplication problems.



13.5.1 Warm-Up: Bell Work

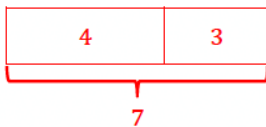
- Two diagrams are shown. One represents $3 + 5 = 8$. The other represents $5 \cdot 3 = 15$.

Complete the table by writing the equation that matches each diagram in the space provided and labeling the length of each diagram.

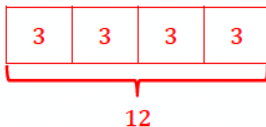
Equation	$5 \cdot 3 = 15$	$3 + 5 = 8$
Diagram		

- Draw a diagram that represents each equation.

a. $4 + 3 = 7$



b. $4 \cdot 3 = 12$



The diagram could also be constructed with three equal segments of 4.



Monitor for and highlight flexibility and fluency in student diagrams. For example, ask one student who drew four groups of three and another who drew three groups of four to display and discuss their equivalence.

13.5.2 Guided Instruction: Making Sense with Models

Component Overview: Students review the difference in algebraic expressions and equations. They then use bar diagrams to represent expressions. Encourage students to pay particular attention to how a bar model represents each expression. Focus on the action in the descriptions rather than memorizing the types of bar models.



13.5.2 Guided Instruction: Making Sense with Models

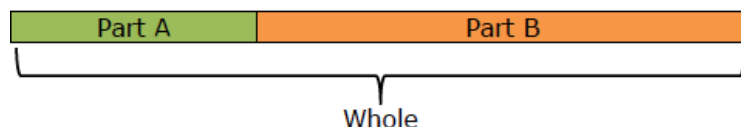
An **algebraic expression** is a mathematical statement containing numbers, operators, and at least one unknown, represented by a variable.

- Complete the statement.

An expression
☐ does
☒ does not
 have an equal sign or inequality symbol.

Bar models help connect the action of a problem to an algebraic expression.

Part-Part-Whole Model



The part-part-whole model can be used to show the following actions.

Action	Equation
Combining or joining	Part A + Part B = Whole
Taking away or separating	Whole - Part A = Part B Whole - Part B = Part A

- "The sum of a number and three" can be shown as the length of the whole bar where one part is n and the other part is 3.

The sum of a number and three



Write an algebraic expression for "The sum of a number and three."

$n + 3$



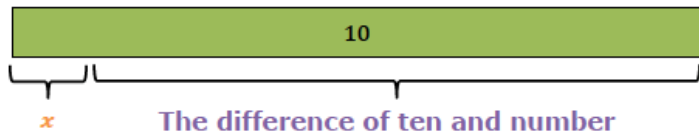
Position students to connect the visual models of the 13.5.1 Warm-Up to the bar model.



Prompt students to discuss the difference between identifying key words and focusing on the action of the story. Only identifying key words can lead to misconceptions; instead, focusing on the action produces more conceptual understanding of real-world problems.

13.5.2 Guided Instruction: Making Sense with Models (continued)

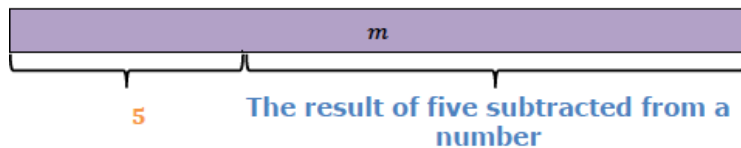
3. "The difference of ten and a number" can be shown where the whole bar is 10 and one part is a number represented by a variable, x . The difference is the other part.



Write an algebraic expression for "The difference of ten and a number."

$10 - x$

4. "Five subtracted from a number" is shown with the whole bar as a number represented by a variable, m , and one part is 5. The other part is the result of 5 being subtracted from the number, m .



Complete the statement.

The algebraic expression for "Five subtracted from a number, m " is $m - 5$.



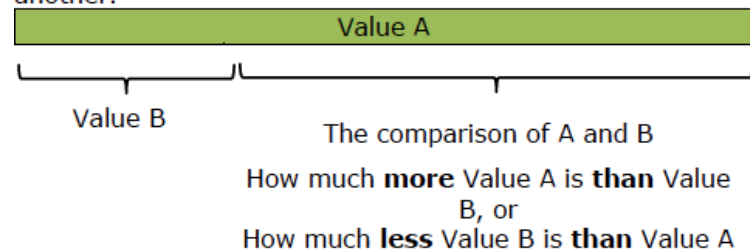
If students are struggling to connect the bar model to an expression, scaffold the instruction further by asking:

- What does "difference" mean?
- What is the whole? What are the parts?
- How can you label the whole on the bar model? How can you label the parts?
- What do you need to do with the parts to find the whole?

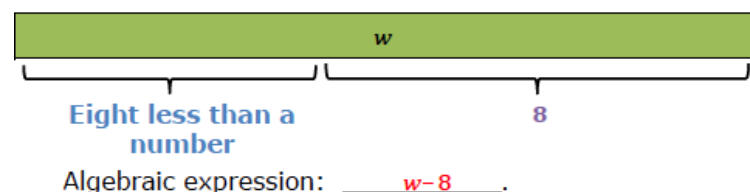
13.5.2 Guided Instruction: Making Sense with Models (continued)

Addition/Subtraction Comparison Model

Comparison models are much like part-part-whole models. They show how much more or less one value is than another.

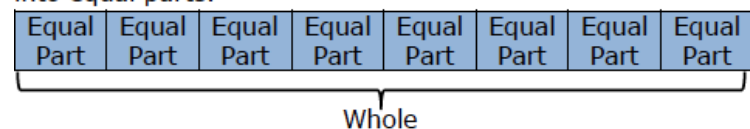


5. Write an algebraic expression for the bar model shown.



Equal Parts – Whole Model

The Equal Parts – Whole Model shows the whole sectioned into equal parts.



The equal parts-whole model can be used to show the following actions.

Action	Equation
Repeated addition (multiplication)	Equal part + ... + equal part = whole Equal part $\times$ number of parts = whole
Equal sharing	Whole $\div$ number of parts = equal part
Partitioning	Whole $\div$ equal part = number of parts



Students may struggle with the lack of detail in the description subtracted from a number. Rephrase to a question “What is five subtracted from your age?” Then, ask students if the answer would be same if you asked that question to your parents or guardian. Next, discuss where a variable should be used in the expression and why. Ask students how the whole could be represented. Finally, discuss the parts and how they could be represented.



Review the meanings of part and whole and use diagrams or images to explain the concepts. Allow students to use different lengths of rods to explore the concept of comparison.

Encourage students to verbalize their comparisons, such as, “The red rod is twice as long as the blue rod.”

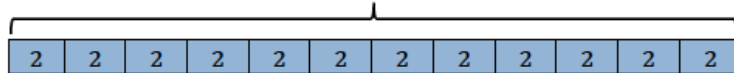
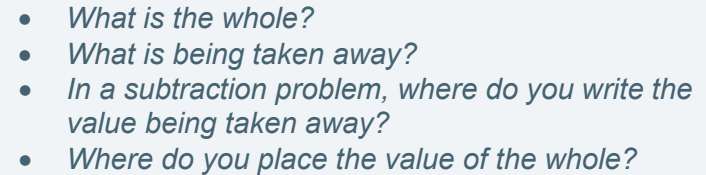
[illegible]

9a

A number, p



A number, p

 $p \div 2 \text{ and } \frac{p}{2}$ 

13.5.3 Collaboration: Matching Activity

Component Overview: Students will collaborate to match bar models to written descriptions in addition to composing algebraic expressions to represent specific situations. Both activities review skills developed in the previous lesson.

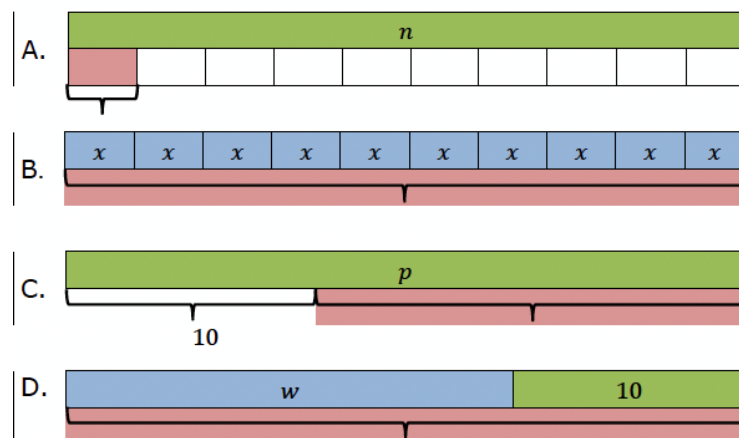


13.5.3 Collaboration: Matching Activity

Part 1

Match each written description to the appropriate bar model, where the red portion is the result of the written description. Write the letter of the bar model beside the corresponding word description.

Written Description	Bar Model
1. Ten subtracted from a number	<i>C</i>
2. A number divided by ten	<i>A</i>
3. The sum of a number and ten	<i>D</i>
4. A number times ten	<i>B</i>



Circulate as students work on 13.5.3 Collaboration, noting any gaps in learning or misconceptions that students have. Alternately, use information and observations from Lessons 1-4 to determine if small group intervention should be used during 13.5.3 Collaboration.



This section provides visual entry points through the use of bar models to represent each situation. Encourage students to think about and orally describe how A and B relate to each other to provide auditory entry points.



Activate prior knowledge to help students make connections between the written descriptions and bar models. Remind students that in a bar model, one bar represents the whole and the other bar represents the parts. Ask probing questions like:

- In C, what represents the whole? What are the parts?
- How do you know that A shows division?
- What algebraic expression could represent the bar model in B?
- What subtraction expression could D represent?

13.5.3 Collaboration: Matching Activity (continued)

Part 2

Turn and talk with your partner to compare your matches. Then, work with your partner to write an algebraic expression that represents the written description and bar model.

Written Description	Algebraic Expression
1. Ten subtracted from a number	$p - 10$
2. A number divided by ten	$\frac{n}{10}$
3. The sum of a number and ten	$w + 10$
4. A number times ten	$10x$



Students may struggle with the abstraction of variables. Consider giving students a relatable example to help them determine where the variable should go. Encourage students to substitute an actual number for the words “a number,” use that number to write the expression, and then replace the number with a variable.



Discuss how each written description is represented by an algebraic expression and a corresponding bar model. Encourage students to verbally explain the relationships seen within each grouping. For example, “10 subtracted from a number can be shown as the expression $p - 10$ because p represents the unknown number that 10 is subtracted from. In the bar model, the green bar represents the total that 10 is subtracted from. The pink bar shows the amount remaining.”

13.5.4 Guided Instruction: Writing Algebraic Expressions from Descriptions of Real-World Contexts

Component Overview: Students will write algebraic expressions to represent a given situation without using a concrete bar model. This encourages students to transition to thinking abstractly about algebraic expressions.



13.5.4 Guided Instruction: Writing Algebraic Expressions from Descriptions of Real-World Contexts

Algebraic expressions can be used to represent real-world situations.

1. Misty has 34 pairs of shoes arranged on a number of shelves. The same number of shoes are on each shelf. She has h number of shelves.
 - a. Write an algebraic expression that represents this situation.

$$\frac{34}{h}$$
 - b. What does the algebraic expression represent?
The expression represents the number of pairs of shoes on each shelf.
2. Martin buys 8 pounds of apples at a price per pound to make pie. Each pound costs p dollars.
 - a. Write an algebraic expression to represent this situation.
 $8p$
 - b. What does the algebraic expression represent?
The expression represents the total cost of the apples.
3. Paige is 6 inches taller than her brother. Paige's brother is m inches tall.
 - a. Write an algebraic expression to represent this situation.
 $m + 6$
 - b. What does the algebraic expression represent?
The expression represents Paige's height.



Three Reads

Implement a Three Reads routine here for each real-world situation.



Use the first read to help students understand context. Ask "What is this situation about?"

After the second read, ask students "What are the quantities in the situation?"

After the third read, ask students to brainstorm possible strategies to connect the situation with the appropriate expression.

Encourage students to repeat this routine themselves for each situation. This helps students connect the language in the question with the expression while keeping the intended level of cognitive demand.



If students are struggling to create algebraic expressions, allow them to use the middle step of first drawing a bar model to represent each situation. Also encourage students to verbally describe the action in each scenario before writing an algebraic expression for it. These modifications provide visual and auditory entry points.

13.5.5 Try It: Writing Algebraic Expressions

Component Overview: Students will practice what they have learned about creating algebraic expressions to represent specific situations. Remind students to look for the action instead of depending on key words when formulating an expression.



13.5.5 Try It: Writing Algebraic Expressions

Write an algebraic expression to represent each situation.

1. Joe is 9 years younger than Drew, who is n years old.
 $n - 9$

2. Jenna's salary, s , is lowered by \$120.
 $s - 120$

3. Maria has 12 cupcakes and wants to split them among f friends.
 $\frac{12}{f}$

4. Target has 8 cookie packages with c cookies in each.
 $8c$



Consider allowing students to work in pairs and then compare their expressions with those created by another student pair. Encourage students to verbally explain why they created each algebraic expression as they did and to describe which part of an expression matches which part of the written situation.

13.5.6 Wrap-Up: Reflection

Component Overview: Students will assess their understanding of creating an algebraic expression to represent a written description. Encourage students to thoroughly explain what they do and do not understand from the lesson.



13.5.6 Wrap-Up: Reflection

- Choose the emoji that reflects your understanding of translating a written description into an algebraic expression.

Answers will vary.



0 1 2 3 4 5 6 7 8 9 10

- Complete the statement.

I chose this emoji because _____

Answers will vary.



There is no right or wrong answer to the Reflection. Encourage students to thoughtfully assess their understanding of the lesson and to provide a concrete example as part of their explanation of why they chose that emoji.

13.6 Writing Descriptions of Real-World Contexts From Algebraic Expressions

Lesson Introduction

 Benchmark	MA.6.AR.1.1 Given a mathematical or real-world context, translate written descriptions into algebraic expressions and translate algebraic expressions into written descriptions.		 Learning Target	I can write a description of a real-world context from an algebraic expression.
 Benchmark Coherence	Building On		Working Toward	
	MA.5.AR.2.1 Translate written real-world and mathematical descriptions into numerical expressions and numerical expressions into written mathematical descriptions.		MA.7.AR.2.2 Write and solve two-step equations in one variable within a mathematical or real-world context, where all terms are rational numbers.	
 Essential Questions	- How can an algebraic expression represent a real-world situation? - How do you create a real-world situation to describe an algebraic expression? - Can an algebraic expression describe more than one real-world situation?			
 Lesson Narrative	This lesson challenges students to make sense of algebraic expressions by creating real-world situations that represent the expressions. Students will have the opportunity to work with their partner to determine if any errors have been made and identify similarities between real-world situations. Students will practice composing expressions to represent real-world situations. Students need to read each story carefully to formulate an expression to represent it.			
 Component Summary	13.6.1 Warm-Up (~5 min.)	Math Story: Madam C.J. Walker (<i>SE p. 291</i>)	13.6.3 Collaboration (10-15 min.)	Create real-world contexts based on algebraic expressions (<i>SE p. 292</i>)
	13.6.2 Activity (10 min.)	Match real-world contexts to expressions (<i>SE p. 292</i>)	13.6.4 Activity (15 min.)	Participate in a gallery walk to create more real-world contexts based on expressions (<i>SE p. 293</i>)
	13.6.5 Wrap-Up (~5 min.)	Explain the difference between two expressions (<i>SE p. 294</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> Algebraic expression		(Optional) Crayons (Optional) Gallery Walk for 13.6.4	- During the second half of the 13.6.3 Collaboration, collect the scenarios that students did not use. Write those scenarios on chart paper and hang them around the room for the 13.6.4 Activity. - Another option is to create your own scenarios before class and hang them on the walls while students work on the second half of the 13.6.3 Collaboration.

 Teacher Tips	Common Misconceptions	Things to Consider	
	Students may struggle to differentiate between multiplication and division or addition and subtraction. Students who only look for key words but do not thoroughly read the intent of the question may only use a key word like “per” or “give” and assume multiplication and addition even though both those key words could mean any operation.	- Before beginning the lesson, consider briefly brainstorming with students some situations in which they use math in their daily lives. List general ideas on the board so students can refer to them during the lesson. - The MA.6.AR.1.1 benchmark is extended to having students write instead of just translating. This is an example of taking something that is of moderate complexity and extending it to a higher complexity. By extending student thinking to applications the student’s mind makes stronger connections to the contexts and their algebraic representations. Be prepared with scaffolding for students who may struggle but encourage students to persevere through using a growth mindset. - Consider doing small-group intervention with students who have been struggling while other students are doing the gallery walk.	
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard	
	Collect and Display While pairs are working, circulate, collect, and make a visual display of vocabulary, phrases, and representations students use as they solve each situation. Make connections between how similar ideas might be communicated and represented in different ways. This helps students use mathematical language during paired and whole-group discussions.	MA.K12.MTR.7.1: Real-World Contexts Students will be working to create real-world scenarios that an algebraic expression could represent. Clarifications within this standard emphasize connecting mathematical concepts to everyday experiences. Use the lesson to provide opportunities for students to create models and compare their models with those created by other students.	
 Differentiate Instruction	Consider pulling small groups during the 13.6.3 Collaboration and 13.6.4 Activity for students who struggle with the 13.6.2 Activity.		
Lesson Notes:			

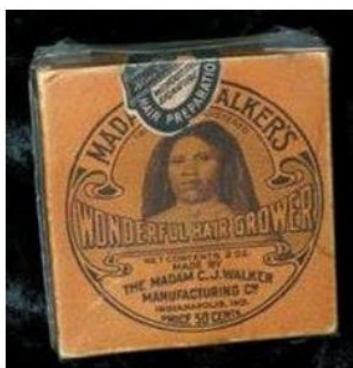
13.6.1 Warm-Up: Self-Made Millionaire

Component Overview: Students are introduced to Madam C.J. Walker and write their own algebraic expressions that could represent her revenue.



13.6.1 Warm-Up: Self-Made Millionaire

Madam C.J. Walker was the first African American self-made millionaire. She made her fortune by inventing and selling specialized hair products, including an innovative Edge Control product that sold for \$0.35 each beginning in 1905.



1. Create an algebraic expression to describe the total revenue Madam C.J. Walker earned for selling any number of Edge Control products, e .

$0.35e$



Consider an extension project for students to complete to accompany their work throughout the unit.

- For each lesson, students must provide and explain one example from that lesson that would fit a context for her edge-control business.
- Partner with an English or social studies teacher and complete a biography on her life and her resilience in her story.
- Watch a clip from “Self-Made” and facilitate a class discussion on the clip chosen.



Students could produce a variety of answers, so prompt them to define their variable and explain the constants they have included in the equation.

13.6.2 Activity: Matching Real-World Contexts to Expressions

Component Overview: Students match the relationship of a real-world situation to an algebraic expression that represents it. Remind students that they practiced this skill in the previous lesson.



13.6.2 Activity: Matching Real-World Contexts to Expressions

- Match each real-world situation with the correct expression.

A. Ryan has 20 songs. His friend bought him x more songs. How many songs does Ryan have?	1. $20x$
B. Karen has x flowers. Helen has 20 times more flowers than Karen. How many flowers does Helen have?	2. $x + 20$
C. George earned 20 points in a card game. Bruno earned x fewer points than George. How many points did Bruno earn?	3. $\frac{20}{x}$
D. Maria has 20 books and wants to put an equal number of books on x bookshelves. How many books go on each shelf?	4. $20 - x$



Consider having early finishers create a visual representation of one or more of the situations/expressions. Encourage students to think of ways to creatively convey the situation(s) as an image.



For each situation/expression pair, discuss the following questions as time allows:

- How did you figure out which expression matches the scenario?
- How did you determine the operation described in the scenario?

13.6.3 Collaboration: Writing Real-World Contexts

Component Overview: Students work with a partner or small group to compose a real-world scenario for each algebraic expression. Remind students that this is the opposite of what they did in the previous lesson when they wrote an algebraic expression to represent a real-world situation. After students select their four scenarios and while they are completing the second table, collect the scenarios they did not use. These scenarios can be used for the 13.6.4 Activity.



13.6.3 Collaboration: Writing Real-World Contexts

- For each expression, create a real-world scenario that the algebraic expression could represent.

Expression	Your Real-World Scenario	What does the variable represent?
$r - 10$	Answers will vary. Ana has r number of ribbons. She gives 10 to a friend. How many ribbons does Ana have left?	the number of ribbons
$21y$	Answers will vary. Zara is making bracelets that contain 21 beads per bracelet. She is making y bracelets for several friends. How many beads will she need?	the number of bracelets she is making
$\frac{100}{a}$	Answers will vary. The class has 100 pieces of dried fruit that they are equally splitting into a bags. How many pieces of fruit are in each bag?	the number of bags
$17 + s$	Answers will vary. Sean has 17 toy cars. For his birthday, he gets s more cars. How many cars does Sean have in all?	the number of cars Sean gets for his birthday
$57 - r$	Answers will vary. Mrs. Ruth has 57 thumbtacks. She gives r thumbtacks to another teacher. How many thumbtacks does she have left?	the number of thumbtacks Mrs. Ruth gives to another teacher
$48w$	Answers will vary. There are 48 pencils in a pack. Mr. Suarez bought w packs of pencils. How many pencils does he have?	the number of packs of pencils Mr. Suarez bought



If students struggle with math vocabulary, then creating a real-world scenario will be difficult. Consider providing students with a vocabulary reference that is visible during the class period. Include words such as “variable, difference, product, sum, splitting equally,” and “quotient.”



Before students begin, it may be helpful to discuss what it means for a variable to represent something. Remind students that they are working with algebraic expressions, not trying to solve a problem for a mathematical answer. Point out that in some situations, the variable may represent a whole amount, while in other situations, the variable may represent part of the problem. If students struggle to begin, consider working through the first one or two scenarios as a whole group.

13.6.3 Collaboration: Writing Real-World Contexts (continued)

2. With your partner, pick four expressions from Question 1 and use them to complete the following table.

Expression	Your Partner's Scenario	Does the Expression Accurately Represent the Scenario?	Similarities Between Your and Your Partner's Scenarios for the Expression
<i>Answers will vary. Students should use the scenarios they created in the previous problem.</i>	<i>Answers will vary. Students should use the scenarios they created in the previous problem.</i>	☐ Yes ☐ No	<i>Answers will vary. Students should use the scenarios they created in the previous problem.</i>
		☐ Yes ☐ No	
		☐ Yes ☐ No	
		☐ Yes ☐ No	



Encourage students to look for similarities and differences in the scenarios. Consider having students use the following questions to discuss the scenarios with their partner:

- Do both scenarios accurately represent the algebraic expression?
- How are the scenarios different?
- How are the scenarios similar?

13.6.4 Activity: Gallery Walk

Component Overview: Students can rotate through the Gallery Walk individually, in pairs, or in small groups. In this Activity, students write algebraic expressions to represent real-world situations. Before students begin, tell them which way to rotate around the room and how and when you will indicate when it is time to move to the next scenario. Either use the unused scenarios students created earlier and/or use the optional black line master. Number each scenario or place a label on each so students can record when expressions go with which scenario on their paper. Students should not be expected to write the scenario on their paper but have the scenario number and its corresponding expression on their chart.



13.6.4 Activity: Gallery Walk

Several scenarios are displayed. For each scenario, write an algebraic expression to match.

Scenario	Expression
Answers will vary. Scenarios will be the ones students wrote in Problem 1 but did not use in Problem 2, or other scenarios that you choose to create.	



Use the following questions in student pairs or as a whole group to discuss the algebraic expressions that students wrote during the Gallery Walk:

- Why does that algebraic expression represent the scenario?
- How did you decide which operation to use?
- How did you decide where to place the variable and the numerical value?

13.6.5 Wrap-Up: Ticket Out the Door

Component Overview: In the Wrap-Up, students will describe the difference in the position of the minuend and the subtrahend in a subtraction expression. Remind students that they are explaining how and why the expression is constructed, not finding a solution for a mathematical equation or problem.



13.6.5 Wrap-Up: Ticket Out the Door

1. Frank has a plant that measured h inches tall last week. This week the plant is 8 inches tall.

- a. Circle the expression that represents the number of inches, h , the plant grew in a week.

$$8 - h$$

$$h - 8$$

- b. Explain your choice.

To determine how many inches the plant grew last week, we start with how tall the plant is this week, 8 inches, and then subtract the number of inches, h , the plant grew last week.

- c. For the expression not chosen, describe a real-world situation that the expression might represent.

Answers will vary. Frank made h hacky sacks to sell at the school carnival. On the first day, he sold 8 hacky sacks. How many hacky sacks does Frank have left to sell?



For students who might struggle representing their conceptual understanding through writing, consider alternative ways to assess or measure learning for this lesson. Students could choose to record themselves explaining the answer verbally, drawing and labeling, or providing an example. All forms can show an equal level of rigor and conceptual understanding.



Interpreting algebraic expressions from a context has been scaffolded over the last several lessons. Use the ticket out the door to inform student understanding as well as student growth for these lessons. Consider writing students positive notes about their progress.

13.7 Adding and Subtracting Linear Expressions With Integer Coefficients

Lesson Introduction

 Benchmark	MA.7.AR.1.1 Apply properties of operations to add and subtract linear expressions with rational coefficients.		 Learning Target	I can use algebra tiles to add and subtract linear expressions.
 Benchmark Coherence	Building On		Working Toward	
	MA.6.AR.1.4 Apply the properties of operations to generate equivalent algebraic expressions with integer coefficients.		MA.8.AR.1.2 Apply properties of operations to multiply two linear expressions with rational coefficients.	
 Essential Questions	- How can you represent algebraic expressions using algebra tiles? - How do you determine whether terms are like or unlike? - How could you use zero pairs when simplifying expressions?			
 Lesson Narrative	Students simplify algebraic expressions using algebra tiles by combining like terms and eliminating zero pairs. Students have prior experience using integer chips to add and subtract rational numbers. They will connect this knowledge to algebra tiles and combining like terms to write equivalent algebraic expressions.			
 Component Summary	13.7.1 Warm-Up (~5 min.)	Students learn about engineers as a Career Profile (video) (<i>SE p. 295</i>)	13.7.5 Guided Instruction (10 min.)	Model algebraic expressions using algebra tiles, identify like terms, eliminate zero pairs, and write in simplest form (<i>SE p. 299</i>)
	13.7.2 Refresher (10 min.)	Exploring zero pairs using algebra tiles (<i>SE p. 295</i>)	13.7.6 Guided Practice (5 min.)	Practice adding and subtracting expressions with algebra tiles and writing expressions in simplest form (<i>SE p. 300</i>)
	13.7.3 Guided Instruction (5 min.)	Introduction to academic vocabulary related to algebraic expressions (<i>SE p. 296</i>)	13.7.7 Your Turn! (5 min.)	Apply understanding of combining like terms to write equivalent algebraic expressions (<i>SE p. 300</i>)
	13.7.4 Exploration (5-10 min.)	Applying understanding of zero pairs and like terms with algebra tiles to algebraic expressions (<i>SE p. 297</i>)	13.7.8 Wrap-Up (~5 min.)	Demonstrate understanding of the key vocabulary from the lesson (<i>SE p. 301</i>)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> - Zero pair - Term - Expression - Coefficient - Like terms - Constant <u>Additional Terms:</u> N/A		- Career Profile video - Algebra tiles (concrete and/or virtual) (Optional) Chart paper (Optional) Marker	
			Additional Preparation	
			13.7.1 Warm-Up contains a video to accompany the question prompts (three-minute video). - If using concrete algebra tiles, determine how many tiles are needed to model all of the examples in the lesson and organize student sets in a container or plastic bag.	

 Teacher Tips	Common Misconceptions		Things to Consider	
	- Students may try to combine terms that are not like terms, for example, $2 + 3x = 5x$ or $3x + 2y = 5xy$. - Students may not recognize the coefficient of 1 when adding like terms, for example, $x + 2x + 3x = 5x$ rather than $6x$.		- Determine if your students need the support of all or part of the refresher. It is possible this component does not need to be used as students did this in Unit 11. - Consider making student anchor charts, Frayer Models, or a word wall to reinforce the vocabulary terms presented in the lesson. - Notice the intentional spiraling throughout these lessons. Students do not need mastery yet, but instead have opportunities to continue practicing adding and subtracting over the next two lessons. - If time is an issue, consider assigning 13.7.7 Your Turn! for homework.	
	Literacy and Language Routines		Mathematical Thinking and Reasoning (MTR) Standard	
	Notice and Wonder Make use of this routine in 13.7.2 Refresher in order to position students to make sense of the rationale behind why variables like x and y are not considered like terms. The visual representation of algebra tiles enhances this routine.		MA.K12.MTR.2.1: Multiple Representations Multiple portions of this lesson provide opportunities for students to represent algebraic expressions in various ways. In 13.7.2 Refresher, students have the opportunity to see multiple ways to represent numeric expressions both using algebra tiles and creating other equivalent expressions.	
 Differentiate Instruction	Consider providing algebra tiles, having students sketch algebra tiles, or providing access to the digital version of these manipulatives as an alternative method to capturing how students are processing the meaning of zero pairs and simplifying expressions.			
Lesson Notes:				

13.7.1 Warm-Up: Career in Robotics

Component Overview: Play the Career Profile video for the entire class. After students watch this two-minute video, prompt the students to read and answer the questions in the Warm-Up independently.



13.7.1 Warm-Up: Career in Robotics

Tessa Lau is the Chief Robot Whisperer for a robotics company. The company creates robot helpers for the service industry. Some of their robots perform tasks like delivering items to guests at a hotel. Tessa has studied applied



physics, computer science, artificial intelligence, and human-computer interaction. She loves hands-on experiments, building things, and the speed of creating with computers. Through programming and engineering, one of her primary jobs is to look at a robot in context and make sure the robots behave well in human environments.

1. Have you ever considered a career as an engineer?
Answers will vary.
2. How do you think a robot could be helpful to you in your everyday life?
Answers will vary.
3. Tessa's advice is "Don't listen to someone who tells you that you can't do something." Describe a time someone told you that you were not able to do something.
Answers will vary.



Do you know someone who is an engineer or computer programmer? Consider bringing in a visitor for your class to ask questions and learn from. Additionally, consider having students return to this lesson later for project-based learning around this Career Profile.



Ask students to explain why they have or have not considered the profession. Consider specifying how many examples students should come up with. Is one example okay or should students write more than one example. Is a text-based response sufficient or would you like students to include a diagram (of their robot invention and/or the task it performs)?

13.7.2 Refresher: Adding and Subtracting Integers with Algebra Tiles

Component Overview: Students connect prior knowledge of integer chips to using algebra tiles to represent zero pairs and algebraic expressions.



13.7.2 Refresher: Adding and Subtracting Integers with Algebra Tiles

Algebra tiles, like integer chips, can be used to model adding and subtracting integers. Each small square represents a positive 1 (blue) or a negative 1 (red).

Integer Chips	Algebra Tiles
 	 

Adding a positive 1 (blue) and a negative 1 (red) represents a zero pair because $1 + -1 = 0$, as shown.

$$\begin{array}{|c|c|} \hline 1 & -1 \\ \hline \end{array} = 0$$

- Describe why this diagram represents the value 0.

1	-1
1	-1
1	-1

Zero pairs result from the sum of each of the three sets of opposites. Algebraically:

$$1 + (-1) + 1 + (-1) + 1 + (-1) = 0 + 0 + 0 = 0.$$

A **zero pair** is a pair of numbers whose sum is zero.



Choose a student to read the definition of zero pairs out loud and briefly discuss how zero pairs are represented using algebra tiles. Students then complete a table to model numeric expressions with algebra tiles. Before answering question 2b and 2c, consider displaying student work that used zero pairs and student work that did not to elicit a discussion about when utilizing zero pairs is necessary and effective.

- How are the representations of zero pairs with integer chips and algebra tiles the same?
- How are they different?



Consider the different entry points and ways to engage with key vocabulary, such as zero pair. Word walls, vocabulary cards, and sentence stems are all strategies that can serve multiple learning styles with vocabulary content.

13.7.2 Refresher: Adding and Subtracting Integers with Algebra Tiles (continued)

2. Complete the following problems.

a. Complete the table.

Numeric Expression	Algebra Tiles Representation	Value
$-7 + 2$		-5
$*6 - 8$		-2
$*-1 - 5$		-6
$*4 - (-3)$		7

- b. For some of the representations with algebra tiles, zero pairs had to be added to the representation. Put an asterisk (*) next to the expressions where this was the case.
- c. Explain why zero pairs had to be added to the representation in these cases.

For $6 - 8$, two zero pairs were added to be able to subtract positive 8.

For $-1 - 5$, five zero pairs were added to be able to subtract positive 5.

For $4 - (-3)$, three zero pairs were added to be able to subtract negative 3.



Students can either place their algebra tiles in each blank table or sketch a depiction of their algebra tiles that represent each expression.

2. Complete the following problems.

a. Complete the table.

Numeric Expression	Algebra Tiles Representation	Value
$-7 + 2$		



Notice and Wonder

The purpose of this activity is to refresh prior learning with adding and subtracting integers before moving into adding and subtracting algebraic expressions.

- For which expressions did zero pairs need to be added to the model to find their value?
- Why was it necessary to add zero pairs in these cases?



13.7.3 Guided Instruction: Algebra Vocabulary

Component Overview: Students review key definitions necessary for working with algebraic expressions. Read each definition out loud and provide an example, then have students share another example or rephrase the definition in their own words.



13.7.3 Guided Instruction: Algebra Vocabulary

There are some fundamental definitions that are necessary to know for working with algebraic expressions.

A **term** is a single number or variable, or numbers and variables multiplied together. For example, the expression $0.5 - 4y$ has two terms.

The expression $0.5 - 4y$ has two terms.

0.5	-4y
term 1	term 2

One of the terms in this expression is a constant term.

1. Which term is the constant term? Explain.

The term 0.5 is the constant because it is not multiplied by a variable.



An **expression** is a mathematical statement containing numerals, operators, grouping symbols, and symbols or variables for unknown values. An expression does not contain an equal sign or inequality symbol.

For example, $4 + 5$, $3x - 2$, and $6x - 7y + 11$ are all expressions.



A **coefficient** is the number or constant that multiplies a variable in an algebraic expression. If no number is specified, the coefficient is 1.

In the expression $4xy$, 4 is the coefficient.

In the expression x , 1 is the coefficient.

In the expression $-8y$, -8 is the coefficient.

Variable terms have coefficients while constant terms do not.



Consider a variety of strategies to incorporate this math vocabulary into your classroom.

- **Visual Scaffolding:** Learning tools like foldables or a Frayer Model provide examples and nonexamples for students to make sense in context.
- **Positive Affirmation:** Consider using a token economy or form of positive affirmation (snaps) when a student correctly utilizes glossary terms in context when responding.
- **Vocab Captain:** Nominate a different student per day to monitor opportunities in which students could have replaced an informal word with a math glossary term. Consider giving the class a homework pass for each class period in which there were zero misses.
- **Visual Word Wall:** Create a visual word wall where students see the vocabulary term and a corresponding example.

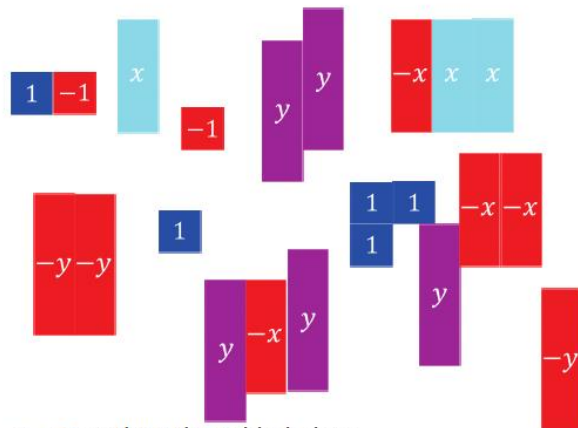
13.7.4 Exploration: What Are Like Terms?

Component Overview: Students categorize algebra tiles and use definitions from 13.7.3 Guided Instruction to create an algebraic expression to match the tiles. Students combine zero pairs to simplify the expression. Listen for students utilizing the key vocabulary introduced in the previous activity.



13.7.4 Exploration: What are Like Terms?

1. Here is a collection of algebra tiles.



- a. Complete the table below.

Type of Algebra Tile	Number of Tiles
x	3
$(-x)$	4
y	5
$(-y)$	3
1	5
(-1)	2

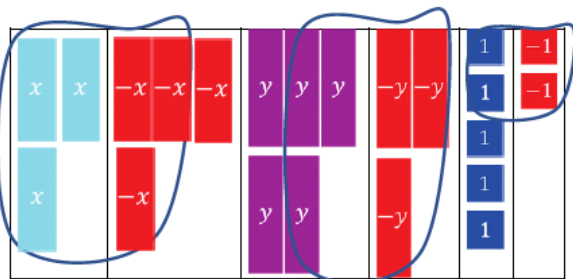


Consider providing algebra tiles for students or groups of students as kinesthetic scaffolding.

13.7.4 Exploration: What Are Like Terms? (continued)

- b. Use the definitions of coefficient and term to complete the expression by writing the coefficient or constant term that matches the number of tiles of each type.

$$3x + (-4x) + 5y + (-3y) + 5 + (-2)$$



- c. Draw the number of algebra tiles of each type under the term in the expression.
- d. Circle the zero pairs in your drawing.
- e. Complete the expression below after eliminating the zero pairs.

$$-x + 2y + 3$$

Like terms are terms with the exact same variable(s) raised to the same exponent.

A **constant** is a fixed numerical value with no variables. All constants are like terms.



Provide space for students to explore and make mistakes as they begin developing their understanding of variables, coefficients, and like terms.

As students explore, monitor for students who might interpret the three light blue algebra tiles as $x + 3$ instead of $3x$. Consider asking those students to describe what $x + 3$ might look like with tiles and compare that to what they see here. (Notice the answer key in 13.7.6 Guided Practice is a sample of what this could look like.)

13.7.4 Exploration: What Are Like Terms? (continued)

2. Answer the following questions about like terms.

a. Classify the following terms.

	Like Terms	Unlike Terms
y and 6	☐	☒
$3x$ and 3	☐	☒
$5x$ and $-3x$	☒	☐
$10, -2, 7.5$	☒	☐



Consider a Turn and Talk for question 2 in which students first answer parts a-c on their own before sharing with a partner to refine their responses.

b. How did you determine whether the terms were like or unlike?

Answers will vary. If the terms had the exact same variable(s) raised to the same exponent, they are like terms. If the terms did not, they were unlike.

c. Compare your answers with your partner.

13.7.5 Guided Instruction: Adding and Subtracting Expressions with Algebra Tiles

Component Overview: Teacher models how to use algebra tiles to eliminate zero pairs and create an equivalent algebraic expression. Students represent a given algebraic expression using algebra tiles and then create an equivalent expression after eliminating zero pairs.



13.7.5 Guided Instruction: Adding and Subtracting Expressions with Algebra Tiles

The algebra tiles shown represent x and $-x$ as rectangles, and integers as squares. Red represents the negative of that variable or constant, while the blue shades represent the positive.



Variables also form zero pairs, as shown.



- Complete the table by crossing out the zero pairs and recording how many there are. Write the resulting algebraic expression after eliminating the zero pairs.

Algebra Tiles	Number of Zero Pairs	Algebraic Expression
Representation: 	2 zero pairs of x	$x - 2$



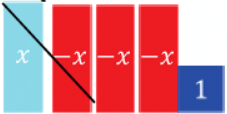
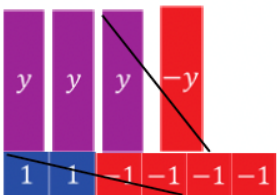
For students new to algebra tiles, consider discussing the dimensions of the tiles, noting that “like” terms have “like” sizes. This would be important to note for any students sketching tiles.



Consider providing algebra tiles for students or groups of students as kinesthetic scaffolding.

13.7.5 Guided Instruction: Adding and Subtracting Expressions with Algebra Tiles (continued)

2. Draw a representation of the following expressions with algebra tiles. Cross out any zero pairs. Then write an equivalent algebraic expression after eliminating any zero pairs.

Algebraic Expression and Representation	Equivalent Algebraic Expression
$x - 3x + 1$ Representation: 	$-2x + 1$
$-4 + 3y - y + 2$ Representation: 	$2y + (-2)$ <i>or</i> $2y - 2$



As students transition from numeric expressions to algebraic expressions, it becomes more essential to highlight the differences in size as an indicator for the like terms. Transition student language from “same size” to “like terms” for effective transitions to algebraic expressions.

13.7.6 Guided Practice: Adding and Subtracting Expressions with Algebra Tiles

Component Overview: Students model given expressions using algebra tiles and determine if they are equivalent.



13.7.6 Guided Practice: Adding and Subtracting Expressions with Algebra Tiles

1. Consider the expressions $2x$ and $x + 2$.
 - a. Draw a model of $2x$ and $x + 2$ with algebra tiles.

Expression	Algebra Tiles Model
$2x$	
$x + 2$	

- b. How are they different?

The expression $2x$ represents the sum of two like terms: $x + x$. Equivalently, it represents the product: $2x$. On the other hand, $x + 2$ represents the sum of two terms that are not like terms, x and 2 , so they cannot be combined.

- c. Are $2x$ and $x + 2$ equivalent algebraic expressions?
No, see part b for justification.

$$x + x \neq x + 2$$



Consider a Turn and Talk or Think-Pair-Share for students or classrooms who learn best with a partner.

- How are these expressions the same? How are they different?
- If the two expressions were added together, what would be the resulting expression?



Consider challenging students to provide mathematical justification by substituting a value for x to prove they are not equivalent.

13.7.7 Your Turn!

Component Overview: Students combine like terms to create equivalent algebraic expressions.



13.7.7 Your Turn!

Write an equivalent expression by combining like terms.

1. $2x + 4x =$ 6x

2. $-y + 7 + 4y - 1 =$ 3y + 6

3. $x - x + x - x + 1 =$ 1

4. $8x - 4 - 4x =$ 4x - 4

5. $2x - 4x =$ -2x



Consider giving students the opportunity to use or sketch algebra tiles or use the digital manipulatives as a method of producing their answers.

13.7.8 Wrap-Up: Cool Down

Component Overview: Students reflect on their progress and provide examples as evidence.



13.7.8 Wrap-Up: Cool Down

- For each key word, check the box in the column that describes your knowledge of the word. Then answer the questions or complete the statements in the last column.

Answers will vary.

Key Word	I already knew the meaning of this word.	I've heard this word before, but I was not sure what it meant.	Questions
Expression	☐	☐	An example of a numeric expression is <u>3</u> . An example of an algebraic expression is <u>$2x + 4$</u> .
Term	☐	☐	In the expression $8x - 9 + 2y$, how many terms are there? <u>3</u>
Coefficient	☐	☐	In the expression $8x - 9 + 2y$, what are the coefficients? <u>8 and 2</u>
Equivalent	☐	☐	When working with algebraic expressions the word "equivalent" means <u>two expressions have the same value when the variables are set to the same numbers</u> .



Consider utilizing students' responses to inform how to scaffold instruction for the next lesson when adding linear expressions.



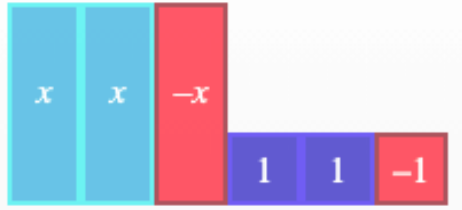
If students state that -9 is a coefficient, further ask them why. Theoretically it is a coefficient of x^0 . If a student does not use that reasoning, then the student has the misconception that any numeric value is a coefficient.

13.8 Adding Linear Expressions With Rational Coefficients

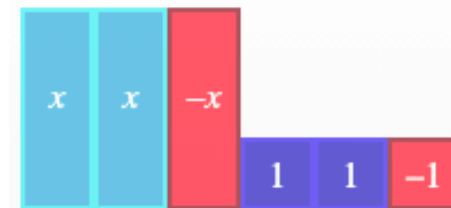
Lesson Introduction

 Benchmark	MA.7.AR.1.1 Apply properties of operations to add and subtract linear expressions with rational coefficients.		 Learning Target	I can add linear expressions with rational coefficients.
 Benchmark Coherence	Building On		Working Toward	
	MA.6.AR.1.4 Apply the properties of operations to generate equivalent algebraic expressions with integer coefficients.		MA.8.AR.1.2 Apply properties of operations to multiply two linear expressions with rational coefficients.	
 Essential Questions	- How can you represent algebraic expressions using algebra tiles? - How is adding linear expressions algebraically similar to adding linear expressions using algebra tiles?			
 Lesson Narrative	This lesson builds on Lesson 7 with a focus on adding linear expressions with rational numbers and exploring the effect of parentheses when adding linear expressions with algebra tiles. Students deepen their understanding of combining like terms and find the sum of algebraic expressions using the fewest possible terms.			
 Component Summary	13.8.1 Warm-Up (~5 min.)	Explore, describe, and make predictions based on a visual pattern of squares (<i>SE p. 302</i>)	13.8.4 Guided Instruction (10 min.)	Teacher models how to find the sum of linear expressions using the fewest terms possible without using algebra tiles (<i>SE p. 304</i>)
	13.8.2 Exploration (10 min.)	Consider four expressions and explain which one does not belong using math reasoning (<i>SE p. 302</i>)	13.8.5 Collaboration (10 min.)	Work with a partner to find the sum of linear expressions algebraically using the fewest terms possible (<i>SE p. 304</i>)
	13.8.3 Collaboration (10 min.)	Students add linear expressions using algebra tiles (<i>SE p. 303</i>)	13.8.6 Your Turn! (10 min.)	Find the sum of linear expressions using the fewest terms possible (<i>SE p. 306</i>)
	13.8.7 Wrap-Up (~5 min.)	Error analysis for adding expressions (<i>SE p. 307</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> Coefficient		- Algebra tiles (Optional) Notecards (Optional) Chart paper (Optional) Markers	- If using concrete algebra tiles, determine how many tiles are needed to model all the examples in the lesson and organize student sets in a container or Ziplock bag. - If using virtual algebra tiles, practice using the manipulative and create student-friendly directions for how to use.

 Teacher Tips	Common Misconceptions - Students may struggle with adding rational numbers as constant or coefficients. - Students may try to combine terms inside of parentheses first that cannot be combined, like rewriting $(x + 2)$ as $2x$.	Things to Consider - In 13.8.3 Collaboration, discuss the value of the parentheses in adding linear expressions after students model a few problems with algebra tiles. Students should conclude that the parentheses can be dropped and then combine like terms. - Mathematically, the meaning of “simplest form” is debated, so use caution with how frequently this language is used. It should be varied with language such as “equivalent forms” and “using the fewest terms possible.”
	Literacy and Language Routines Which One Doesn’t Belong Prompt students to share multiple answers and pathways for other students in 13.8.2 Exploration’s Which One Doesn’t Belong. There is no one “correct” answer, just like neither of two equivalent expressions is more “correct” in representation (MA.K12.MTR.2.1: Multiple Representations).	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.2.1: Multiple Representations Students progress from modeling the problems concretely with algebra tiles in 13.8.3 Collaboration to adding linear algebraic expressions with rational coefficients algebraically in the subsequent components of the lesson. Highlight multiple representations throughout the lesson, both in language and in models like algebra tiles, to continue reinforcing concepts of equivalence in multiple representations.
	 Differentiate Instruction	

The use of algebra tiles in 13.8.2 Exploration provides concrete and visual representations of the concepts. Consider using algebra tiles throughout this lesson for students who may struggle with adding like terms and finding the sums, where possible/appropriate, with particular emphasis on zero pairs.	
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Lesson Notes:



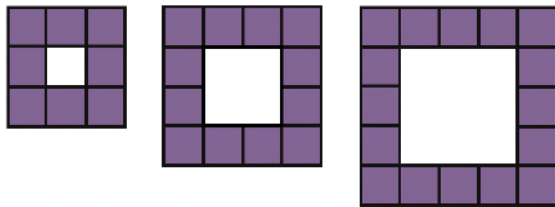
13.8.1 Warm-Up: Visual Patterns

Component Overview: Students explore, describe, and make predictions based on a visual pattern of blocks.



13.8.1 Warm-Up: Visual Patterns

Three figures are shown.



1. What pattern do you see?

Answers will vary.

I see small purple squares making larger squares with a white square in the middle. The square gets bigger every time. The length of the square increases by 1 unit each step in the pattern. The number of purple squares increases by 4 each step.

2. Explain how many small purple squares you think will be in the next figure.

Answers will vary.

The number of small purple squares increases by 4 each time. Since there were 16 small purple squares in the third figure, there will be 20 in the next figure.



MA.K12.MTR.5.1: Patterns and Structure

Students who successfully exhibit mathematical thinking and reasoning recognize patterns in what they see and develop generalizations for similar problems. Highlight student moves that embody these qualities.

The patterns and mathematical thinking students exhibit may not exactly match the key but may still be correct mathematical thinking. For example, a student might write the sequence “8, 12, 16” and be able to predict the next figure will have 16 squares. Students do not have to be able to express the pattern algebraically to demonstrate mathematical thinking.

13.8.2 Exploration: Which One Doesn't Belong?

Component Overview: Students explore four expressions and explain which one does not belong with the other three. Encourage students to explore different possible correct answers and determine their rationale.



13.8.2 Exploration: Which One Doesn't Belong?

- Which expression does not belong with the other three? Explain your reasoning.

Answers will vary.

Expression C, because the others are equivalent to $-5x + 4$ while Expression C is equivalent to $-5x + 3$.

Expression B because it's the only expression with fractions.

Expression C because it's the only expression with decimals.

Expression A $7 - 5x - 3$ $-5x + 4$	Expression B $\frac{3}{2}x + 6 - 2 - 6\frac{1}{2}x$ $-5x + 4$
Expression C $-5x + 7.25 - 4.25$ $-5x + 3$	Expression D $-2x - 3x + 4$ $-5x + 4$



Which One Doesn't Belong

This language routine purposefully provides four different “correct” answers in order to increase entry points and amplify mathematical discussion.

Consider the following possible responses for each expression:

- Expression A: “Only one that begins with a constant” or “Only one that begins with a positive integer.”
- Expression B: “Only one with fractions,” “Only one that begins and ends with a variable,” or “Only one with four terms.”
- Expression C: “Only one with decimals” or “Only one in which the simplified form is $-5x+3$.”
- Expression D: “Only one with only one constant.”

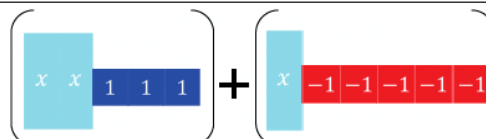
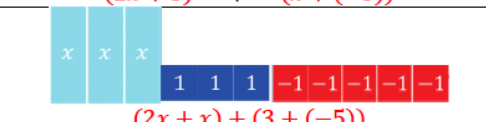
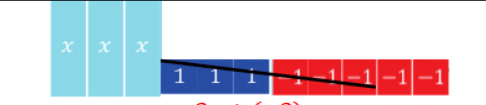
13.8.3 Collaboration: Adding Expressions with Algebra Tiles

Component Overview: Working with a partner, students add linear expressions using algebra tiles. After students have found the sum of each expression, engage in a whole class discussion to verify answers and address any potential misconceptions that may have arisen.

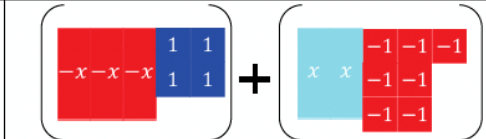
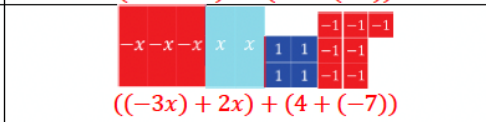
13.8.3 Collaboration: Adding Expressions with Algebra Tiles

Algebra tile representations of an expression are given. Work with your partner to find the sum by following the steps below.

1.

On the lines, write the algebraic expression for each representation.	 $(2x + 3) + (x + (-5))$
Rearrange the tiles to group like tiles. Draw the representation.	 $(2x + x) + (3 + (-5))$
Eliminate any zero pairs to find the sum. Write the sum.	 $3x + (-2)$

2.

On the lines, write the algebraic expression for each representation.	 $(-3x + 4) + (2x + (-7))$
Rearrange the tiles to group like tiles. Draw the representation.	 $((-3x) + 2x) + (4 + (-7))$
Eliminate any zero pairs to find the sum. Write the sum.	 $-x + (-3)$



Consider providing algebra tiles for kinesthetic learners interacting with this activity. Digital algebra tiles are also available in the Student and Teacher Areas.



Notice the visual representation of the tiles. A common misconception might be the constants interpreted as coefficients. For example, the first set of tiles may be interpreted as $3x^2$ instead of $2x + 3$.



Notice the use of “zero pairs” when finding the sum.

13.8.4 Guided Instruction: Adding Linear Expressions

Component Overview: This component is intentionally scaffolded to model the process for combining like terms without the visual aid of algebra tiles. The teacher models how to rearrange the terms to put like terms together and then combine them to find the simplified expression, similar to how students have done this with algebra tiles up until this point.



13.8.4 Guided Instruction: Adding Linear Expressions

When adding two expressions, rearrange the terms so that like terms are grouped together. Then add the coefficients of the like variable terms together and add the constants. Zero pairs are eliminated in this process.



Subtracting an integer is the same as adding the opposite. For example, the expression $9 - 13$ can be rewritten as $9 + -13$.

- Find the sum of the two expressions.

a.

Original Expression	$(2x + 6) + (5x + 4)$
Regrouped Expression: Rearrange the terms so that like terms are together.	$2x + 5x + 6 + 4$
Simplified Expression: Add the coefficients of the like variable terms and add the constants.	$7x + 10$

b.

Original Expression	$(-2x + 6) + (5x - 4)$
Regrouped Expression: Rearrange the terms so that like terms are together.	$-2x + 5x + 6 + (-4)$
Simplified Expression: Add the coefficients of the like variable terms and add the constants.	$3x + 2$



Consider using language such as, “Rearrange the terms so like terms are together...this is regrouping the expression,” and similarly making connections to the second phrase in each table to “simplifying” so that those connections are clear when students move on to the guided practice with a partner.



This lesson only prioritizes adding linear expressions, so do not spend time on modeling or explaining the relationship between addition and subtraction. Lesson 9 will cover this.

13.8.5 Collaboration: Adding Linear Expressions

Component Overview: Working with a partner, students find the sum of linear expressions using the fewest terms possible.

13.8.5 Collaboration: Adding Linear Expressions

1. Work with your partner to find the sum of the two expressions in simplest form.

a.

Original Expression	$4 + (8 - 0.4x)$
Regrouped Expression	$(-0.4x) + (8 + 4)$
Simplified Expression	$(-0.4x) + 12$

b.

Original Expression	$\left(3\frac{1}{2}x - \frac{3}{5}\right) + \left(2\frac{1}{4}x + \frac{1}{3}\right)$
Regrouped Expression	$\left(3\frac{1}{2}x + 2\frac{1}{4}x\right) + \left(\left(-\frac{3}{5}\right) + \frac{1}{3}\right)$
Simplified Expression	$5\frac{3}{4}x + \left(-\frac{4}{15}\right) = 5\frac{3}{4}x - \frac{4}{15}$

c.

Original Expression	$(-5y + 2) + (5y - 8)$
Regrouped Expression	$((-5y) + 5y) + (2 + (-8))$
Simplified Expression	-6



Notice the prompts with “regrouped” and “simplified” to represent the steps students can take to adding expressions. Use these terms in your question stems as you are monitoring students’ practice.



Students can also use colors and/or shapes to code like terms when simplifying linear expressions. It may also be helpful to encourage students to circle like terms with different shapes or colors.

13.8.5 Collaboration: Adding Linear Expressions (continued)

d.

Original Expression	$(0.7 - 2.4x) + (4.1x - 5.32)$
Regrouped Expression	$((-2.4x) + 4.1x) + (0.7 + (-5.32))$
Simplified Expression	$1.7x + (-4.62) = 1.7x - 4.62$



Notice the use of decimals and fractions in parts d and e. Encourage students to either rewrite -5.5 as a fraction or the other fractions as decimals.

e.

Original Expression	$\left(\frac{3}{4}w - 5.5\right) + \left(-\frac{1}{2}w + 4\right)$
Regrouped Expression	$\left(\frac{3}{4}w + \left(-\frac{1}{2}w\right)\right) + ((-5.5) + 4)$
Simplified Expression	$\frac{1}{4}w + (-1.5) = \frac{1}{4}w - 1.5$



Lesson 8 is all about adding, while the next lesson begins exploring subtraction. For students who are ready for a challenge, consider priming them with questions like:

- How would the simplified expression be different if the operation between parentheses was subtraction?

13.8.6 Your Turn!

Component Overview: Students work independently to find the sum of linear expressions using the fewest terms possible.



13.8.6 Your Turn!

1. Find the sum of each expression using the fewest terms possible.

a. $(x + 7) + (x - 10)$
 $2x + (-3)$
 $= 2x - 3$

b. $(3x - 5) + (5 + 6x)$
 $9x$

c. $(1\frac{6}{7} - y) + (4y + \frac{1}{7})$
 $3y + 2$

d. $(4w - 6) + (-6w - 1)$
 $(-2w) + (-7)$
 $= -2w - 7$

e. $(9 - 4.6x) + (-5.3x - 7)$
 $-9.9x + 2$

f. $(-\frac{5}{2}k - 7) + (\frac{1}{5}k + 4)$
 $= -\frac{23}{10}k + (-3)$
 $= -\frac{23}{10}k - 3$
 $= -2\frac{3}{10}k + (-3)$
 $= -2\frac{3}{10}k - 3$



Notice the entry points for questions 1a, 1b, and 1d (whole numbers) compared to 1c, 1e, and 1f (fractions and decimals). Consider partitioning practice so that students can choose which set of questions to engage with first.



Review the sum of each expression before moving on to the Wrap-Up and use this as an opportunity to clarify any lingering misconceptions.

13.8.7 Wrap-Up: Error Analysis

Component Overview: Students analyze a student's work finding the sum of two linear expressions. Students then write an explanation of why the work is correct or not.



13.8.7 Wrap-Up: Error Analysis

- Your friend finds the sum of $(-2\frac{1}{2}x + 7.1) + (4x - 5.2)$ to be $-\frac{3}{2}x + 1.9$.

Write a detailed note to your friend explaining why their answer is correct or why it is incorrect.

The answer is incorrect because the sign of the coefficient of x should be positive, not negative.

$$-2\frac{1}{2} + 4 = 1\frac{1}{2} = \frac{3}{2}, \text{ so}$$

$$\left(-2\frac{1}{2}x + 4x\right) + (7.1 + (-5.2)) = \frac{3}{2}x + 1.9.$$



MA.K12.MTR.4.1: Error Analysis

Consider highlighting clarification language in MA.K12.MTR.4.1 that students exhibit in their responses. If student response compares the larger positive value as reasoning behind their analysis, then select, sequence, and present student work to advance and deepen understanding of correct and increasingly efficient methods.

If student response shows evidence of different response using mathematical justification, then develop students' ability to justify methods and compare their responses to the responses of their peers.

13.9 Subtracting Linear Expressions with Rational Coefficients

Lesson Introduction

 Benchmark	MA.7.AR.1.1 Apply properties of operations to add and subtract linear expressions with rational coefficients.		 Learning Targets	- I can subtract linear expressions with rational coefficients. - I can apply properties of operations to add and subtract linear expressions with rational coefficients.	
 Benchmark Coherence	Building On MA.6.AR.1.4 Apply the properties of operations to generate equivalent algebraic expressions with integer coefficients.			Working Toward MA.8.AR.1.2 Apply properties of operations to multiply two linear expressions with rational coefficients.	
 Essential Questions	- What is different about subtracting linear expressions from adding linear expressions? - How do you subtract linear expressions? - How can you apply what you have learned about combining two linear expressions to combining three or more?				
 Lesson Narrative	Students extend their understanding of combining like terms from the previous lesson by exploring four different methods or strategies of subtracting linear expressions. Students analyze examples that start with integer coefficients and constants and build rational constants and coefficients in subtraction expressions with the fewest possible terms.				
 Component Summary	13.9.1 Warm-Up (~5 min.)	Bell work adding linear expressions (<i>SE p. 308</i>)	13.9.5 Guided Practice (10 min.)	Students work with a partner to find the differences of linear expressions (<i>SE p. 311</i>)	
	13.9.2 Exploration (5 min.)	Students use algebra tiles to find the difference of linear expressions (<i>SE p. 308</i>)	13.9.6 Collaboration (10 min.)	Error analysis for analyzing student work on operations with expressions (<i>SE p. 312</i>)	
	13.9.3 Collaboration (10 min.)	Students collaborate to analyze student work using the distributive property and then relate this to algebra tiles (<i>SE p. 309</i>)	13.9.7 Your Turn! (5 min.)	Combine like terms to simplify three linear expressions with rational coefficients (<i>SE p. 313</i>)	
	13.9.4 Guided Instruction (5-10 min.)	Teacher models how to find the differences of linear expressions by adding the opposite or distributing a negative (<i>SE p. 310</i>)	13.9.8 Wrap-Up (~5 min.)	Reflection on a mistake from today's lesson (<i>SE p. 313</i>)	
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms</u> N/A <u>Additional Terms:</u> - Additive inverse - Distributive property		- Algebra tiles (Optional) Notecards (Optional) Chart paper (Optional) Markers		- If using concrete algebra tiles, determine how many tiles are needed to model all the examples in the lesson and organize student sets in a container or ziplock bag. - Determine which students will work together for 13.9.6 Collaboration.

 Teacher Tips	Common Misconceptions - Students may forget to distribute subtraction, or (-1), to all terms in the parentheses (like in the 13.9.2 Exploration). - Students may not simplify the expression completely or have trouble combining all of the like terms in the expressions.	Things to Consider - When adding decimals, consider providing a model or anchor chart for rewriting problems vertically and line up the decimal places. When adding fractions, students may need to be reminded to obtain a common denominator. - If time is an issue, then consider assigning 13.9.6 Collaboration and/or 13.9.7 Your Turn! for homework.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder 13.9.4 Guided Instruction provides an opportunity for students to Notice and Wonder about two methods for subtracting linear expressions. Students complete a table using the additive inverse method and applying the distributive property method. Engage the class in a Notice and Wonder routine to draw conclusions and connections between the methods.	MA.K12.MTR.2.1: Multiple Representations In 13.9.4 Guided Instruction, students apply two different methods to subtract linear equations. Use this as an opportunity for students to make connections between the two methods and understand how to solve linear subtraction in multiple ways.
 Differentiate Instruction	Consider providing Math Nation’s On-Ramp video for operations with fractions as scaffolding for students when operating with mixed numbers or fractions greater than one (as seen in 13.9.5 Guided Practice and 13.9.7 Your Turn!). Avoid a previous learning target becoming a barrier to conceptual understanding for subtracting expressions by providing scaffolding.	
Lesson Notes:		

13.9.1 Warm-Up: Bell Work

Component Overview: Students combine like terms in algebraic expressions to find their sums.



13.9.1 Warm-Up: Bell Work

Combine like terms.

1. $(3y + 4.6) + (-5y - 3.25)$

$$= (3y + (-5y)) + (4.6 - 3.25) = -2y + 1.35$$

2. $(6\frac{3}{4}x - 7) + (-\frac{5}{2}x + 7)$

$$= (\frac{27}{4}x + (-\frac{10}{4}x)) + ((-7) + 7)$$

$$= \frac{17}{4}x \text{ or } 4\frac{1}{4}x$$



Consider having students circle or underline like terms before combining. This will provide a visual representation that can aid in understanding of what like terms are.

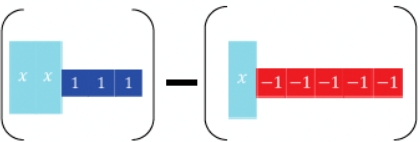
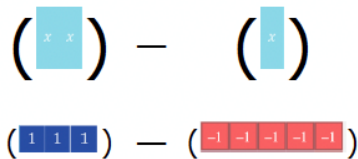
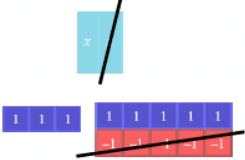
13.9.2 Exploration: Subtracting Expressions with Algebra Tiles

Component Overview: Students use algebra tiles to find the difference of the algebraic expression from the previous lesson's 13.8.3 Collaboration.



13.9.2 Exploration: Subtracting Expressions with Algebra Tiles

- Algebra tile representations are given below. Use the algebra tiles to find the difference.

On the lines, write the algebraic expression for each representation.	 $(2x + 3) - (x - 5)$
Rearrange the tiles to show subtraction of like tiles together. Draw the representation.	
If necessary, create any zero pairs needed for subtraction.	
Find the difference by subtracting (taking away) like tiles. Write the algebraic representation of the difference.	 $x + 8$



Make note of which students recognize that both x and -5 are being subtracted from $2x + 3$ compared to students who only subtract x and then add -5 . Instead of formalizing and explaining the misconceptions, use this activity as an opportunity to instead notice the difference when comparing with peers.



- Some students found the difference to be $x - 2$ while others found it to be $x + 8$. Why do you think that is? Which is correct?
- What is being subtracted from $2x + 3$?

13.9.3 Collaboration: Subtracting Expressions

Component Overview: Students work with a partner to evaluate the given work of four students. Read through each student's statement and work as a class, then partner students together to begin their evaluation. Make sure students not only find the error but also describe how to correct it.



13.9.3 Collaboration: Subtracting Expressions

1. Noah, Lin, Jada, and Andre are rewriting the expression $(x + 8) - (4 - 9x)$ as an equivalent expression with fewer terms. Each student described their work.

Discuss with your partner which students made an error. Describe each student's error and correct the error.

- a. Noah says, "I dropped the parentheses, combined like terms, and ended up with $-8x + 4$."

$$\begin{aligned}(x + 8) - (4 - 9x) \\ x + 8 - 4 - 9x \\ -8x + 4\end{aligned}$$

Noah did not distribute (-1) to both terms. After distributing (-1) to both terms, the result is $x + 8 - 4 + 9x = (x + 9x) + (8 - 4) = 10x + 4$.

- b. Lin says, "I simplified inside the parentheses and ended up with $13x$."

$$\begin{aligned}(x + 8) - (4 - 9x) \\ (8x) - (-5x) \\ 8x + 5x \\ 13x\end{aligned}$$

Lin combined terms that were not like terms. Lin should group the like terms together $(x + 9x) + (8 - 4)$ and then combine like terms to obtain $10x + 4$.

- c. Jada says, "I know that subtracting is adding the opposite, and ended up with $4 + 10x$."

$$\begin{aligned}(x + 8) - (4 - 9x) \\ (x + 8) + (-4 + 9x) \\ x + 8 - 4 + 9x \\ 4 + 10x\end{aligned}$$

Jada has a correct answer. The explanation should include intermediate steps.



This activity acts as a primer for 13.9.4 Guided Instruction. Monitor student responses while they collaborate and make note of any responses that you can use to build on when formalizing approaches in 13.9.4. Students should analyze and compare student responses before writing what students may have done incorrectly, as seeing the different examples will help them solidify their initial thoughts.



Consider providing sentence stems for the written explanations, such as:

- "Noah's answer is correct/incorrect. I know this because..."
- "The mistake he/she made was..."
- "It can be corrected by..."

This is a Collaboration, so prompt students to refine their responses with their partners.

13.9.3 Collaboration: Subtracting Expressions (continued)

- d. Andre says, "I used the distributive property since there is an invisible (-1) in front of the second parentheses to distribute, but I ended up with $10x + 4$."

$$\begin{aligned}(x + 8) - (4 - 9x) \\(x + 8) - 1(4 - 9x) \\x + 8 - 4 + 9x \\10x + 4\end{aligned}$$

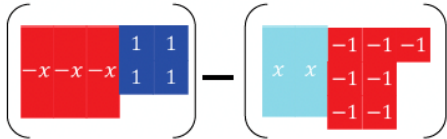
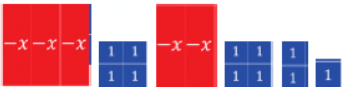
Andre has a correct explanation and answer.



Consider probing questions for students or partners that are struggling to identify the differences in student work.

- Compare Noah and Jada's work. What is the same? What is different?
- Compare Jada's work to Andre's work. What is the same? What is different?

2. When subtracting expressions, use the distributive property to add the opposite. With the algebra tile representations below, model finding the difference by adding the opposite.

On the lines, write the algebraic expression for each representation.	 $(-3x + 4) - (2x - 7)$
Rewrite subtracting to adding the opposite by changing (or flipping) the sign of the 2 nd expression. Draw the representation.	 $(-3x + 4) + (-2x + 7)$
Combine any zero pairs to find the sum. Write the sum.	$-5x + 11$

13.9.4 Guided Instruction: Subtracting Linear Expressions

Component Overview: Review the statements as a class, then the teacher models how to use both methods to complete the table. Have students engage in a Notice and Wonder routine before answering the written response question.



13.9.4 Guided Instruction: Subtracting Linear Expressions

- To subtract linear expressions,
 - add the *opposite*, or additive inverse of each term in the expression.

OR

- distribute a *negative one* to each term in the expression.
- Complete the table below to show both methods.

Additive Inverse Method	Applying the Distributive Property
$(4x + 5) - (-2x + 6)$ $= (4x + 5) + (2x + (-6))$ $= (4x + 2x) + (5 + (-6))$ $= 6x + (-1)$ $= 6x - 1$	$(4x + 5) - (-2x + 6)$ $= 4x + 5 + 2x - 6$ $= (4x + 2x) + (5 + (-6))$ $= 6x + (-1)$ $= 6x - 1$

- When adding and subtracting, parentheses are used to group expressions. Why is the use of parentheses especially important when subtracting?

The use of parentheses ensures that the entire expression is subtracted. This means the negative 1 is distributed to all terms in the expression.



Consider using the following questions to extend student thinking and formalize their understanding of each approach following their written responses:

- How is the expression $(4x + 5) - (-2x + 6)$ different from the expression $4x + 5 - -2x + 6$?
- What do the parentheses communicate about what is being subtracted?

13.9.5 Guided Practice: Subtracting Linear Expressions with Rational Coefficients

Component Overview: Students find the differences in linear expressions. Students work with their partners after seeing the teacher model strategies in the Guided Instruction. If, after the Guided Instruction, it appears students need an additional example, considering working through one of the Guided Practice problems as a whole class before releasing to partners.



13.9.5 Guided Practice: Subtracting Linear Expressions with Rational Coefficients

1. Find the difference of each expression.

a. $(3y + 3.5) - (8.2y - 6)$ $= (3y + 3.5) - 8.2y + 6$ $= (3y + (-8.2y)) + (3.5 + 6)$ $= -5.2y + 9.5$	b. $(m - \frac{3}{8}) - (3\frac{1}{4} - m)$ $= (m - \frac{3}{8}) - 3\frac{1}{4} + m$ $= (m + m) + ((-\frac{3}{8}) + (-3\frac{1}{4}))$ $= 2m - 3\frac{5}{8}$
c. $(-2 + \frac{9}{2}y) - (5.3y - 5\frac{1}{2})$ $= (-2 + \frac{9}{2}y) - 5.3y + 5\frac{1}{2}$ $= (\frac{9}{2}y + (-5.3y)) + (-2 + 5\frac{1}{2})$ $= -0.8y + 3.5 \text{ or } -\frac{4}{5}y + 3\frac{1}{2}$	d. $(-7.5t - 3) - (7 + 3\frac{1}{8}t)$ $= (-7.5t - 3) - 7 - 3\frac{1}{8}t$ $= -7.5t - 3\frac{1}{8}t + (-3 + -7)$ $= -7.5t - 3\frac{1}{8}t + (-10)$ $= -10\frac{5}{8}t - 10$

2. What is the result when $-3x + 4.48$ is subtracted from $-2.2x + 3$?

- Write the algebraic expression.
 $(-2.2x + 3) - (-3x + 4.48)$
- Rewrite the expression without parentheses.
 $-2.2x + 3x + 3 + (-4.48)$
- Simplify the expression by rewriting it using the fewest terms possible.
 $0.8x - 1.48$



Consider pulling a small group for students who would benefit from a model or think-aloud that highlights the usefulness of explicitly writing the coefficient -1 for better visual scaffolding of the distribution.



Notice the phrase “subtracted from” in question 2. Students might invert the order. Consider providing this prompt as a self-check for students:

I see some students found the difference to be $0.8x - 1.48$ while others found it to be $-0.8x - 1.48$.

- Which is correct?
- Which expression is being subtracted?

13.9.6 Collaboration: Adding and Subtracting Linear Expressions

Component Overview: Students work with a partner to analyze, discuss, and correct students' errors in three linear expressions with integer coefficients. Circulate as students complete the problem together and make note of common misconceptions to address whole class with. Review student responses as a class and clarify misconceptions noted during circulation.



13.9.6 Collaboration: Adding and Subtracting Linear Expressions

1. Two students made errors when rewriting the expression $(6x + 9) - (7 - x) + (8x + 11)$.

Zoe's Work	Patricia's Work
$(6x + 9) - (7 - x) + (8x + 11)$	$(6x + 9) - (7 - x) + (8x + 11)$
$(6x + 9) - (7x + 18)$	$(5x + 2) + (8x + 11)$
$-x - 9$	$13x + 13$

- a. Working with your partner, write an explanation to Zoe that describes her error.

Zoe, it looks like you did not follow the order of operations or recognize that a -1 was being distributed to $(7 - x)$. You subtracted x from $8x$ and added 7 and 11 , but you can't combine these terms until you have first distributed the negative to $(7 - x)$. Once that is done, then you can combine like terms from left to right.

- b. Write an explanation to Patricia that describes her error.

Patricia, it looks like you did not distribute -1 to both terms in the expression $(7 - x)$. You subtracted 7 from 9 , but you should have added x to $6x$ rather than subtracting it because subtracting $-x$ is the same as adding x .

- c. Simplify the expression $(6x + 9) - (7 - x) + (8x + 11)$.
 $= 6x + 9 - 7 + x + 8x + 11$
 $= 15x + 13$



Consider providing an additional entry point for verbal processors by intentionally scaffolding the error analysis as a Think-Pair-Share in which students first analyze independently before sharing responses with a partner for more precise responses.



Following question 1b, give students an opportunity to formalize their learning before applying in question 1c.

- When have you made this error before?
- What do you look for to avoid this error in the future?
- What will that look like in question 1c?

13.9.7 Your Turn!

Component Overview: Students work independently to combine like terms and simplify three linear expressions with rational coefficients.



13.9.7 Your Turn!

1. Rewrite each expression by combining like terms.

$$\begin{aligned} \text{a. } & (1 - 4y) - (2.5y + 7) - (-0.3y + 3) \\ & = (-4y - 2.5y + 0.3y) + (1 - 7 - 3) \\ & = -6.2y - 9 \end{aligned}$$

$$\begin{aligned} \text{b. } & \left(3\frac{1}{2}x - 4\right) + (x - 3.2) - \left(-\frac{1}{4}x + 9\right) \\ & = \left(3\frac{1}{2}x + x + \frac{1}{4}x\right) + (-4 - 3.2 - 9) \\ & = 4\frac{3}{4}x - 16\frac{1}{5} \text{ or } 4.75x - 16.2 \end{aligned}$$

$$\begin{aligned} \text{c. } & -(2r + 5) - (6 - 3r) + (10r - 4) \\ & = (-2r + 3r + 10r) + (-5 - 6 - 4) \\ & = 11r - 15 \end{aligned}$$



For extension and error analysis practice (MA.K12.MTR.4.1), prompt students to determine what a student did wrong if they got the following answers:

- $-1.2y - 9$ (added $+2.5y$ instead of $-2.5y$)
- $4\frac{3}{4}x + 1.8$ (added $+9$ instead of -9)
- $5r - 15$ (added $-2r - 3r$ instead of $+3r$)

13.9.8 Wrap-Up: Ticket Out the Door

Component Overview: Students combine like terms to simplify two linear expressions with rational coefficients.



13.9.8 Wrap-Up: Ticket Out the Door

Write the following expressions using the fewest possible terms.

1. $(5x - 13\frac{1}{4}) - (4\frac{1}{2} - 5x)$

$$= 5x + 5x - 13\frac{1}{4} - 4\frac{1}{2}$$

$$= 10x - 17\frac{3}{4}$$

2. $(0.7 - 2.4x) + (-5.32 + 4\frac{1}{10}x)$

$$= 0.7 - 5.32 - 2.4x + 4\frac{1}{10}x$$

$$= -4.62 + 1.7x$$



Students may get $0x - 17\frac{3}{4}$ or $-17\frac{3}{4}$ for the first linear expression. Remind students that writing the coefficient of -1 in front of the second parenthesis can help them remember to distribute.



While the lesson's learning target is focused on subtracting linear expressions, this ticket out the door is intended to be used as formative assessment for both adding and subtracting linear expressions following the first three lessons of the unit. Use the informal data gathered from this formative assessment to plan for adjustments or remediation prior to the end of the unit.

13.10 Equivalent Linear Expressions

Lesson Introduction

Lesson Introduction				
 Benchmark	MA.7.AR.1.2 Determine whether two linear expressions are equivalent.		 Learning Target	I can verify using substitution that linear expressions are equivalent.
 Benchmark Coherence	Building On		Working Toward	
	MA.6.AR.1.4 Apply the properties of operations to generate equivalent algebraic expressions with integer coefficients.		MA.8.AR.1.3 Rewrite the sum of two algebraic expressions having a common monomial factor as a common factor multiplied by the sum of two algebraic expressions.	
 Essential Questions	- What is the difference between equivalence and equality? - How can you determine whether two algebraic expressions are equivalent?			
 Lesson Narrative	Students extend their understanding of equivalent expressions to include substituting rational numbers into variable expressions and evaluating them to determine equivalence. The concept of equal versus equivalent is explored, and students debate what values are ideal to use when proving equivalence. The focus of this lesson should be on substitution, not simplifying expressions to prove equivalence.			
 Component Summary	13.10.1 Warm-Up (~5 min.)	Reflect on the origins of the word “algebra” (<i>SE p. 314</i>)	13.10.4 Guided Instruction (10 min.)	Determine whether algebraic expressions are equivalent using substitution (<i>SE p. 316</i>)
	13.10.2 Refresher (5 min.)	Evaluate expressions for given values of the variable using substitution (<i>SE p. 315</i>)	13.10.5 Guided Practice (10 min.)	Use substitution to verify if expressions are equivalent (<i>SE p. 318</i>)
	13.10.3 Collaboration (10 min.)	Evaluate expressions for multiple values to determine if the expressions are equivalent (<i>SE p. 315</i>)	13.10.6 Your Turn! (5 min.)	Determine which expressions are equivalent by substituting values for the variable (<i>SE p. 319</i>)
	13.10.7 Wrap-Up (~5 min.)	Self-assessments of understanding of lesson’s learning targets (<i>SE p. 320</i>)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> - Equivalent - Equal <u>Additional Terms:</u> N/A		(Optional) Colored pencils (Optional) Chart paper (Optional) Markers	
			Additional Preparation	
			Consider providing an anchor chart, word wall, or other visual representation of the key terms in this lesson for visual learners.	

 Teacher Tips	Common Misconceptions - Students may add substituted values with their coefficients rather than multiplying. For example, in question 1 in 13.10.2 Refresher, students may add $-3(5)$ to get 2 rather than multiplying to get -15. - Students may describe equivalent expressions as “equal” to each other.	Things to Consider - Lesson 12 will revisit the use of properties as a method of determining equivalence, while this lesson targets opportunities for substitution, with the 13.10.7 Wrap-Up giving students choice in which method they choose. - Determine if your students need the support of all or part of the 13.10.2 Refresher - If time is an issue, then consider assigning 13.10.6 Your Turn! for homework.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Stronger and Clearer Students extend their understanding of equivalence from numeric to algebraic expressions, particularly making this connection during 13.10.3 Collaboration. Encourage students to engage in a Stronger and Clearer routine for articulating their responses in question 3, taking multiple attempts and getting feedback from their partner to make each attempt better, stronger, and clearer.	MA.K12.MTR.1.1: Participate in Effortful Learning Notice the elements of collaboration embedded throughout this lesson that provide opportunities for students to collectively contribute toward learning targets. Monitor student responses as they discuss their thinking in 13.10.3 Collaboration, 13.10.4 Guided Instruction, and 13.10.5 Guided Practice, and use talk moves to facilitate student discussion as they refine their responses and learn from each other.
 Differentiate Instruction	Collaboration and discussion are key in this lesson. Utilize sentence frames to enhance student discussion to target key learning opportunities as students collaborate throughout the lesson. Sentence frames provide low-floor, high-ceiling entry points for learners to either enter the conversation or extend their thinking to incorporate mathematical concepts. Expression _____ Is equivalent to _____. I used a method of _____ to show that _____ substitution evaluating properties equivalent	
Lesson Notes:		

13.10.1 Warm-Up: Origins of Algebra

Component Overview: Students reflect on the origin of the word “algebra” and how it relates to algebraic expressions and equations.



13.10.1 Warm-Up: Origins of Algebra

The word **algebra** comes from the Arabic word الجبر, or al-jabr, which means, “the restoring of broken parts.” The word algebra comes from the title of a book written by a Persian mathematician named Muhammad ibn Musa al-Khwarizmi in the 9th century.



Students will provide a variety of creative answers for this prompt. Consider what connections could be made to content in this unit or in previous units.

1. Based on what you know about algebraic expressions and equations, explain why you think this meaning of the Arabic word makes sense to describe algebra.

Answers will vary.

“Restoring of broken parts” makes sense for algebra because algebra includes solving equations and combining parts (terms) that are alike.

13.10.2 Refresher: Evaluating Algebraic Expressions

Component Overview: Students work to evaluate expressions for given values of the variable. Quickly review the values of each expression before moving on to ensure students have checked their work and are given time to ask clarifying questions.



13.10.2 Refresher: Evaluating Algebraic Expressions

Evaluate each expression for the given value of x .

1. $-3x - 8$ for $x = 5$

$$\begin{aligned} &= -3(5) - 8 \\ &= -15 - 8 \\ &= -23 \end{aligned}$$

2. $4 - 2(x^2 - 7)$ for $x = -1$

$$\begin{aligned} &= 4 - 2(1 - 7) \\ &= 4 - 2(-6) \\ &= 4 + 12 \\ &= 16 \end{aligned}$$

3. $(4.5 + x) \cdot 3$ for $x = -\frac{5}{2}$

$$\begin{aligned} &= (4.5 + (-\frac{5}{2})) \cdot 3 \\ &= (4.5 - 2.5) \cdot 3 \\ &= 2 \cdot 3 \\ &= 6 \end{aligned}$$



Students continue practicing throughout the lesson. Avoid reteaching operations and substitution. Instead, make note of which students could use additional support and pull a small group during 13.10.3 Collaboration or 13.10.5 Guided Practice based on your observations.

13.10.3 Collaboration: Solving Equivalent Expressions Using Substitution

Component Overview: Students work with a partner to evaluate expressions for given values and determine whether they are equivalent, reasoning that the expressions are equivalent because they have the same value when evaluated for multiple values. Consider engaging the class in a Notice and Wonder routine before completing question 4 and revisiting their wonderings after they have simplified both expressions.



13.10.3 Collaboration: Solving Equivalent Expressions Using Substitution

- With your partner, complete the table by choosing who will be Partner A or Partner B to determine if the expressions are equivalent.

Partner and Value	$8x + 4x - 2$	$2(6x - 1)$	Equivalent?
A $x = 2$	$= 8(2) + 4(2) - 2$ $= 16 + 8 - 2$ $= 22$	$= 2(12 - 1)$ $= 2(11)$ $= 22$	☐ yes ☐ no
B $x = 5$	$= 8(5) + 4(5) - 2$ $= 40 + 20 - 2$ $= 58$	$= 2(30 - 1)$ $= 2(29)$ $= 58$	☐ yes ☐ no
A $x = -5$	$= 8(-5) + 4(-5) - 2$ $= -40 + -20 + -2$ $= -62$	$= 2(-30 + -1)$ $= -62$	☐ yes ☐ no
B $x = (-2)$	$= 8(-2) + 4(-2) - 2$ $= -16 + -8 + -2$ $= -26$	$= 2(-12 + -1)$ $= -26$	☐ yes ☐ no



Consider having students who finish early make a prediction about future expressions.

- How do the properties of operations relate to these expressions and their equivalence?
- How could properties of operations be used to determine equivalence without evaluating using substitution?

This rationale and method of thinking builds proficiency in fluency (MA.K12.MTR.3.1).

- Complete the statement to describe the results in the last column of the table.

Depending on the value being substituted into the expressions, the evaluated expressions had the same value as each other.

- ☐ always
☐ sometimes
☐ never

- What do the results tell you about the expressions $8x + 4x - 2$ and $2(6x - 1)$?

They are equivalent expressions.

- Rewrite $8x + 4x - 2$ by combining the like terms.

$12x - 2$

- Use the distributive property to rewrite $2(6x - 1)$.

$12x - 2$

- Discuss with your partner how the two expressions verify the results from your table.



Stronger and Clearer

Question 3 sample response: "Just like numeric expressions can look different but be the same, linear expressions can also look different but actually have the same value."

13.10.4 Guided Instruction: Determining Equivalent Expressions Using Substitution

Component Overview: Students review the definitions and difference between equivalence and equality. Then students will determine whether expressions are equivalent by substituting different values for x . Students will be guided through why the expressions are not equivalent and consider why using zero as a variable value can be both beneficial and problematic.



13.10.4 Guided Instruction: Determining Equivalent Expressions Using Substitution

The two expressions $8x + 4x - 2$ and $2(6x - 1)$ are equivalent, but not equal. The expressions $12x - 2$ and $12x - 2$ are equal.

1. What is the difference between **equivalence** and **equality**?



Algebraic expressions are **equivalent** if they have different terms but are equal when evaluated using the same value for the variables. Algebraic expressions are **equal** if they have the same terms and operations.

2. Determine whether $5 - 3(2x + 1)$ and $4x + 2$ are equivalent using substitution of numerical values.
 - a. Complete the table.

x	$5 - 3(2x + 1)$	$4x + 2$
What is the value when $x = 0$?	$= 5 - 3(0 + 1)$ $= 5 + -3(1)$ $= 5 + -3$ $= 2$	$= 0 + 2$ $= 2$
What is the value when $x = \frac{1}{2}$?	$= 5 + -3(1 + 1)$ $= 5 + -3(2)$ $= 5 + -6$ $= -1$	$= 2 + 2$ $= 4$
Choose another value for x to substitute. <i>Answers will vary. For example, if we let $x = -2$.</i>	$= 5 + -3(-4 + 1)$ $= 5 + -3(-3)$ $= 5 + 9$ $= 14$	$= -8 + 6$ $= -2$



This is a discussion opportunity. Students do not need to write an explanation here. Listen as students discuss their thinking in response to this prompt to determine student thinking to highlight or misconceptions to address.



Use talk moves to prompt students to make sense of the difference between equivalent and equal. Consider a Think-Pair-Share or similar routine and prompt students to include an example that illustrates their working definitions. Sample exemplar response: "Equivalent expressions have the same value. Equal expressions are exactly the same in appearance."

13.10.4 Guided Instruction: Determining Equivalent Expressions Using Substitution (continued)

- b. Explain if the expressions are equivalent.
No, the expressions are not equivalent. They do not have the same value for all values substituted in for the variable.
- c. Explain why the expressions were equal when "0" was substituted for x .
They were equal when 0 was substituted in for the variable, because the other terms of the expression are equivalent.
- d. Discuss with your partner the advantages and disadvantages of using "0" to substitute for a variable to check for equivalence.
The advantages are that the calculations are easy. The disadvantages are that it may appear that the expressions are equivalent when they aren't.

3. Determine whether $3 + x(4 - 2)$ and $x(3 - 7) + 5$ are equivalent using substitution of numerical values.

- a. Complete the table.

x	$3 + x(4 - 2)$	$x(3 - 7) + 5$
What is the value when $x = 1$?	$= 3 + (2)$ $= 5$	$= 1(-4) + 5$ $= 1$
What is the value when $x = 0$?	$= 3 + 0$ $= 3$	$= 0 + 5$ $= 5$
Choose another value for x to substitute. <i>Answers will vary. One example is $x = -2$.</i>	$= 3 + -2(4 - 2)$ $= 3 + -2(2)$ $= 3 + -4$ $= -1$	$= -2(3 - 7) + 5$ $= -2(-4) + 5$ $= 8 + 5$ $= 13$

- b. How many values are best to use when substituting values in for the variable to check for equivalency?
It depends, but I like to substitute at least one negative and one positive number. If you only test zero, your calculations may be easy. However, the expressions may appear equivalent, when they are not.



Notice the multiple entry points for learning in this component. Utilize all entry points for students to engage visually with the table or verbally by explaining and discussing.



Consider asking students the following:

- What are the advantages and disadvantages of substituting zero in for the variable?
- How many values do you need to substitute in for the variable to determine if the expressions are equivalent?
- Should you always use zero to determine equivalence? Why or why not?

13.10.5 Guided Practice: Determining Equivalent Expressions Using Substitution

Component Overview: Students work independently or in pairs to determine equivalence.



13.10.5 Guided Practice: Determining Equivalent Expressions Using Substitution

1. Consider the three expressions below. Determine whether the expressions are equivalent by substituting the given value of x .

a. Complete the table.

x	$4x - \frac{1}{2}(6 + 8x)$	$8x - 3$	$-(4x + 3) + 4x$
$x = 3$	$= 12 + -1/2(6 + 24)$ $= 12 + -1/2(30)$ $= 12 + -15$ $= -3$	$= 24 - 3$ $= 21$	$= -1(12 + 3)$ $= -12 - 3$ $= -15$ $= -3$
$x = -2.75$	$= 4(-2.75) + -1/2(6 + 8(-2.75))$ $= -11 + -1/2(6 + -22)$ $= -11 + -1/2(-16)$ $= -11 + 8$ $= -3$	$= -22$ $- 3$ $= -25$	$= -1(-11 + 3)$ $= 11 - 3$ $= 8$ $= -3$
Answers will vary. For example, $x = 1$.	$= 4 + -1/2(6 + 8)$ $= 4 + -1/2(14)$ $= 4 + -7$ $= -3$	$= 8 - 3$ $= 5$	$= -1(4 + 3) + 4$ $= -7 + 4$ $= -3$

- b. What conclusion can you draw about the expressions from the information in the table?

The expressions $4x - \frac{1}{2}(6 + 8x)$ and $-(4x + 3) + 4x$ are equivalent. The expression $8x - 3$ is not equivalent to either expression.



Consider a small group or student choice based on the needs of your students.

- **Small Group:** Pull a small group to model or think-aloud operations and properties with students identified from 13.10.2 Refresher.
- **Choice:** Allow students to choose which row to complete then find a partner around the room who completed the same row to compare responses. Additionally, students could choose their own value for x and substitute to prove equivalence.



Ask advanced learners to rewrite the expression that is not equivalent to the other two expressions so that it is equivalent. Challenge them to include decimals and fractions and at least four different operations in their expression.

13.10.5 Guided Practice: Equivalent Expression Using Substitution (continued)

2. Determine whether the expressions $\frac{2}{5}(15x - 30y) + 10x$ and $4(4x - 2y)$ are equivalent.

x and y	$\frac{2}{5}(15x - 30y) + 10x$	$4(4x - 2y)$
$x = -2$ $y = 0$	$= \frac{2}{5}(-30) + -20$ $= -12 + -20$ $= -32$	$= 4(-8)$ $= -32$
Answers will vary. For example, $x = 1$ $y = 1$	$= \frac{2}{5}(15 - 30) + 10$ $= \frac{2}{5}(-15) + 10$ $= -6 + 10$ $= 4$	$= 4(4 - 2)$ $= 4(2)$ $= 8$

No, they are not equivalent.



Review the table and student responses to question 1b to ensure students are mastering the content. Correct any errors as necessary before moving on to question 2. This is an opportunity to encourage student precision in multiplication. Encourage students to check their work.

13.10.6 Your Turn!

Component Overview: Students work independently to determine which expressions are equivalent by substituting values for the variable.



13.10.6 Your Turn!

1. Determine which expressions are equivalent by substituting the same values for the variable.

A $(5x - 3) - \frac{3}{4}(8 + 2x)$ $ \begin{aligned} & x = 2 \\ & = (10 - 3) - \frac{3}{4}(8 + 4) \\ & = 7 - \frac{3}{4}(12) \\ & = 7 - 9 \\ & = -2 \end{aligned} $	B $-5\frac{1}{4}(4x - 1)$ $ \begin{aligned} & x = 2 \\ & = -\frac{21}{4}(8 - 1) \\ & = -\frac{21}{4}(7) \\ & = -\frac{147}{4} \\ & = -36\frac{3}{4} \end{aligned} $
C $9 - 6\left(\frac{-2}{3}x + 4\right)$ $ \begin{aligned} & x = 2 \\ & = 9 + -6\left(-\frac{4}{3} + 4\right) \\ & = 9 + 8 + -24 \\ & = -7 \end{aligned} $	D $\frac{3}{2}x - 9 + 2x$ $ \begin{aligned} & x = 2 \\ & = 3 - 9 + 4 \\ & = -6 + 4 \\ & = -2 \end{aligned} $
E $\left(\frac{30}{2}\right)(-1) + 4x$ $ \begin{aligned} & x = 2 \\ & = -15 + 8 \\ & = -7 \end{aligned} $	F $6 - 12\frac{1}{2}x - \frac{3}{4} - 8.5x$ $ \begin{aligned} & x = 2 \\ & = 6 + (-25) + -\frac{3}{4} + (-17) \\ & = -36\frac{3}{4} \end{aligned} $



Consider providing students with which expressions are equivalent as scaffolding. This acts as a self-check for students who need to build confidence and fluency in their calculations.

2. Complete each statement using the expressions from the table.

A is equivalent to D

C is equivalent to E

B is equivalent to F

13.10.7 Wrap-Up: Reflection

Component Overview: Students complete a self-assessment of their understanding of the lesson's learning targets using an emoji scale. Then students provide a written explanation to support their choice of emoji.

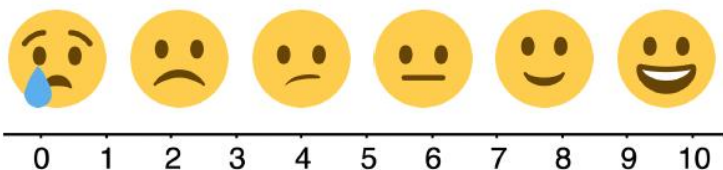


13.10.7 Wrap-Up: Reflection

1. Rate yourself using the scale below on today's learning target.

Answers will vary.

I can verify using substitution that linear expressions are equivalent.



2. Write an explanation of why you rated yourself where you did.

Answers will vary.

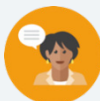


Consider allowing students to mark-up text and label each activity with an emoji. Students could also draw their own emoji for each lesson component.

13.11 What Values Make an Equation or Inequality True?

Lesson Introduction

 Benchmark	MA.6.AR.2.1 Given an equation or inequality and a specified set of integer values, determine which values make the equation or inequality true or false.		 Learning Target	I can determine if a given integer value makes an equation or inequality true.
 Benchmark Coherence	Building On MA.5.AR.2.3 Determine and explain whether an equation involving any of the four operations is true or false.		Working Toward MA.7.AR.2.1 Write and solve one-step inequalities in one variable within a mathematical context and represent solutions algebraically or graphically.	
 Essential Questions	- How are solutions to equations and inequalities the same? How are they different? - What is the difference between a strict inequality and an inequality? - How can you use mental math to determine whether inequalities are true or false?			
 Lesson Narrative	In this lesson, students determine if equations and inequalities are true by substituting values into the equation or inequality. Students learn the meaning of the equal sign and the inequality symbols. Students use mental math strategies to determine whether inequalities are true or false.			
 Component Summary	13.11.1 Warm-Up (~5 min.)	Day at Work – Film Composer (<i>SE p. 321</i>)	13.11.4 Guided Instruction (15 min.)	Determine if inequalities are true (<i>SE p. 323</i>)
	13.11.2 Guided Instruction (10 min.)	Determine if equations are true for a given value (<i>SE p. 321</i>)	13.11.5 Your Turn! (5 min.)	Practice using substitution to determine if inequalities are true for given values (<i>SE p. 326</i>)
	13.11.3 Guided Practice (5 min.)	Practice determining if equations are true (<i>SE p. 323</i>)	13.11.6 Wrap-Up (~5 min.)	Self-assess understanding of the learning target (<i>SE p. 326</i>)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> - Equation - Solution - Inequality - Strict inequality <u>Additional Terms:</u> N/A		Career Profile video	
			Additional Preparation	
			The 13.11.1 Warm-Up video can be found in the online Teacher Area to present for students before they answer questions in the Warm-Up.	

 Teacher Tips	Common Misconceptions		Things to Consider	
	- Students may believe the symbol $<$ can only mean or represent “less than,” even though an expression like $x < 3$ can be read as “3 is greater than x.” - Students may struggle with operations with integers.		- Avoid using common strategies like alligator inequalities that perpetuate misconceptions. Such shortcut methods that are not mathematically based, often confuse students. Instead, provide space and opportunity for students to articulate the meanings of symbols in their own words and give examples. - Consider giving students number lines to help them evaluate if the inequality is true or false. - If time is an issue, then consider assigning 13.11.5 Your Turn! for homework.	
	Literacy and Language Routines		Mathematical Thinking and Reasoning (MTR) Standard	
	True or False 13.11.4 Guided Instruction provides an opportunity for students to consider mental math strategies for determining whether inequalities are true.		MA.K12.MTR 6.1: Reasonableness Throughout the 13.11.2 Guided Instruction, 13.11.3 Guided Practice, 13.11.4 Guided Instruction, and 13.11.5 Your Turn!, students assess the reasonableness of their responses. Students are encouraged to use mental math to determine whether inequalities are true, encouraging mathematical reasoning.	
 Differentiate Instruction	Consider allowing students who need additional scaffolding access to the 6 th Grade On-Ramp video titled True or False Equations.			
Lesson Notes:				

13.11.1 Warm-Up: Day at Work – Film Composer

Component Overview: Play the “Day at Work: Film Composer” video for students. Students then reflect on the video and how the composer’s skills apply to mathematics.



13.11.1 Warm-Up: Day at Work – Film Composer

Shohan Cagle is a film composer. He writes music for commercials and soundtracks for TV shows and movies.



1. What did you notice about what he does to write a soundtrack?

Answers will vary. May include watching the entire movie/show. May include talking to the producer of the movie/show.

2. Shohan said that most people say they don’t normally hear the soundtrack. Name a TV show or a movie whose soundtrack you remember.

Answers will vary. The Wizard of Oz. Elf.

3. Shohan mentioned that one of his challenges is a short deadline. He said that his strategy to get his work done is “to sit and focus and get in the right mindset”. How can you use Shohan’s strategy?

Answers will vary. Before I start my assignment, I look over my notes. Before I start my assignment, I turn off all distractions around me.



Students may not be familiar with what a soundtrack is. Provide students with examples of soundtracks to familiarize them with the word. Challenge students to research how the careers they are interested in pursuing use math knowledge and skills.

13.11.2 Guided Instruction: Determining Values That Make an Equation True or False

Component Overview: Students begin by reviewing the definition of equation. Students use Guess, Check, and Revise to determine the number that produces the same value for the expression on the left and right side of the equal sign. This leads students to review the formal definition of a solution. Finally, students substitute values into equations to determine if the values are solutions to the equations.



13.11.2 Guided Instruction: Determining Values That Make an Equation True or False



An **equation** is a mathematical relation statement where two equivalent expressions and values are separated by an equal sign.

- In the equation $4x + 5 = 8x - 23$, what are the two expressions?
 $4x + 5$
 $8x - 23$
- Work with your partner to guess a value for x and then substitute it into both expressions. A high and a low guess have been provided for you in the table.

x	$4x + 5$	$8x - 23$	Are the expressions equal?
-3	$4(-3) + 5 = -7$	$8(-3) - 23 = -47$	No
14	$4(14) + 5 = 61$	$8(14) - 23 = 89$	No
	<i>Answers</i>	<i>will</i>	<i>vary</i>
10	$4(10) + 5 = 45$	$8(10) - 23 = 57$	No
7	$4(7) + 5 = 33$	$8(7) - 23 = 33$	Yes

For an equation to be **true**, both sides of the equation must have the same value. For example, the equation $78 - 56 = 2 \cdot 11$ is true because both expressions, $78 - 56$ and $2 \cdot 11$, are equal to 22.

The **solution** of an equation is the value(s) that, when substituted in for the variable(s), makes the equation true.



This component is intentionally scaffolded to build conceptual understanding of equations and solutions. After reviewing the definitions, students apply the definitions in the subsequent exercises. Check for student understanding as you circulate while students work. Debrief the activity, asking students to share their work.



Consider an inquiry-based approach to verifying solutions algebraically using substitution.

- Is -3 a solution to the equation? How do you know?
- How many solutions did the equation have?

13.11.2 Guided Instruction: Determining Values That Make an Equation True or False (continued)

3. Based on your guesses, what is the solution to the equation $4x + 5 = 8x - 23$?

$$x = 7$$

4. Use substitution to determine if the equation $2x + 4x + 7 = 19$ is true for each value.

x	$2x + 4x + 7 = 19$	Is the equation true?
-1	$2(-1) + 4(-1) + 7 = 19$ $-2 - 4 + 7 = 19$ $1 \neq 19$	No
0	$2(0) + 4(0) + 7 = 19$ $0 + 0 + 7 = 19$ $7 \neq 19$	No
2	$2(2) + 4(2) + 7 = 19$ $4 + 8 + 7 = 19$ $19 = 19$	Yes

5. The solution to the equation $2x + 4x + 7 = 19$ is $x = 2$ because when this value is substituted for x , the equation is *true*.



An auditory entry point is provided through class and partner discussion. Consider providing a visual entry point by having students use colored pencils to substitute values so they can keep track of substitutions.



Students may think that all equations will only have one solution. In Grade 7 Accelerated, students find solutions to $x^2 = p$, which has two solutions. If students think that equations will always only have one solution point ahead to what students will learn in Grade 7 Accelerated and beyond.

13.11.3 Guided Practice: Determining Which Values Make an Equation True or False

Component Overview: Students substitute integers into equations to determine which values make the equations true. Reinforce that students should show their work. If students have an incorrect answer, they can retrace the steps in their work to identify their mistakes.



13.11.3 Guided Practice: Determining Which Values Make an Equation True or False

From the specified set of integers, determine which, if any, make the equation true. Show your work.

1. $18 + 7w = 10w - 9$ $\{-9, -2, 4, 9, 11\}$

$$18 + 7(-9) = 10(-9) - 9 \quad 18 + 7(-2) = 10(-2) - 9$$

$$18 - 63 = -90 - 9 \quad 18 - 14 = -20 - 9$$

$$-45 \neq -99 \quad 4 \neq -29$$

Not True *Not True*

$$18 + 7(4) = 10(4) - 9 \quad 18 + 7(9) = 10(9) - 9$$

$$18 + 28 = 40 - 9 \quad 18 + 63 = 90 - 9$$

$$46 \neq 31 \quad 81 = 81$$

Not True *True*

$$18 + 7(11) = 10(11) - 9$$

$$18 + 77 = 110 - 9$$

$$95 \neq 101$$

Not True

2. $-2(3 - p) = 4p + 10$ $\{-8, -3, 0, 10\}$

$$-2(3 - (-8)) = 4(-8) + 10 \quad -2(3 - 0) = 4(0) + 10$$

$$-2(11) = -32 + 10 \quad -2(3) = 0 + 10$$

$$-22 = -22 \quad -6 \neq 10$$

True *False*

$$-2(3 - (-3)) = 4(-3) + 10 \quad -2(3 - 10) = 4(10) + 10$$

$$-2(6) = -12 + 10 \quad -2(-7) = 50$$

$$-12 \neq -2 \quad 14 \neq 50$$

False *False*



Consider selecting and assigning some (not all) of the given values to students. This will give students more time to work on each equation.



Use probing questions to encourage students to use mental math strategies and improve their fluency:
Do you always have to determine the value for each side of the equation to determine if the equation is true? Why or why not?

What strategies do you use to determining the value that make the equation true?

13.11.4 Guided Instruction: Determining Values That Make an Inequality True or False

Component Overview: Students review the definitions of inequality and strict inequality. Students use substitution to evaluate statements and determine if each inequality is true for the value of x . They also analyze a student's work to see how the student used mental math to determine if an inequality was true.



13.11.4 Guided Instruction: Determining Values That Make an Inequality True or False

A **strict inequality** is a mathematical relationship between two expressions that are not equal. A strict inequality uses the symbols, $<$, $>$, or $\neq$.

An **inequality** is a mathematical relationship between two expressions that are not equal ($<$, $>$, or $\neq$) or are not always equal ($\leq$ or $\geq$).

1. Use substitution to evaluate the mathematical statement. Then determine if each inequality is true for the value of x .

x	Strict Inequality $9x - 5 < 13$	Inequality $9x - 5 \leq 13$	Which inequality is true?
5	$9(5) - 5 < 13$ $40 < 13$	$9(5) - 5 \leq 13$ $40 \leq 13$	Neither
2	$9(2) - 5 < 13$ $13 < 13$	$9(2) - 5 \leq 13$ $13 \leq 13$	$\leq$ Inequality
-3	$9(-3) - 5 < 13$ $-32 < 13$	$9(-3) - 5 \leq 13$ $-32 \leq 13$	Both

2. Answer the following.

- a. The inequality $x < 5$, means any number that is

☐ smaller
☐ larger

than 5 will make the inequality true.

- b. List four numbers that are not integers that make the inequality true.

Answers will vary. $-\frac{3}{4}$, -0.5 , $\frac{3}{2}$, 3.25



Initially, students may struggle to understand the difference between inequalities and equations or may not understand how to read the symbols correctly. Students may also struggle to conceptually understand the difference between an inequality and a strict inequality. Ask guiding questions to help students make the conceptual connections:

- Do inequalities always express relationships between expressions that are not equal? Why or why not?
- What is the difference between a strict inequality and an inequality?



Utilize number lines as an instructional strategy for comparing values.

13.11.4 Guided Instruction: Determining Values That Make an Inequality True or False (continued)

3. Answer the following.

a. The inequality, $-11 \geq x$, means any number that is

- ☐ smaller than or equal to
- ☐ larger than or equal to

-11 will make the

inequality true.

b. List four numbers that are not integers that make the inequality true.

Answers will vary. $-\frac{83}{5}, -15.5, -\frac{64}{5}, -12.75$

4. Use substitution to determine if the inequality $16 \geq 7m - 5$ is true for each value specified for m .

m	$16 \geq 7m - 5$	Is the inequality true?
-2	$16 \geq 7(-2) - 5$ $16 \geq -14 - 5$ $16 \geq -19$	Yes
0	$16 \geq 7(0) - 5$ $16 \geq -5$	Yes
3	$16 \geq 7(3) - 5$ $16 \geq 21 - 5$ $16 \geq 16$	Yes
6	$16 \geq 7(6) - 5$ $16 \geq 42 - 5$ $16 \geq 37$	No



True or False

Provide an opportunity for students to consider mental math strategies for determining whether inequalities are true.

13.11.4 Guided Instruction: Determining Values That Make an Inequality True or False (continued)

5. Answer the following.

- a. Determine the integer(s) from the specified set that make the inequality **true**. Show your work.

$$-x + 3 + 5x \geq -8x - 7 \quad \{-12, -5, 5, 16\}$$

$$-(-12) + 3 + 5(-12) \geq -8(-12) - 7 \quad -(-5) + 3 + 5(-5) \geq -8(-5) - 7$$

$$12 + 3 - 60 \geq 96 - 7$$

$$5 + 3 - 25 \geq 40 - 7$$

$$15 - 60 \geq 89$$

$$-17 \geq 33$$

$$-45 \geq 89$$

False

False

$$-(5) + 3 + 5(5) \geq -8(5) - 7$$

$$-16 + 3 + 5(16) \geq -8(16) - 7$$

$$-5 + 3 + 25 \geq -47$$

$$-16 + 3 + 80 \geq -128 - 7$$

$$23 \geq -47$$

$$67 \geq -135$$

True

True

- b. When Diego substituted 16 in $-x + 3 + 5x \geq -8x - 7$, he stopped after writing $-16 + 3 + 5(16) \geq -8(16) - 7$ because he said he knew it was true. When his teacher asked him how he knew, he said it was because the left side would be a positive number and the right side would be a negative number.

With your partner, discuss how Diego knew the left side of the inequality would be a positive number and the right side of the inequality would be a negative number.

$$-16 + 3 + 5(16)$$

Since $5(16) + 3 > -16$ the left side is positive.

$$-8(16) - 7$$

Both values in this expression are negative, therefore their sum is a negative number.



As students discuss Diego's response, use guiding questions to connect their work in the previous question to Diego's work.

- *How did Diego determine if the inequality was true? Do you agree with this method?*
- *How did you use mental math to evaluate whether inequalities were true? How is this similar to Diego's strategy?*

13.11.4 Guided Instruction: Determining Values That Make an Inequality True or False (continued)

6. Determine which values from the specified set make the inequality **false**. Show your work.

$$12 < 2(13y - 1) \quad \{-14, 0, 1, 9\}$$

$$12 < 2(13(-14) - 1)$$

$$12 < 2(-182 - 1)$$

$$12 < 2(-183)$$

$$12 < -366$$

False

$$12 < 2(13(0) - 1)$$

$$12 < 2(0 - 1)$$

$$12 < 2(-1)$$

$$12 < -2$$

False

$$12 < 2(13(1) - 1)$$

$$12 < 2(13 - 1)$$

$$12 < 2(12)$$

$$12 < 24$$

True

$$12 < 2(13(9) - 1)$$

$$12 < 2(117 - 1)$$

$$12 < 2(116)$$

$$12 < 232$$

True

The values from the specified set that make the inequality $12 < 2(13y - 1)$, **false** are -14 and 0.



Ensure students read the directions for question 6 carefully, as they are finding values that make the inequality false. This is not a misconception so much as a potential error that students may make if they do not read carefully.

13.11.5 Your Turn!

Component Overview: Students independently use substitution to determine if inequalities are true or false for given integer values. Circulate while students work to monitor students' progress, support struggling students, and determine which students you will ask to share during the debrief.



13.11.5 Your Turn!

- Use substitution to determine if the inequality $86 \geq 17a - 8a - 23$ is true or false for each integer value specified for a .

a	$86 \geq 17a - 8a - 23$	True or False
-8	$86 \geq 17(-8) - 8(-8) - 23$ $86 \geq -136 + 64 - 23$ $86 \geq -95$	True
0	$86 \geq 17(0) - 8(0) - 23$ $86 \geq -23$	True
7	$86 \geq 17(7) - 8(7) - 23$ $86 \geq 119 - 56 - 23$ $86 \geq 40$	True
12	$86 \geq 17(12) - 8(12) - 23$ $86 \geq 204 - 96 - 23$ $86 \geq 85$	True
21	$86 \geq 17(21) - 8(21) - 23$ $86 \geq 357 - 168 - 23$ $86 \geq 166$	False



Consider assigning students a smaller subset of values to substitute into the inequalities to give them more time to evaluate each inequality and check their work.



For an extension, challenge students with inequalities or equations that have no solution or infinite solutions. For example: $4(7 - 3x) = 5x - 17x + 28$, $2(5x - 7) = 5(2x - 1)$, $8(3x + 7) > 6(4x - 3)$, $-2(5 - 6x) \leq 4(3x - 11)$.

13.11.6 Wrap-Up: Reflection

Component Overview: Students self-assess, identifying one question they have and any topics they are unsure about after the lesson. Students also brainstorm a plan for filling the gap in their understanding of the lesson's learning targets.



13.11.6 Wrap-Up: Reflection

1. One question I still have after today's lesson is ...

Answers will vary.

2. Today I am still unsure about ...

Answers will vary.

3. To fill in this gap I intend to ...

Answers will vary.



This 13.11.6 Wrap-Up requires a written explanation. Consider allowing students to explain their answers orally or record their responses using an app like FlipGrid.

13.12 Equivalent Expressions Using Properties of Operations

Lesson Introduction

 Benchmark	MA.7.AR.1.2 Determine whether two linear expressions are equivalent.		 Learning Targets	- I can use properties of operations to write equivalent forms of an algebraic expression. - I can determine whether two linear expressions are equivalent.
 Benchmark Coherence	Building On MA.6.AR.1.4 Apply the properties of operations to generate equivalent algebraic expressions with integer coefficients.		Working Toward MA.8.AR.1.3 Rewrite the sum of two algebraic expressions having a common monomial factor as a common factor multiplied by the sum of two algebraic expressions.	
 Essential Questions	How can you prove linear expressions are equivalent using properties of operations?			
 Lesson Narrative	Students extend the use of properties from numeric to algebraic expressions. Students begin by refining their working definitions of commutative, associative, and distributive properties in the context of numeric expressions before exploring what those properties look like through the lens of algebraic expressions. This lesson includes more spiraled practice on fluency with fractions.			
 Component Summary	13.12.1 Warm-Up (~5 min.)	Fact or Fiction? equivalent expressions (<i>SE p. 327</i>)	13.12.4 Guided Practice (10 min.)	Write equivalent algebraic expressions using the properties of operations (<i>SE p. 330</i>)
	13.12.2 Refresher (10 min.)	Reviewing associative, commutative, and distributive properties of numeric expressions to apply to algebraic expressions (<i>SE p. 328</i>)	13.12.5 Your Turn! (10 min.)	Students choose properties to rewrite equivalent algebraic expressions (<i>SE p. 330</i>)
	13.12.3 Guided Instruction (10 min.)	Extending properties from numeric to algebraic expressions (<i>SE p. 329</i>)	13.12.6 Wrap-Up (~5 min.)	Complete a self-assessment of ability to use properties of operations to write equivalent forms of algebraic expressions (<i>SE p. 331</i>)
 Resources	Glossary		Materials	
	Additional Preparation			
<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Associative property - Distributive property - Commutative property		(Optional) Note cards (Optional) Chart paper (Optional) Markers		Consider a Gallery Walk with chart paper for 13.12.2 Refresher.

 Teacher Tips	Common Misconceptions - When rewriting expressions using the distributive property, students may forget to distribute to all terms. - Students may add instead of multiply coefficient and substituted values, like $3x \rightarrow 3(2) \rightarrow 5$, or fail to use parentheses and interpret this as 32.	Things to Consider - Similarly, for variable expressions, encourage students to substitute values into the expressions and evaluate each expression to verify that the expressions are equivalent. - Colors may be helpful when substituting values in for the variables. - Determine if your students need the support of all or part of the 13.12.2 Refresher.
	Literacy and Language Routines Think-Pair-Share In 13.12.4 Guided Practice, students create equivalent expressions using properties of operations. Prior to independently recording the properties, ask students to engage in a Think-Pair-Share to swap their expressions with a partner and determine the properties their partners used.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.4.1: Discussion and Reflection After reviewing the associative, commutative, and distributive properties, students have multiple opportunities to apply, extend, and practice the properties. During partner work, encourage students to reflect on their own choice of properties to rewrite the expressions and compare that to those of their partner.
	 Differentiate Instruction	
13.12.5 Your Turn! offers student choice and multiple pathways as students apply any of the properties to different expressions. Consider a Four Corners activity in which each property is assigned a corner. Students must find a classmate from a different corner, show their work, and verify the work of another. Alternatively, students can hold up a number from one to three, with each number representing which property they applied. Students then must pair up with a classmate of the same number to verify their answers before finding a new partner with a different number and comparing their work.		
Lesson Notes:		

13.12.1 Warm-Up: Fact or Fiction?

Component Overview: Students recall the properties of operations to determine whether the numeric expression on the left side of an equation is equivalent to the numeric expression on the right side of the equation. To verify that numeric expressions are equivalent, encourage students to evaluate the expressions and review quickly as a class before moving on to 13.12.2 Refresher.



13.12.1 Warm-Up: Fact or Fiction?

- Determine whether each equation below is fact (true) or fiction (false).

Equation	Fact or Fiction
$(30 + 8) \cdot 9 = 270 + 72$	☐ Fact ☐ Fiction
$4 + 5(9 - 15) = 9(9 - 15)$	☐ Fact ☐ Fiction
$2 \cdot (7 + 8) = (2 \cdot 7) + 8$	☐ Fact ☐ Fiction
$(5 + 4) - 10 = 5 + (4 - 10)$	☐ Fact ☐ Fiction



Consider prompting questions to compare methods of evaluation and properties.

- Do you have to evaluate (or simplify) to determine if both sides of the equation are equivalent?
- Are the sides of each equation equal or equivalent?

13.12.2 Refresher: Rewriting Expressions Using Properties of Operations

Component Overview: Students apply the associative, commutative, and distributive properties to rewrite and evaluate expressions, connecting prior knowledge with numeric expressions to algebraic expressions. Then students work with a partner to discuss why the associative and commutative properties don't hold for all operations and create examples to prove their findings.



13.12.2 Refresher: Rewriting Expressions Using Properties of Operations

1. Fill in the missing blanks for each property.

- Using the **commutative property of addition**, $3 + 12$ can be rewritten as $\underline{12 + 3}$. Both expressions have a sum of $\underline{15}$.
- Using the **commutative property of addition**, $4x + 5$ can be rewritten as $\underline{5 + 4x}$. Both expressions will have the same sum regardless of the value of x .
- Using the **commutative property of multiplication**, $(4)(-2)$ can be rewritten as $\underline{(-2)(4)}$. Both expressions have a product of $\underline{-8}$.
- Using the **commutative property of multiplication**, $(-3y)(0.5)$ can be rewritten as $\underline{(0.5)(-3y)}$. Both expressions will have the same product regardless of the value of y .
- Using the **associative property of addition**, $-5 + (6 + 1)$ can be rewritten as $\underline{(-5 + 6) + 1}$. Both sums are equal to $\underline{2}$.
- Using the **associative property of addition**, $9x + (-2x + 5)$ can be rewritten as $\underline{(9x + -2x) + 5}$. Both expressions will have the same sum regardless of the value of x .
- Using the **associative property of multiplication**, $2 \cdot (7 \cdot 3)$ can be rewritten as $\underline{(2 \cdot 7) \cdot 3}$. Both products are equal to $\underline{42}$.
- Using the **associative property of multiplication**, $(-6)(4y)$ can be rewritten as $\underline{(-6 \cdot 4)y}$. Both expressions will have the same product regardless of the value of y .



Data-Driven Instruction

Allow data and informal observations from previous lessons in this unit to drive decision-making.

Small Groups: Utilize a teacher-led small group if you have a handful of students who need additional support.

Gallery Walk: Create three sets of all three properties (nine total so each poster only has three to four students at one time) on chart paper. Students must visit each property and write at least one example of that property. After they write one example for each property, they then must find a peer's examples and verify by writing a check mark if correct.

Utilize independent practice if students need time meaningfully engaging in producing examples. These are best utilized in context, so prioritize question 2 more than question 1.

13.12.2 Refresher: Rewriting Expressions Using Properties of Operations (continued)

- i. Using the **distributive property of multiplication over addition**, $5(6 + 2)$ can be rewritten as $(5 \cdot 6) + (5 \cdot 2)$. Both expressions have a value of 40.
 - j. Using the **distributive property of multiplication over addition**, $7(3y + 8)$ can be rewritten as $(7 \cdot 3y) + (7 \cdot 8)$ or $21y + 56$. Both expressions will have the same product regardless of the value of y .
2. Discuss with your partner why the associative property and the commutative property do not hold true for subtraction and division. Give an example of each to show mathematically that these properties do not apply to subtraction and division.
- The associative property does not hold true for subtraction. For example: $(10 - 5) - 2 = 3$ is not equal to $10 - (5 - 2) = 7$.*
- The associative property does not hold true for division. For example: $(50 \div 5) \div 2 = 5$ is not equal to $50 \div (5 \div 2) = 20$.*
- The commutative property does not hold true for subtraction. For example: $10 - 5 = 5$ is not equal to $5 - 10 = -5$.*
- The commutative property does not hold true for division. For example: $10 \div 5 = 2$ is not equal to $5 \div 10 = \frac{1}{2}$.*
3. Jay gave the example $6 + ^{-}11$ is equal to $^{-}11 + 6$ to show that subtraction is commutative. Explain to Jay why his example does not show that subtraction is commutative.
- Jay's problem shows that the commutative property holds for addition.*



Consider providing sentence starters for students who may struggle with written explanations. Alternatively, allow students to explain their responses orally to provide an additional entry point.

13.12.3 Guided Instruction: Equivalent Expressions Using Properties of Operations

Component Overview: Students apply the associative, commutative, and distributive properties to determine if algebraic expressions are equivalent.



13.12.3 Guided Instruction: Equivalent Expressions Using Properties of Operations

Algebraic expressions are considered equivalent if when evaluating the expression by substituting values in for variables, the resulting values are the same.

Algebraic expressions are also considered equivalent when one expression is rewritten using one or more of the properties of operations.

1. Complete each statement.

- a. The ☐ associative
☐ commutative
☒ distributive property can be used to

state that $5(x + 2) - 3x$ is equivalent to $5x + 10 - 3x$.

- b. The ☐ associative
☒ commutative
☐ distributive property can be used to

state that $\frac{2}{3}x + \left(\frac{3}{8} + \frac{5}{6}x\right)$ is equivalent to $\frac{2}{3}x + \left(\frac{5}{6}x + \frac{3}{8}\right)$.

2. Which properties can be used to show that $2(x - 4) + 5(3 + 6x)$ is equivalent to $2x + 30x - 8 + 15$?

The distributive property of multiplication over addition and over subtraction were used to obtain $2x - 8 + 15 + 30x$. Then, the commutative property was used to rearrange the terms and obtain $2x + 30x - 8 + 15$.



How can you use properties of operations to determine whether algebraic expressions are equivalent?



For students who may finish early, encourage them to rewrite the expression in question 2 in its simplest form and explain how they know it is in simplest form.

13.12.4 Guided Practice: Rewriting Expressions Using Properties of Operations

Component Overview: Students rewrite equivalent algebraic expressions using the properties of operations.



13.12.4 Guided Practice: Rewriting Expressions Using Properties of Operations

- Write an equivalent expression using a property of operations. State the property used.

Answers will vary.

Expression	Equivalent Expression	Property Used
$-6(2 - 3x) + 1 - \frac{x}{2}$	$= -12 + 18x + 1 - \frac{1}{2}x$ $= 18x - \frac{1}{2}x - 12 + 1$ $= 17\frac{1}{2}x - 11$	Distributive & Commutative Property
$12x + 10 + 4x - 17$	$= 16x - 7$	Commutative Property
$5x + (3x + 9) + 2$	$= 8x + 11$	Associative Property

- Several expressions are given below. Determine which expressions are equivalent using properties of operations.

$35x + (7x + 56)$	$5(8x + 7)$	$35x + 56$
$(8x + 7) \cdot 5$	$7(5x + 8)$	$(35x + 7x) + 56$

Complete the table by writing the expressions that are equivalent based on the property used.

Equivalent Expressions		Property Used
$35x + (7x + 56)$	$(35x + 7x) + 56$	Associative Property
$5(8x + 7)$	$(8x + 7) \cdot 5$	Commutative Property
$35x + 56$	$7(5x + 8)$	Distributive Property



Think-Pair-Share

Encourage mathematical discourse by implementing a Think-Pair-Share to position students as property experts in creating, verifying, and justifying responses.



Challenge students to write an equivalent expression that uses all three properties of operations (the associative, commutative, and distributive properties).

13.12.5 Your Turn!

Component Overview: Students write equivalent algebraic expressions using the properties of operations. There is more than one correct equivalent expression.



13.12.5 Your Turn!

Write an equivalent expression using the associative, commutative, or distributive property. State the property used.

Answers will vary.

1. $10x + \frac{2}{5}(-30 + 15x)$
 $= 10x + -12 + 6x$
Distributive property of multiplication over addition
2. $(18x + 15) + 4x - 17$
 $= (15 + 18x) + 4x - 17$
Commutative property of addition
3. $(-\frac{1}{5}x + 6) \cdot 5$
 $= 5 \cdot (-\frac{1}{5}x + 6)$
Commutative property of multiplication
4. $\frac{5}{4}n + \frac{7}{3} - \frac{11}{2}n - 12\frac{1}{3}$
 $= \frac{5}{4}n + -\frac{11}{2}n + \frac{7}{3} + -12\frac{1}{3}$
Commutative property of addition
5. $-3(\frac{1}{2}x + 4) - \frac{3}{5}(-1 + 10x)$
 $= \frac{-3}{2}x - 12 - \frac{3}{5}(-1 + 10x)$
Distributive property of multiplication over addition



There is more than one correct equivalent expression. Ask students to share and justify how they used properties to write an equivalent expression and consider having students evaluate the examples they have come up with as a class to encourage analysis. Correct student responses could be used to create a Card Sort as a review activity before the unit assessment.



Consider providing a specific property to use to rewrite each expression. This will encourage students to practice all three properties and may provide an entry point for students who may struggle getting started on their own.

13.12.6 Wrap-Up: Ticket Out the Door

Component Overview: Students use properties of operations to determine if two linear expressions are equivalent.



13.12.6 Wrap-Up: Ticket Out the Door

Use properties of operations to determine if the expressions are equivalent.

1. $\frac{4}{5}(x - 1) - \frac{1}{8}(2x + 1)$ and $\frac{11}{20}x - \frac{37}{40}$

The expressions are equivalent.

$$\begin{aligned} & \frac{4}{5}x - \frac{4}{5} - \frac{2}{8}x - \frac{1}{8} \\ &= \left(\frac{4}{5}x - \frac{2}{8}x\right) + \left(-\frac{4}{5} - \frac{1}{8}\right) \\ &= \left(\frac{11}{20}x\right) + \left(-\frac{37}{40}\right) \\ &= \frac{11}{20}x - \frac{37}{40} \end{aligned}$$

2. $1\frac{3}{10}x + \frac{1}{5}x - \frac{1}{10} - x - \frac{1}{10}$ and $1\frac{1}{10} - \frac{4}{5}x$

The expressions are not equivalent.

$$\begin{aligned} & \left(1\frac{3}{10}x + \frac{1}{5}x - x\right) + \left(-\frac{1}{10} - \frac{1}{10}\right) \\ &= \left(\frac{3}{10}x + \frac{2}{10}x\right) + \left(-\frac{2}{10}\right) \\ &= \left(\frac{5}{10}x\right) + \left(-\frac{2}{10}\right) \\ &= \frac{1}{2}x - \frac{1}{5} \end{aligned}$$



Consider prompting students to provide more information to justify their reasoning:

- Students can show their work and label each step of the work with a property.
- Prompt students to show their work and write a one sentence summary describing which property they used to justify equivalence.